Potential Energy

$$U = -C\left[\sum_I \frac{(\xi_I N_I - 1)\xi_I}{N_I} K_{II} + \sum_{I\neq J} \xi_I \xi_J K_{IJ} + \frac{4}{3}\sum_{I,J} \bar{\xi}_I \xi_J K_{\bar{I}J}\right]$$

Where,

$$C = \frac{18\xi^2}{5(5\eta-1)} \; 16\pi^2 \left(\frac{8\alpha_s}{3a}\right) \frac{R^2}{(6\pi^2)^2}$$

or, $$-\frac{U}{C} = \sum_I \frac{(\xi_I N_I - 1)\xi_I}{N_I} K_{II} + \sum_{I\neq J} \xi_I \xi_J K_{IJ} + \frac{4}{3}\sum_{I,J} \bar{\xi}_I \xi_J K_{\bar{I}J}$$

or, $$-\frac{U}{C} = \frac{(\xi_1 N_1 - 1)\xi_1}{N_1} K_{11} + \frac{(\xi_2 N_2 - 1)\xi_2}{N_2} K_{22} + \xi_1\xi_2 K_{12} + \xi_2\xi_1 K_{21}$$
$$+ \frac{4}{3}\bar{\xi}_1 \xi_1 K_{\bar{1}1} + \frac{4}{3}\bar{\xi}_1 \xi_2 K_{\bar{1}2} + \frac{4}{3}\bar{\xi}_2 \xi_1 K_{\bar{2}1}$$
$$+ \frac{4}{3}\bar{\xi}_2 \xi_2 K_{\bar{2}2}$$

Since, there are no light antiquarks, $\overline{f_1|v} = 0$.
It follows,

$$K_{\bar{I}1} = K_{\bar{I}2} = 0$$

$$\Rightarrow \boxed{-\frac{U}{C} = \frac{(\xi_1 N_1 - 1)\xi_1}{N_1} K_{11} + \frac{(\xi_2 N_2 - 1)\xi_2}{N_2} K_{22} + \xi_1\xi_2 K_{12} + \xi_2\xi_1 K_{21} + \frac{4}{3}\bar{\xi}_2 \xi_1 K_{\bar{2}1} + \frac{4}{3}\bar{\xi}_2 \xi_2 K_{\bar{2}2}}$$

Our job is now to find the values of $K_{11}, K_{22}, K_{12}, K_{21}, K_{\bar{2}1}$ & $K_{\bar{2}2}$, so that we can put these values in above equation and get the final expression of potential energy.

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Calculation of K_{11}

$$K_{11} = \int_0^{x_1} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{3/2} + \int_0^{x_1} dx \sqrt{x} (f_1(x))^{3/2} \int_x^{x_1} dx \frac{(f_1(x'))^{3/2}}{\sqrt{x'}}$$

$$\Rightarrow K_{11}\cdot 1 = \int_0^{x_1} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{3/2}$$

$$= \int_0^{\bar{x}_2} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{3/2} + \int_{\bar{x}_2}^{x_2} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{3/2}$$

$$+ \int_{x_2}^{x_1} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{3/2}$$

$$= (\beta_{1\bar{2}})^{3/2} \int_0^{\bar{x}_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} \left[\frac{(\beta_{1\bar{2}})^{3/2}}{Q_2} \left\{ x \frac{df_2(x)}{dx} - f_2(x) \right\} \right] + (\nu_{12})^{3/2} \int_{\bar{x}_2}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}}$$

$$\left[\frac{N_2}{3\eta} (\beta_{1\bar{2}})^{3/2} - \frac{N_2}{3\eta} (\nu_{12})^{3/2} (\beta_{2\bar{2}}) \frac{Q_2}{Q_2} + \frac{(\nu_{12})^{3/2}}{Q_2} \left\{ x \frac{df_2(x)}{dx} - f_2(x) \right\} \right]$$

$$+ \int_{x_2}^{x_1} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}} \left[\frac{N_2}{3\eta} (\beta_{1\bar{2}})^{3/2} + \frac{N_2}{3\eta} (\nu_{12})^{3/2} - \frac{N_2}{3\eta} (\nu_{12}) \frac{Q_2}{Q_1} - \frac{N_2}{3\eta} (\nu_{12})^{3/2} (\beta_{2\bar{2}})^{3/2} \right.$$

$$\left. - \frac{N_2}{3\eta} (\beta_{2\bar{2}}) (\nu_{12}) \frac{Q_2}{Q_1} + \frac{N_2}{3\eta} (\beta_{2\bar{2}})^{3/2} (\nu_{12}) \frac{Q_2}{Q_1} + \frac{1}{Q_1} \left\{ x \frac{df_1(x)}{dx} - f_1(x) \right\} \right]$$

$$= \frac{(\beta_{1\bar{2}})^3}{Q_2} \int_0^{\bar{x}_2} dx \sqrt{x} (f_2(x))^{3/2} \frac{df_2(x)}{dx} - \frac{(\beta_{1\bar{2}})^3}{Q_2} \int_0^{\bar{x}_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}} + \frac{N_2}{3\eta} (\beta_{1\bar{2}})^{3/2} (\nu_{12})^{3/2}$$

$$\int_{\bar{x}_2}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} - \frac{N_2}{3\eta} (\nu_{12})^3 (\beta_{2\bar{2}}) \frac{Q_2}{Q_2} \int_{\bar{x}_2}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} + \frac{(\nu_{12})^3}{Q_2} \int_{\bar{x}_2}^{x_2} dx \sqrt{x} (f_2(x))^{3/2} \frac{df_2(x)}{dx}$$

$$- \frac{(\nu_{12})^3}{Q_2} \int_{\bar{x}_2}^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}} + \left\{ \frac{N_2}{3\eta} (\beta_{1\bar{2}})^{3/2} + \frac{N_2}{3\eta} (\nu_{12})^{3/2} - \frac{N_2}{3\eta} (\nu_{12}) \frac{Q_2}{Q_1} \right.$$

$$\left. - \frac{N_2}{3\eta} (\nu_{12})^{3/2} (\beta_{2\bar{2}})^{3/2} - \frac{N_2}{3\eta} (\beta_{2\bar{2}}) (\nu_{12}) \frac{Q_2}{Q_1} + \frac{N_2}{3\eta} (\nu_{12}) (\beta_{2\bar{2}})^{3/2} \frac{Q_2}{Q_1} \right\}$$

$$\int_{\bar{x}_2}^{x_1} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}} + \frac{1}{Q_1} \int_{\bar{x}_2}^{x_1} dx \sqrt{x} (f_1(x))^{3/2} \frac{df_1(x)}{dx} - \frac{1}{Q_1} \int_{\bar{x}_2}^{x_1} dx \frac{(f_1(x))^{5/2}}{\sqrt{x}}$$

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$$= \frac{(\beta_{12})^3}{\varphi_2}\int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}\frac{df_2(x)}{dx} + \frac{(r_{12})^3}{\varphi_2}\int_{x_2}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}\frac{df_2(x)}{dx}$$

$$+\frac{1}{\varphi_1}\int_{x_2}^{x_1} dx\sqrt{x}\,(f_1(x))^{3/2}\frac{df_1(x)}{dx} - \frac{(\beta_{12})^3}{\varphi_2}\int_0^{x_2} dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}} - \frac{(r_{12})^3}{\varphi_2}\int_{x_2}^{x_2} dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

$$-\frac{1}{\varphi_1}\int_{x_2}^{x_1} dx\,\frac{(f_1(x))^{5/2}}{\sqrt{x}} + \left\{\frac{N_2}{3\eta}(\beta_{12})^{3/2}(r_{12})^{3/2} - \frac{N_2}{3\eta}(r_{12})^3(\beta_{22})\frac{\varphi_2}{\varphi_2}\right\}$$

$$\int_{x_2}^{x_2} dx\,\frac{(f_2(x))^{3/2}}{\sqrt{x}} + \left\{\frac{N_2}{3\eta}(\beta_{12})^{3/2} + \frac{N_2}{3\eta}(r_{12})^{3/2} - \frac{N_2}{3\eta}(r_{12})\frac{\varphi_2}{\varphi_1} - \frac{N_2}{3\eta}(r_{12})^{3/2}\right.$$

$$(\beta_{22})^{3/2} - \frac{N_2}{3\eta}(\beta_{22})(r_{12})\frac{\varphi_2}{\varphi_1} + \left.\frac{N_2}{3\eta}(r_{12})(\beta_{22})^{3/2}\frac{\varphi_2}{\varphi_1}\right\}\int_{x_2}^{x_1} dx\,\frac{(f_1(x))^{3/2}}{\sqrt{x}}$$

$$= \underbrace{-\frac{(\beta_{12})^3}{\varphi_2}\int_0^{x_2} dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}} - \frac{(r_{12})^3}{\varphi_2}\int_{x_2}^{x_2} dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}} - \frac{1}{\varphi_1}\int_{x_2}^{x_1} dx\,\frac{(f_1(x))^{5/2}}{\sqrt{x}}}$$

$$+ \underbrace{\left\{\frac{N_2}{3\eta}(\beta_{12})^{3/2}(r_{12})^{3/2} - \frac{N_2}{3\eta}(r_{12})^3(\beta_{22})\frac{\varphi_2}{\varphi_2}\right\}\int_{x_2}^{x_2} dx\,\frac{(f_2(x))^{3/2}}{\sqrt{x}}}$$

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$$+ \underbrace{\left\{\frac{N_2}{3\eta}(\beta_{12})^{3/2} + \frac{N_2}{3\eta}(r_{12})^{3/2} - \frac{N_2}{3\eta}(r_{12})\frac{\varphi_2}{\varphi_1} - \frac{N_2}{3\eta}(r_{12})^{3/2}(\beta_{22})^{3/2} - \frac{N_2}{3\eta}(\beta_{22})(r_{12})\frac{\varphi_2}{\varphi_1}\right.}$$

$$\left.+\frac{N_2}{3\eta}(r_{12})(\beta_{22})^{3/2}\frac{\varphi_2}{\varphi_1}\right\}\int_{x_2}^{x_1} dx\,\frac{(f_1(x))^{3/2}}{\sqrt{x}}$$

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Above equation is evaluated in three sections separately

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$$-\frac{(\beta_{12})^3}{\varphi_2}\int_0^{\bar{x}_2} dx\,\frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}} - \frac{(\gamma_{12})^3}{\varphi_2}\int_{\bar{x}_2}^{x_2} dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}} - \frac{1}{\varphi_1}\int_{x_2}^{x_1} dx\,\frac{(f_1(x))^{5/2}}{\sqrt{x}}$$

Substituting values of above integrations from appendix

$$= -\frac{(\beta_{12})^3}{\varphi_2}\left[\frac{4}{7}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\bar{x}_2} - \frac{5\overline{N_2}}{2\eta}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}\right] - \frac{(\gamma_{12})^3}{\varphi_2}\left[\frac{4}{7}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}\right.$$

$$-\frac{4}{7}(\beta_{2\bar{2}})^{5/2}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\bar{x}_2} - \frac{5}{7\varphi_2}\left\{\frac{N_2\varphi_2}{3\eta} + \beta_{2\bar{2}}\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta} - (\beta_{2\bar{2}})^{3/2}\frac{N_2\varphi_2}{3\eta}\right\}$$

$$\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5}{7\varphi_2}(\beta_{2\bar{2}})^2\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}\frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2}\Big] - \frac{1}{\varphi_1}\left[\frac{4}{7}\sqrt{x}\,(f_1(x))^{5/2}\Big|_{x_1}\right.$$

$$-\frac{4}{7}(\gamma_{12})^{5/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} - \frac{5}{7\varphi_1}\left\{\frac{N_1\varphi_1}{3\eta} + \gamma_{12}\frac{N_2\varphi_2}{3\eta} - (\gamma_{12})^{3/2}\frac{N_2\varphi_1}{3\eta}\right.$$

$$+\frac{\overline{N_2}\varphi_1}{3\eta}\left((\gamma_{12})^{3/2}(\beta_{2\bar{2}})^{3/2} - (\beta_{12})^{3/2}\right) - (\gamma_{12})(\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}$$

$$\left.+(\gamma_{12})(\beta_{2\bar{2}})\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}\right]\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{5}{7\varphi_1}(\gamma_{12})^2\left[\frac{N_2\varphi_2}{3\eta} + \beta_{2\bar{2}}\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}\right.$$

$$\left.-(\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}\right]\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= -\frac{4}{7}\frac{(\beta_{12})^3}{\varphi_2}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\bar{x}_2} + \frac{5\overline{N_2}}{2\eta}\frac{(\beta_{12})^3}{\varphi_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2}$$

$$-\frac{4}{7}\frac{(\gamma_{12})^3}{\varphi_2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} + \frac{4}{7}(\beta_{2\bar{2}})^{5/2}\frac{(\gamma_{12})^3}{\varphi_2}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\bar{x}_2}$$

$$+\frac{5}{7\varphi_2}\frac{N_2\varphi_2}{3\eta}\frac{(\gamma_{12})^3}{\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5}{7\varphi_2}\beta_{2\bar{2}}\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}\frac{(\gamma_{12})^3}{\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$-\frac{5}{7\varphi_2}\frac{\overline{N_2}\varphi_2}{3\eta}(\beta_{2\bar{2}})^{3/2}\frac{(\gamma_{12})^3}{\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5}{7\varphi_2}\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}(\beta_{2\bar{2}})^2$$

$$\frac{(\gamma_{12})^3}{\varphi_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2} - \frac{4}{7\varphi_1}\sqrt{x}\,(f_1(x))^{5/2}\Big|_{x_1} + \frac{4}{7\varphi_1}(\gamma_{12})^{5/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}$$

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$$+\frac{1}{Q_1}\frac{5}{7Q_1}\frac{N_1Q_1}{3\eta}\frac{df_1(x)}{dx}\Big|_{x_1}+\frac{5}{7Q_1}\frac{\mu_{12}N_2Q_2}{3\eta}\frac{1}{Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$-\frac{1}{Q_1}\frac{5}{7Q_1}(\mu_{12})^{3/2}\frac{N_2Q_1}{3\eta}\frac{df_1(x)}{dx}\Big|_{x_1}+\frac{1}{Q_1}\frac{5}{7Q_1}\frac{N_2Q_1}{3\eta}(\mu_{12})^{3/2}(\beta_{22})^{3/2}$$

$$\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{1}{Q_1}\frac{5}{7Q_1}\frac{N_2Q_1}{3\eta}(\beta_{12})^{3/2}\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{1}{Q_1}\frac{5}{7Q_1}(\mu_{12})(\beta_{22})^{3/2}$$

$$\frac{N_2Q_2}{3\eta}\frac{df_1(x)}{dx}\Big|_{x_1}+\frac{1}{Q_1}\frac{5}{7Q_1}(\mu_{12})(\beta_{22})\frac{N_2Q_2}{3\eta}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$-\frac{1}{Q_1}\frac{5}{7Q_1}(\mu_{12})^2N_2Q_2\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{1}{Q_1}\frac{5}{7Q_1}(\mu_{12})^2(\beta_{22})\frac{N_2Q_2}{3\eta}$$

$$\frac{df_2(x)}{dx}\Big|_{x_2}+\frac{1}{Q_1}\frac{5}{7Q_1}(\mu_{12})^2(\beta_{22})^{3/2}\frac{N_2Q_2}{3\eta}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$=\frac{4}{7Q_2}(\beta_{22})^{5/2}(\mu_{12})^3\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\overline{x_2}}-\frac{4}{7Q_2}(\beta_{12})^3\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\overline{x_2}}$$

$$+\frac{4}{7Q_1}(\mu_{12})^{5/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}-\frac{4}{7Q_2}(\mu_{12})^3\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}$$

$$-\frac{4}{7Q_1}\sqrt{x}\,(f_1(x))^{5/2}\Big|_{x_1}+\frac{5N_2}{21\eta Q_2}(\beta_{12})^3\frac{df_2(x)}{dx}\Big|_{\overline{x_2}}-\frac{5N_2Q_2}{21\eta Q_2^2}$$

$$(\beta_{22})^2(\mu_{12})^3\frac{df_2(x)}{dx}\Big|_{\overline{x_2}}+\frac{5N_2}{21\eta Q_2}(\mu_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+\frac{5N_2Q_2}{21\eta Q_2^2}(\beta_{22})(\mu_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{5N_2}{21\eta Q_2}(\beta_{22})^{3/2}(\mu_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$-\frac{5N_2Q_2}{21\eta Q_1^2}(\mu_{12})^2\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{5N_2Q_2}{21\eta Q_1^2}(\mu_{12})^2(\beta_{22})\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+\frac{5N_2Q_2}{21\eta Q_1^2}(\mu_{12})^2(\beta_{22})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2}+\frac{5N_2}{21\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$+\frac{5N_2Q_2}{21\eta Q_1^2}(\mu_{12})\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{5N_2}{21\eta Q_1}(\mu_{12})^{3/2}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$+\frac{5N_2}{21\eta Q_1}(\mu_{12})^{3/2}(\beta_{22})^{3/2}\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{5N_2}{21\eta Q_1}(\beta_{12})^{3/2}\frac{df_1(x)}{dx}\Big|_{x_1}$$

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$$- \frac{5\overline{N_2}\Phi_2}{2\pi\eta\Phi_1^2}(r_{12})(\beta_{22})^{3/2}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{5\overline{N_2}\,\overline{\Phi_2}}{2\pi\eta\Phi_1^2}(r_{12})(\beta_{22})\frac{df_1(x)}{dx}\Big|_{x_1} \quad \text{— (A)}$$

$K_{11}-5$

$$\left\{\frac{N_2}{3\eta}(\beta_{1\bar{2}})^{3/2}(r_{12})^{3/2} - \frac{N_2}{3\eta}(r_{12})^3(\beta_{2\bar{2}})\frac{\varphi_2}{\varphi_2}\right\}\int_{\bar{x}_2}^{x_2} dx\,\frac{(f_2(x))^{3/2}}{\sqrt{x}}$$

$$= \frac{1}{\varphi_2}\left\{\frac{N_2}{3\eta}(\beta_{1\bar{2}})^{3/2}(r_{12})^{3/2} - \frac{N_2}{3\eta}(r_{12})^3(\beta_{2\bar{2}})\frac{\varphi_2}{\varphi_2}\right\}\int_{\bar{x}_2}^{x_2} dx\,\frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{\varphi_2}\left\{\frac{N_2}{3\eta}(\beta_{1\bar{2}})^{3/2}(r_{12})^{3/2} - \frac{N_2}{3\eta}(r_{12})^3(\beta_{2\bar{2}})\frac{\varphi_2}{\varphi_2}\right\}\left[\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{df_2(x)}{dx}\Big|_{\bar{x}_2}\right]$$

$$= \frac{N_2}{3\eta\varphi_2}(\beta_{1\bar{2}})^{3/2}(r_{12})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{N_2}{3\eta}(r_{12})^3(\beta_{2\bar{2}})\frac{\varphi_2}{\varphi_2^2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{N_2}{3\eta\varphi_2}(\beta_{1\bar{2}})^{3/2}(r_{12})^{3/2}(\beta_{2\bar{2}})\frac{df_2(x)}{dx}\Big|_{\bar{x}_2} + \frac{N_2}{3\eta}(r_{12})^3(\beta_{2\bar{2}})^2\frac{\varphi_2}{\varphi_2^2}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}$$

Section - III (1)

$$\left\{\frac{N_2}{3\eta}(\beta_{1\bar{2}})^{3/2} + \frac{N_2}{3\eta}(r_{12})^{3/2} - \frac{N_2}{3\eta}(r_{12})\frac{\varphi_2}{\varphi_1} - \frac{N_2}{3\eta}(r_{12})^{3/2}(\beta_{2\bar{2}})^{3/2}\right.$$
$$\left. - \frac{N_2}{3\eta}(\beta_{2\bar{2}})(r_{12})\frac{\varphi_2}{\varphi_1} + \frac{N_2}{3\eta}(r_{12})(\beta_{2\bar{2}})^{3/2}\frac{\varphi_2}{\varphi_1}\right\}\int_{x_2}^{x_1} dx\,\frac{(f_1(x))^{3/2}}{\sqrt{x}}$$

$$= \frac{1}{\varphi_2}\left\{\frac{N_2}{3\eta}(\beta_{1\bar{2}})^{3/2} + \frac{N_2}{3\eta}(r_{12})^{3/2} - \frac{N_2}{3\eta}(r_{12})\frac{\varphi_2}{\varphi_1} - \frac{N_2}{3\eta}(r_{12})^{3/2}(\beta_{2\bar{2}})^{3/2}\right.$$
$$\left. - \frac{N_2}{3\eta}(\beta_{2\bar{2}})(r_{12})\frac{\varphi_2}{\varphi_1} + \frac{N_2}{3\eta}(r_{12})(\beta_{2\bar{2}})^{3/2}\frac{\varphi_2}{\varphi_1}\right\}\int_{x_2}^{x_1} dx\,\frac{d^2 f_1(x)}{dx^2}$$

$$= \frac{1}{\varphi_2}\left\{\frac{N_2}{3\eta}(\beta_{1\bar{2}})^{3/2} + \frac{N_2}{3\eta}(r_{12})^{3/2} - \frac{N_2}{3\eta}(r_{12})\frac{\varphi_2}{\varphi_1} - \frac{N_2}{3\eta}(r_{12})^{3/2}(\beta_{2\bar{2}})^{3/2}\right.$$
$$\left. - \frac{N_2}{3\eta}(\beta_{2\bar{2}})(r_{12})\frac{\varphi_2}{\varphi_1} + \frac{N_2}{3\eta}(r_{12})(\beta_{2\bar{2}})^{3/2}\frac{\varphi_2}{\varphi_1}\right\}\left[\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{df_1(x)}{dx}\Big|_{x_2}\right]$$

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$$= \frac{N_2}{3\eta Q_1}(\beta_{12})^{3/2} + \frac{N_2}{3\eta Q_1}(M_{12})^{3/2} - \frac{N_2}{3\eta}(M_{12})\frac{Q_2}{Q_1^2} - \frac{N_2}{3\eta Q_1}(M_{12})^{3/2}(\beta_{22})^{3/2}$$

$$- \frac{N_2}{3\eta}(\beta_{22})(M_{12})\frac{Q_2}{Q_1^2} + \frac{N_2}{3\eta}(M_{12})(\beta_{22})^{3/2}\frac{Q_2}{Q_1^2}\Big\}\frac{df_1(\alpha)}{d\alpha}\Big|_{\alpha_1}$$

$$- \Big\{\frac{N_2}{3\eta Q_1}(\beta_{12})^{3/2}(M_{12}) + \frac{N_2}{3\eta Q_1}(M_{12})^{5/2} - \frac{N_2}{3\eta}(M_{12})^2\frac{Q_2}{Q_1^2} - \frac{N_2}{3\eta Q_1}(M_{12})^{5/2}(\beta_{22})^{3/2}$$

$$- \frac{N_2}{3\eta}(\beta_{22})(M_{12})^2\frac{Q_2}{Q_1^2} + \frac{N_2}{3\eta}(\beta_{22})^{3/2}(M_{12})^2\frac{Q_2}{Q_1^2}\Big\}\frac{df_2(\alpha)}{d\alpha}\Big|_{\alpha_2}$$

Now, adding all three sections

$$K_{11} \cdot 1 = \frac{4}{7Q_2}(\beta_{22})^{5/2}(M_{12})^3\sqrt{\alpha}\,(f_2(\alpha))^{5/2}\Big|_{\alpha_2} - \frac{4}{7Q_2}(\beta_{12})^3\sqrt{\alpha}\,(f_2(\alpha))^{5/2}\Big|_{\alpha_2}$$

$$+ \frac{4}{7Q_1}(M_{12})^{5/2}\sqrt{\alpha}\,(f_2(\alpha))^{5/2}\Big|_{\alpha_2} - \frac{4}{7Q_2}(M_{12})^3\sqrt{\alpha}\,(f_2(\alpha))^{5/2}\Big|_{\alpha_2} - \frac{4}{7Q_1}\sqrt{\alpha}\,(f_1(\alpha))^{5/2}\Big|_{\alpha_1}$$

$$+ \frac{5N_2}{2\eta Q_2}(\beta_{12})^3\frac{df_2(\alpha)}{d\alpha}\Big|_{\alpha_2} - \frac{5}{2\eta}\frac{N_2Q_2}{Q_2^2}(\beta_{22})^2(M_{12})^3\frac{df_2(\alpha)}{d\alpha}\Big|_{\alpha_2}$$

$$+ \frac{5N_2}{2\eta Q_2}(M_{12})^3\frac{df_2(\alpha)}{d\alpha}\Big|_{\alpha_2} + \frac{5N_2Q_2}{2\eta Q_2^2}(\beta_{22})(M_{12})^3\frac{df_2(\alpha)}{d\alpha}\Big|_{\alpha_2} - \frac{5N_2}{2\eta Q_2}(\beta_{22})^{3/2}(M_{12})^3$$

$$\frac{df_2(\alpha)}{d\alpha}\Big|_{\alpha_2} - \frac{5N_2Q_2}{2\eta Q_1^2}(M_{12})^2\frac{df_2(\alpha)}{d\alpha}\Big|_{\alpha_2} - \frac{5N_2Q_2}{2\eta Q_1^2}(M_{12})^2(\beta_{22})\frac{df_2(\alpha)}{d\alpha}\Big|_{\alpha_2}$$

$$+ \frac{5N_2Q_2}{2\eta Q_1^2}(M_{12})^2(\beta_{22})^{3/2}\frac{df_2(\alpha)}{d\alpha}\Big|_{\alpha_2} + \frac{5N_1}{2\eta Q_1}\frac{df_1(\alpha)}{d\alpha}\Big|_{\alpha_1} + \frac{5N_2Q_2}{2\eta Q_1^2}(M_{12})\frac{df_1(\alpha)}{d\alpha}\Big|_{\alpha_1}$$

$$- \frac{5N_2}{2\eta Q_1}(M_{12})^{3/2}\frac{df_1(\alpha)}{d\alpha}\Big|_{\alpha_1} + \frac{5N_2}{2\eta Q_1}(M_{12})^{3/2}(\beta_{22})^{3/2}\frac{df_1(\alpha)}{d\alpha}\Big|_{\alpha_1} - \frac{5N_2}{2\eta Q_1}(\beta_{12})^{3/2}\frac{df_1(\alpha)}{d\alpha}\Big|_{\alpha_1}$$

$$- \frac{5N_2Q_2}{2\eta Q_1^2}(M_{12})(\beta_{22})^{3/2}\frac{df_1(\alpha)}{d\alpha}\Big|_{\alpha_1} + \frac{5N_2Q_2}{2\eta Q_1^2}(M_{12})(\beta_{22})\frac{df_1(\alpha)}{d\alpha}\Big|_{\alpha_1}$$

$$+ \Big\{\frac{N_2}{3\eta}(M_{12})^3(\beta_{22})^2\frac{Q_2}{Q_2^2} - \frac{N_2}{3\eta Q_2}(\beta_{12})^{3/2}(M_{12})^{3/2}(\beta_{22})\Big\}\frac{df_2(\alpha)}{d\alpha}\Big|_{\alpha_2}$$

$$+ \Big\{\frac{N_2}{3\eta Q_2}(\beta_{12})^{3/2}(M_{12})^{3/2} - \frac{N_2}{3\eta}(M_{12})^3(\beta_{22})\frac{Q_2}{Q_2^2}\Big\}\frac{df_2(\alpha)}{d\alpha}\Big|_{\alpha_2}$$

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$$+\left\{\frac{\overline{N_2}}{3\eta Q_1}(\beta_{12})^{3/2}+\frac{\overline{N_2}}{3\eta Q_1}(r_{12})^{3/2}-\frac{N_2}{3\eta}(r_{12})\frac{Q_2}{Q_1^2}-\frac{\overline{N_2}}{3\eta Q_1}(r_{12})^{3/2}(\beta_{22})^{3/2}\right.$$

$$-\frac{N_2}{3\eta}(\beta_{22})(r_{12})\frac{Q_2}{Q_1^2}+\frac{\overline{N_2}}{3\eta}(r_{12})(\beta_{22})^{3/2}\frac{Q_2}{Q_1^2}\left.\right\}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$-\left\{\frac{\overline{N_2}}{3\eta Q_1}(\beta_{12})^{3/2}(r_{12})+\frac{\overline{N_2}}{3\eta Q_1}(r_{12})^{5/2}-\frac{N_2}{3\eta}(r_{12})^2\frac{Q_2}{Q_1^2}-\frac{\overline{N_2}}{3\eta Q_1}(r_{12})^{5/2}(\beta_{22})^{3/2}\right.$$

$$-\frac{N_2}{3\eta}(\beta_{22})(r_{12})^2\frac{Q_2}{Q_1^2}+\frac{N_2}{3\eta}(\beta_{22})^{3/2}(r_{12})^2\frac{Q_2}{Q_1^2}\left.\right\}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$=\frac{4}{7Q_2}(\beta_{22})^{5/2}(r_{12})^3\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\overline{x_2}}-\frac{4}{7\overline{Q_2}}(\beta_{12})^3\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\overline{x_2}}$$

$$+\frac{4}{7Q_1}(r_{12})^{5/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}-\frac{4}{7Q_2}(r_{12})^3\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}-\frac{4}{7Q_1}\sqrt{x}\,(f_1(x))^{5/2}\Big|_{x_1}$$

$$+\frac{5\overline{N_2}}{2\eta\overline{Q_2}}(\beta_{12})^3\frac{df_2(x)}{dx}\Big|_{\overline{x_2}}-\frac{5}{2\eta}\frac{\overline{N_2Q_2}}{Q_2^2}(\beta_{22})^2(r_{12})^3\frac{df_2(x)}{dx}\Big|_{\overline{x_2}}$$

$$+\left\{\frac{\overline{N_2}}{3\eta}(r_{12})^3(\beta_{22})^2\frac{\overline{Q_2}}{Q_2^2}-\frac{\overline{N_2}}{3\eta Q_2}(\beta_{12})^{3/2}(r_{12})^{3/2}(\beta_{22})\right\}\frac{df_2(x)}{dx}\Big|_{\overline{x_2}}$$

$$+\frac{5N_2}{2\eta Q_2}(r_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2}+\frac{5\overline{N_2}\,\overline{Q_2}}{2\eta Q_2^2}(\beta_{22})(r_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{5\overline{N_2}}{2\eta Q_2}(\beta_{22})^{3/2}(r_{12})^3$$

$$\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{5N_2Q_2}{2\eta Q_1^2}(r_{12})^2\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{5\overline{N_2}\,\overline{Q_2}}{2\eta Q_1^2}(r_{12})^2(\beta_{22})\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+\frac{5\overline{N_2}}{Q_2\eta}\frac{Q_2}{Q_1^2}(r_{12})^2(\beta_{22})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2}+\left\{\frac{\overline{N_2}}{3\eta Q_2}(\beta_{12})^{3/2}(r_{12})^{3/2}-\frac{\overline{N_2}}{3\eta}(r_{12})^3(\beta_{22})\frac{Q_2}{Q_2^2}\right.$$

$$-\frac{\overline{N_2}}{3\eta Q_1}(\beta_{12})^{3/2}(r_{12})-\frac{\overline{N_2}}{3\eta Q_1}(r_{12})^{5/2}+\frac{N_2}{3\eta}(r_{12})^2\frac{Q_2}{Q_1^2}+\frac{\overline{N_2}}{3\eta Q_1}(r_{12})^{5/2}(\beta_{22})^{3/2}$$

$$+\frac{\overline{N_2}}{3\eta}(\beta_{22})(r_{12})^2\frac{\overline{Q_2}}{Q_1^2}-\frac{\overline{N_2}}{3\eta}(\beta_{22})^{3/2}(r_{12})^2\frac{Q_2}{Q_1^2}\left.\right\}\frac{df_2(x)}{dx}\Big|_{x_2}+\frac{5N_2}{2\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$+\frac{5N_2Q_2}{2\eta Q_1^2}(r_{12})\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{5N_2}{2\eta Q_1}(r_{12})^{3/2}\frac{df_1(x)}{dx}\Big|_{x_1}+\frac{5\overline{N_2}}{2\eta Q_1}(r_{12})^{3/2}(\beta_{22})^{3/2}$$

$$\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{5\overline{N_2}}{2\eta Q_1}(\beta_{12})^{3/2}\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{5\overline{N_2}Q_2}{2\eta Q_1^2}(r_{12})(\beta_{22})^{3/2}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$+\frac{5\overline{N_2}\,\overline{Q_2}}{2\eta Q_1^2}(r_{12})(\beta_{22})\frac{df_1(x)}{dx}\Big|_{x_1}+\left\{\frac{\overline{N_2}}{3\eta Q_1}(\beta_{12})^{3/2}+\frac{\overline{N_2}}{3\eta Q_1}(r_{12})^{3/2}-\frac{N_2}{3\eta}(r_{12})\frac{Q_2}{Q_1^2}\right.$$

$$-\frac{\overline{N_2}}{3\eta Q_1}(r_{12})^{3/2}(\beta_{22})^{3/2}-\frac{\overline{N_2}}{3\eta}(\beta_{22})(r_{12})\frac{\overline{Q_2}}{Q_1^2}+\frac{\overline{N_2}}{3\eta}(r_{12})(\beta_{22})^{3/2}\frac{Q_2}{Q_1^2}\left.\right\}\frac{df_1(x)}{dx}\Big|_{x_1}$$

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$$K_{11}\cdot 2 = \int_0^{x_1} dx\sqrt{x}\,(f_1(x))^{3/2} \int_x^{x_1} dx' \frac{(f_2(x'))^{3/2}}{\sqrt{x'}}$$

$$= \int_0^{\bar{x}_2} dx\sqrt{x}\,(f_1(x))^{3/2} \int_x^{x_1} dx' \frac{(f_1(x'))^{3/2}}{\sqrt{x'}} + \int_{\bar{x}_2}^{x_2} dx\sqrt{x}\,(f_1(x))^{3/2} \int_x^{x_1} dx' \frac{(f_1(x'))^{3/2}}{\sqrt{x'}}$$

$$+ \int_{x_2}^{x_1} dx\sqrt{x}\,(f_1(x))^{3/2} \int_x^{x_1} dx' \frac{(f_1(x'))^{3/2}}{\sqrt{x'}}$$

Substituting values from appendix

$$= (\beta_{1\bar{1}})^{3/2} \int_0^{\bar{x}_2} dx\sqrt{x}\,(\overline{f_2(x)})^{\frac{3}{2}} \left[\frac{(\beta_{1\bar{2}})^{3/2}}{\bar{\varphi}_2} \left\{ \frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2} - \frac{d\overline{f_2(x)}}{dx} \right\} \right.$$

$$\left. + \frac{(r_{12})^{3/2}}{\bar{\varphi}_2} \left\{ \frac{df_2(x)}{dx}\Big|_{x_2} - \beta_{2\bar{2}} \frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2} \right\} + \frac{1}{\varphi_1} \left\{ \frac{df_1(x)}{dx}\Big|_{x_1} - r_{12} \frac{df_2(x)}{dx}\Big|_{x_2} \right\} \right]$$

$$+ (r_{12})^{\frac{3}{2}} \int_{\bar{x}_2}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \left[\frac{(r_{12})^{3/2}}{\varphi_2} \left\{ \frac{df_2(x)}{dx}\Big|_{x_2} - \frac{df_2(x)}{dx} \right\} + \frac{1}{\varphi_1} \left\{ \frac{df_1(x)}{dx}\Big|_{x_1} \right.\right.$$

$$\left.\left. - r_{12} \frac{df_2(x)}{dx}\Big|_{x_2} \right] + \int_{x_2}^{x_1} dx\sqrt{x}\,(f_1(x))^{3/2} \left[\frac{1}{\varphi_1} \left\{ \frac{df_1(x)}{dx}\Big|_{x_1} - \frac{df_1(x)}{dx} \right\} \right]$$

$$= \frac{(\beta_{1\bar{2}})^3}{\bar{\varphi}_2} \int_0^{\bar{x}_2} dx\sqrt{x}\,(\overline{f_2(x)})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2} - \frac{(\beta_{1\bar{2}})^3}{\bar{\varphi}_2} \int_0^{\bar{x}_2} dx\sqrt{x}\,(\overline{f_2(x)})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}$$

(1)

$$+ \frac{(\beta_{1\bar{2}})^{3/2}(r_{12})^{3/2}}{\varphi_2} \int_0^{\bar{x}_2} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2} \frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\beta_{1\bar{2}})^{3/2}(r_{12})^{3/2}(\beta_{2\bar{2}})}{\varphi_2}$$

$$\int_0^{\bar{x}_2} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2} \frac{df_2(x)}{dx}\Big|_{\bar{x}_2} + \frac{(\beta_{1\bar{2}})^{3/2}}{\varphi_1} \int_0^{\bar{x}_2} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2} \frac{df_1(x)}{dx}\Big|_{x_1}$$

$$- \frac{(\beta_{1\bar{2}})^{3/2}(r_{12})}{\varphi_1} \int_0^{\bar{x}_2} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2} \frac{df_2(x)}{dx}\Big|_{x_2} + \frac{(r_{12})^3}{\bar{\varphi}_2} \int_{\bar{x}_2}^{x_2} dx\sqrt{x}$$

(2)

$$(f_2(x))^{3/2} \frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(r_{12})^3}{\bar{\varphi}_2} \int_{\bar{x}_2}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \frac{df_2(x)}{dx} +$$

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$$+\frac{(\gamma_{12})^{3/2}}{\varphi_1}\int_{\bar{x}_2}^{x_2} d\alpha\sqrt{\alpha}\,(f_2(\alpha))^{3/2}\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1} - \frac{(\gamma_{12})^{5/2}}{\varphi_1}\int_{\bar{x}_2}^{x_2} d\alpha\sqrt{\alpha}\,(f_2(\alpha))^{3/2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2}$$

$$+\frac{1}{\varphi_1}\int_{\bar{x}_2}^{x_1} d\alpha\sqrt{\alpha}\,(f_1(\alpha))^{3/2}\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1} - \frac{1}{\varphi_1}\int_{\bar{x}_2}^{x_1} d\alpha\sqrt{\alpha}\,(f_1(\alpha))^{3/2}\frac{df_1(\alpha)}{d\alpha}$$

$$=\frac{(\beta_{1\bar{2}})^3}{\varphi_2}\int_{0}^{\bar{x}_2} d\alpha\sqrt{\alpha}\,(\overline{f_2(\alpha)})^{3/2}\frac{\overline{df_2(\alpha)}}{d\alpha}\Big|_{\bar{x}_2} + \frac{(\beta_{1\bar{2}})^{3/2}(\gamma_{12})^{3/2}}{\varphi_2}\int_{0}^{\bar{x}_2} d\alpha\sqrt{\alpha}\,(\overline{f_2(\alpha)})^{3/2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2}$$

$$-\frac{(\beta_{1\bar{2}})^{3/2}(\gamma_{12})^{3/2}(\beta_{2\bar{2}})}{\varphi_2}\int_{0}^{\bar{x}_2} d\alpha\sqrt{\alpha}\,(\overline{f_2(\alpha)})^{3/2}\frac{\overline{df_2(\alpha)}}{d\alpha}\Big|_{\bar{x}_2} + \frac{(\beta_{1\bar{2}})^{3/2}}{\varphi_1}\int_{0}^{\bar{x}_2} d\alpha\sqrt{\alpha}\,(\overline{f_2(\alpha)})^{3/2}$$

$$\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1} - \frac{(\beta_{1\bar{2}})^{3/2}(\gamma_{12})}{\varphi_1}\int_{0}^{\bar{x}_2} d\alpha\sqrt{\alpha}\,(\overline{f_2(\alpha)})^{3/2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} + \frac{(\gamma_{12})^3}{\varphi_2}\int_{\bar{x}_2}^{x_2} d\alpha$$

$$\sqrt{\alpha}\,(f_2(\alpha))^{3/2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} + \frac{(\gamma_{12})^{3/2}}{\varphi_1}\int_{\bar{x}_2}^{x_2} d\alpha\sqrt{\alpha}\,(f_2(\alpha))^{3/2}\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1}$$

$$-\frac{(\gamma_{12})^{5/2}}{\varphi_1}\int_{\bar{x}_2}^{x_2} d\alpha\sqrt{\alpha}\,(f_2(\alpha))^{3/2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} + \frac{1}{\varphi_1}\int_{x_2}^{x_1} d\alpha\sqrt{\alpha}\,(f_1(\alpha))^{3/2}\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1}$$

$$=\frac{(\beta_{1\bar{2}})^3}{\varphi_2}\frac{\overline{N_2}}{3\eta}\overline{df_2(\alpha)}\Big|_{\bar{x}_2} + \frac{(\beta_{1\bar{2}})^{3/2}(\gamma_{12})^{3/2}}{\varphi_2}\frac{N_2}{3\eta}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} - \frac{(\beta_{1\bar{2}})^{3/2}(\gamma_{12})^{3/2}(\beta_{2\bar{2}})}{\varphi_2}$$

$$\frac{\overline{N_2}}{3\eta}\overline{df_2(\alpha)}\Big|_{\bar{x}_2} + \frac{(\beta_{1\bar{2}})^{3/2}}{\varphi_1}\frac{\overline{N_2}}{3\eta}\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1} - \frac{(\beta_{1\bar{2}})^{3/2}(\gamma_{12})}{\varphi_1}\frac{N_2}{3\eta}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2}$$

$$+\frac{(\gamma_{12})^3}{\varphi_2^2}\left[x\frac{df_2(\alpha)}{d\alpha} - f_2(\alpha)\right]_{x_2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} - \frac{(\gamma_{12})^3(\beta_{2\bar{2}})}{\varphi_2^2}\left[x\overline{\frac{df_2(\alpha)}{d\alpha}} - \overline{f_2(\alpha)}\right]\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2}$$

$$+\frac{(\gamma_{12})^{3/2}}{\varphi_1\varphi_2}\left[x\frac{df_2(\alpha)}{d\alpha} - f_2(\alpha)\right]_{x_2}\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1} - \frac{(\gamma_{12})^{3/2}(\beta_{2\bar{2}})}{\varphi_1\varphi_2}\left[x\overline{\frac{df_2(\alpha)}{d\alpha}} - \overline{f_2(\alpha)}\right]\frac{df_1(\alpha)}{d\alpha}\Big|_{\bar{x}_2}$$

$$-\frac{(\gamma_{12})^{5/2}}{\varphi_1\varphi_2}\left[x\frac{df_2(\alpha)}{d\alpha} - f_2(\alpha)\right]_{x_2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} + \frac{(\gamma_{12})^{5/2}(\beta_{2\bar{2}})}{\varphi_1\varphi_2}\left[x\overline{\frac{df_2(\alpha)}{d\alpha}} - \overline{f_2(\alpha)}\right]\frac{df_2(\alpha)}{d\alpha}\Big|_{\bar{x}_2}$$

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$$+\frac{1}{\varphi_1^2}\left[x\frac{df_1(x)}{dx}-f_1(x)\right]_{x_1}\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{\gamma_{12}}{\varphi_1^2}\left[x\frac{df_2(x)}{dx}-f_2(x)\right]_{x_2}^{x_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$=\frac{\overline{N_2}}{3\eta}\frac{(\beta_{12})^3}{\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}+\frac{\overline{N_2}}{3\eta}\frac{(\beta_{12})^{3/2}(\gamma_{12})^{3/2}}{\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{\overline{N_2}}{3\eta}\frac{(\beta_{12})^{3/2}(\gamma_{12})^{3/2}(\beta_{22})}{\varphi_2}$$

$$\frac{df_2(x)}{dx}\Big|_{x_2}+\frac{\overline{N_2}}{3\eta}\frac{(\beta_{12})^{3/2}}{\varphi_1}\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{\overline{N_2}}{3\eta}\frac{(\beta_{12})^{3/2}(\gamma_{12})}{\varphi_1}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$-\frac{\overline{N_2}\overline{\varphi_2}}{3\eta}\frac{(\gamma_{12})^3(\beta_{22})}{\varphi_2^2}\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{\overline{N_2}\overline{\varphi_2}}{3\eta}\frac{(\gamma_{12})^{3/2}}{\varphi_1\varphi_2}(\beta_{22})\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$+\frac{\overline{N_2}\overline{\varphi_2}}{3\eta}\frac{(\gamma_{12})^{5/2}}{\varphi_1\varphi_2}(\beta_{22})\frac{df_2(x)}{dx}\Big|_{x_2}+\frac{N_2\varphi_2}{3\eta}\frac{(\gamma_{12})^3}{\varphi_2^4}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+\frac{\overline{N_2}\overline{\varphi_2}}{3\eta}(\beta_{22})\frac{(\gamma_{12})^3}{\varphi_2^2}\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{\overline{N_2}\varphi_2}{3\eta}(\beta_{22})^{3/2}\frac{(\gamma_{12})^3}{\varphi_2^4}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+\frac{N_2\varphi_2}{3\eta}\frac{(\gamma_{12})^{3/2}}{\varphi_2\varphi_2}\frac{df_1(x)}{dx}\Big|_{x_1}+\frac{\overline{N_2}\overline{\varphi_2}}{3\eta}(\beta_{22})\frac{(\gamma_{12})^{3/2}}{\varphi_1\varphi_2}\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{\overline{N_2}\varphi_2}{3\eta}$$

$$(\beta_{22})^{3/2}\frac{(\gamma_{12})^{3/2}}{\varphi_1\varphi_2}\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{N_2\varphi_2}{3\eta}\frac{(\gamma_{12})^{5/2}}{\varphi_1\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$-\frac{\overline{N_2}\overline{\varphi_2}}{3\eta}(\beta_{22})\frac{(\gamma_{12})^{5/2}}{\varphi_2\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}+\frac{\overline{N_2}\varphi_2}{3\eta}(\beta_{22})^{3/2}\frac{(\gamma_{12})^{5/2}}{\varphi_1\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$-\frac{N_2\varphi_2}{3\eta}\frac{\gamma_{12}}{\varphi_1^2}\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{\overline{N_2}\varphi_2}{3\eta}(\beta_{22})\frac{(\gamma_{12})}{\varphi_1^2}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$+\frac{\overline{N_2}\varphi_2}{3\eta}(\beta_{22})^{3/2}\frac{(\gamma_{12})}{\varphi_1^2}\frac{df_1(x)}{dx}\Big|_{x_1}+\frac{N_1\varphi_1}{3\eta}\cdot\frac{1}{\varphi_1^4}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$+\frac{N_2\varphi_2}{3\eta}\frac{(\gamma_{12})}{\varphi_1^2}\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{N_2\varphi_2}{3\eta}\frac{(\gamma_{12})^{3/2}}{\varphi_1^4}\frac{df_1(x)}{dx}\Big|_{x_1}+\frac{\overline{N_2}\varphi_2}{3\eta\varphi_1^4}\left\{(\gamma_{12})^{3/2}(\beta_{22})^{3/2}\right.$$

$$\left.-(\beta_{12})^{3/2}\right\}\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{\overline{N_2}\varphi_2}{3\eta\varphi_1^2}(\gamma_{12})(\beta_{22})^{3/2}\frac{df_1(x)}{dx}\Big|_{x_1}+\frac{\overline{N_2}\overline{\varphi_2}}{3\eta\varphi_1^2}(\gamma_{12})(\beta_{22})$$

$$\frac{df_1(x)}{dx}\Big|_{x_1}$$

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$$= \frac{N_2}{3\eta}\frac{(\beta_{12})^3}{Q_2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} - \frac{N_2}{3\eta}\frac{(\beta_{12})^{3/2}(r_{12})^{3/2}(\beta_{22})}{Q_2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} + \frac{N_2}{3\eta Q_2}(\beta_{12})^{3/2}(r_{12})^{3/2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2}$$

$$- \frac{N_2}{3\eta Q_1}(\beta_{12})^{3/2}(r_{12})\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} - \frac{N_2 Q_2}{3\eta Q_2^2}(r_{12})^3(\beta_{22})\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} + \frac{N_2 Q_2}{3\eta}\frac{(r_{12})^{5/2}}{Q_1 Q_2}$$

$$(\beta_{22})\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} + \frac{N_2}{3\eta Q_2}(r_{12})^3\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} + \frac{N_2 Q_2}{3\eta Q_2^2}(\beta_{22})(r_{12})^3\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2}$$

$$- \frac{N_2}{3\eta Q_2}(\beta_{22})^{3/2}(r_{12})^3\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} - \frac{N_2}{3\eta Q_1}(r_{12})^{5/2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} - \frac{N_2 Q_2}{3\eta Q_1 Q_2}(\beta_{22})$$

$$(r_{12})^{5/2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} + \frac{N_2}{3\eta Q_1}(\beta_{22})^{3/2}(r_{12})^{5/2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} + \frac{N_2}{3\eta}\frac{(\beta_{12})^{3/2}}{Q_1}\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1}$$

$$- \frac{N_2 Q_2}{3\eta Q_1 Q_2}(r_{12})^{3/2}(\beta_{22})\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1} + \frac{N_2}{3\eta Q_1}(r_{12})^{3/2}\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1} + \frac{N_2 Q_2}{3\eta Q_1 Q_2}(\beta_{22})$$

$$(r_{12})^{3/2}\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1} - \frac{N_2}{3\eta Q_1}(\beta_{22})^{3/2}(r_{12})^{3/2}\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1} - \frac{N_2 Q_2}{3\eta}\frac{r_{12}}{Q_1^2}\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1}$$

$$- \frac{N_2 Q_2}{3\eta Q_1^2}(\beta_{22})(r_{12})\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1} + \frac{N_2 Q_2}{3\eta Q_1^2}(\beta_{22})^{3/2}(r_{12})\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1}$$

$$+ \frac{N_1}{3\eta Q_1}\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1} + \frac{N_2 Q_2}{3\eta Q_1^2}(r_{12})\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1} - \frac{N_2 (r_{12})^{3/2}}{3\eta Q_1}\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1}$$

$$+ \frac{N_2}{3\eta Q_1}\left\{(r_{12})^{3/2}(\beta_{22})^{3/2} - (\beta_{12})^{3/2}\right\}\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1} - \frac{N_2 Q_2}{3\eta Q_1^2}(r_{12})$$

$$(\beta_{22})^{3/2}\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1} + \frac{N_2 Q_2}{3\eta Q_1^2}(r_{12})(\beta_{22})\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1}$$

$$= \frac{N_2}{3\eta}\frac{(\beta_{12})^3}{Q_2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} - \frac{N_2}{3\eta}\frac{(\beta_{12})^{3/2}(r_{12})^{3/2}(\beta_{22})}{Q_2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} + \frac{N_2}{3\eta Q_2}(\beta_{12})^{3/2}(r_{12})^{3/2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2}$$

$$- \frac{N_2}{3\eta Q_1}(\beta_{12})^{3/2}(r_{12})\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} + \frac{N_2}{3\eta Q_2}(r_{12})^3\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} - \frac{N_2}{3\eta Q_2}(\beta_{22})^{3/2}(r_{12})^3\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2}$$

$$- \frac{N_2}{3\eta Q_1}(r_{12})^{5/2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2} + \frac{N_2}{3\eta Q_1}(\beta_{22})^{3/2}(r_{12})^{5/2}\frac{df_2(\alpha)}{d\alpha}\Big|_{x_2}$$

$$+ \frac{N_1}{3\eta Q_1}\frac{df_1(\alpha)}{d\alpha}\Big|_{x_1} \qquad \text{——— (B)}$$

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Adding $K_{11}\cdot 1$ & $K_{11}\cdot 2$

$K_{11} = K_{11}\cdot 1 \oplus K_{11}\cdot 2$

$$= \frac{4}{7Q_2}(\beta_{2\bar{2}})^{5/2}(r_{12})^3\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} - \frac{4}{7Q_2}(\beta_{1\bar{2}})^3\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}$$

$$+\frac{4}{7Q_1}(r_{12})^{5/2}\sqrt{x}\,(f_1(x))^{5/2}\Big|_{x_2} - \frac{4}{7Q_2}(r_{12})^3\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} - \frac{4}{7Q_1}\sqrt{x}\,(f_1(x))^{5/2}\Big|_{x_1}$$

$$+\frac{5\bar{N}_2}{21\eta\bar{Q}_2}(\beta_{1\bar{2}})^3\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\bar{N}_2\bar{Q}_2}{21\eta Q_2^2}(\beta_{2\bar{2}})^2(r_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+\Big\{\frac{\bar{N}_2}{3\eta}\frac{(\beta_{1\bar{2}})^3}{\bar{Q}_2} - \frac{\bar{N}_2}{3\eta}\frac{(\beta_{1\bar{2}})^{3/2}(r_{12})^{3/2}(\beta_{2\bar{2}})}{Q_2} + \frac{\bar{N}_2}{3\eta}(r_{12})^3(\beta_{2\bar{2}})^2\frac{\bar{Q}_2}{Q_2^2}$$

$$-\frac{\bar{N}_2}{3\eta Q_2}(\beta_{1\bar{2}})^{3/2}(r_{12})^{3/2}(\beta_{2\bar{2}})\Big\}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5N_2}{21\eta Q_2}(r_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+\frac{5\bar{N}_2\bar{Q}_2}{21\eta Q_2^2}(\beta_{2\bar{2}})(r_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\bar{N}_2}{21\eta Q_2}(\beta_{2\bar{2}})^{3/2}(r_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$-\frac{5N_2Q_2}{21\eta Q_1^2}(r_{12})^2\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\bar{N}_2Q_2}{21\eta Q_1^2}(r_{12})^2(\beta_{2\bar{2}})\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+\frac{5\bar{N}_2}{21\eta}\frac{Q_2}{Q_1^2}(r_{12})^2(\beta_{2\bar{2}})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2} + \Big\{\frac{\bar{N}_2}{3\eta Q_2}(\beta_{1\bar{2}})^{3/2}(r_{12})^{3/2} - \frac{\bar{N}_2}{3\eta Q_1}(\beta_{1\bar{2}})^{3/2}(r_{12})$$

$$+\frac{N_2}{3\eta Q_2}(r_{12})^3 - \frac{\bar{N}_2}{3\eta Q_2}(\beta_{2\bar{2}})^{3/2}(r_{12})^3 - \frac{N_2}{3\eta Q_1}(r_{12})^{5/2} + \frac{\bar{N}_2}{3\eta Q_1}(\beta_{2\bar{2}})^{3/2}(r_{12})^{5/2}$$

$$+\frac{\bar{N}_2}{3\eta Q_2}(\beta_{1\bar{2}})^{3/2}(r_{12})^{3/2} - \frac{\bar{N}_2}{3\eta}(r_{12})^3(\beta_{2\bar{2}})\frac{\bar{Q}_2}{Q_2^2} - \frac{\bar{N}_2}{3\eta Q_1}(\beta_{1\bar{2}})^{3/2}(r_{12})$$

$$-\frac{\bar{N}_2}{3\eta Q_1}(r_{12})^{5/2} + \frac{N_2}{3\eta}(r_{12})^2\frac{Q_2}{Q_1^2} + \frac{\bar{N}_2}{3\eta Q_1}(r_{12})^{5/2}(\beta_{2\bar{2}})^{3/2} + \frac{\bar{N}_2}{3\eta}(\beta_{2\bar{2}})(r_{12})^2\frac{\bar{Q}_2}{Q_2^2}$$

$$-\frac{\bar{N}_2}{3\eta}(\beta_{2\bar{2}})^{3/2}(r_{12})^2\frac{Q_2}{Q_1^2}\Big\}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5N_1}{21\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{5N_2Q_2}{21\eta Q_1^2}(r_{12})\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$-\frac{5N_2}{21\eta Q_1}(r_{12})^{3/2}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{5\bar{N}_2}{21\eta Q_1}(r_{12})^{3/2}(\beta_{2\bar{2}})^{3/2}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{5\bar{N}_2}{21\eta Q_1}(\beta_{1\bar{2}})^{3/2}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$-\frac{5\bar{N}_2Q_2}{21\eta Q_1^2}(r_{12})(\beta_{2\bar{2}})^{3/2}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{5\bar{N}_2\bar{Q}_2}{21\eta Q_1^2}(r_{12})(\beta_{2\bar{2}})\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$+\Big\{\frac{N_1}{3\eta Q_1} + \frac{\bar{N}_2}{3\eta Q_1}(\beta_{1\bar{2}})^{3/2} + \frac{\bar{N}_2}{3\eta Q_1}(r_{12})^{3/2} - \frac{N_2}{3\eta}(r_{12})\frac{Q_2}{Q_2^2} - \frac{\bar{N}_2}{3\eta Q_1}(r_{12})^{3/2}(\beta_{2\bar{2}})^{3/2}$$

$$-\frac{\bar{N}_2}{3\eta}(\beta_{2\bar{2}})(r_{12})\frac{\bar{Q}_2}{Q_2^2} + \frac{\bar{N}_2}{3\eta}(r_{12})(\beta_{2\bar{2}})^{3/2}\frac{Q_2}{Q_1^2}\Big\}\frac{df_1(x)}{dx}\Big|_{x_1}$$

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$$= \frac{4}{7}\left[\frac{(\beta_{22})^{5/2}(\gamma_{12})^3}{\varphi_2} - \frac{(\beta_{12})^3}{\varphi_2}\right]\sqrt{x}\,\overline{(f_2(x))^{5/2}}\Big|_{x_2}$$

$$+\frac{4}{7}\left[\frac{(\gamma_{12})^{5/2}}{\varphi_1} - \frac{(\gamma_{12})^3}{\varphi_2}\right]\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} - \frac{4}{7\varphi_1}\sqrt{x}\,(f_1(x))^{5/2}\Big|_{x_1}$$

$$+\left[\frac{N_1}{3\eta}\left\{\frac{(\beta_{12})^3}{\varphi_2} - \frac{(\beta_{12})^{3/2}(\gamma_{12})^{3/2}(\beta_{22})}{\varphi_2} + (\gamma_{12})^3(\beta_{22})^2\frac{\varphi_2}{\varphi_2^2}\right.\right.$$

$$\left.\left. - \frac{(\beta_{12})^{3/2}(\gamma_{12})^{3/2}(\beta_{22})}{\varphi_2}\right\} + \frac{5N_2}{2\eta}\left\{\frac{(\beta_{12})^3}{\varphi_2} - (\beta_{22})^2(\gamma_{12})^3\frac{\varphi_2}{\varphi_2^2}\right\}\right]$$

$$\frac{d\overline{f_2(x)}}{dx}\Big|_{x_2} + \left[\frac{N_2}{3\eta}\left\{\frac{(\beta_{12})^{3/2}(\gamma_{12})^{3/2}}{\varphi_2} - \frac{(\beta_{12})^{3/2}(\gamma_{12})}{\varphi_1} - \frac{(\beta_{22})^{3/2}(\gamma_{12})^3}{\varphi_2}\right.\right.$$

$$+\frac{(\beta_{22})^{3/2}(\gamma_{12})^{5/2}}{\varphi_1} + \frac{(\beta_{12})^{3/2}(\gamma_{12})^{3/2}}{\varphi_2} - (\gamma_{12})^3(\beta_{22})\frac{\varphi_2}{\varphi_2^2} - \frac{(\beta_{12})^{3/2}(\gamma_{12})}{\varphi_1}$$

$$\left.\left. - \frac{(\gamma_{12})^{5/2}}{\varphi_1} + \frac{(\gamma_{12})^{5/2}(\beta_{22})^{3/2}}{\varphi_1} + (\beta_{22})(\gamma_{12})^2\frac{\varphi_2}{\varphi_1^2} - (\beta_{12})^{3/2}(\gamma_{12})^2\frac{\varphi_2}{\varphi_1^2}\right\}\right.$$

$$+\frac{N_2}{3\eta}\left\{\frac{(\gamma_{12})^3}{\varphi_2} - \frac{(\gamma_{12})^{5/2}}{\varphi_1} + (\gamma_{12})^2\frac{\varphi_2}{\varphi_1^2}\right\} + \frac{5}{2\eta}\left\{\frac{N_2}{\varphi_2}(\gamma_{12})^3 + \frac{N_2\varphi_2}{\varphi_2^2}(\beta_{22})(\gamma_{12})^3\right.$$

$$-\frac{N_2}{\varphi_2}(\beta_{22})^{3/2}(\gamma_{12})^3 - \frac{N_2\varphi_2}{\varphi_1^2}(\gamma_{12})^2 - \frac{N_2\varphi_2}{\varphi_1^2}(\gamma_{12})^2(\beta_{22}) + \frac{N_2\varphi_2}{\varphi_1^2}(\gamma_{12})^2$$

$$\left.(\beta_{22})^{3/2}\right\}\left]\frac{d\overline{f_2(x)}}{dx}\right|_{x_2} + \left[\frac{N_1}{3\eta\varphi_1} - \frac{N_2}{3\eta}(\gamma_{12})\frac{\varphi_2}{\varphi_1^2} + \frac{N_2}{3\eta\varphi_1}\left\{(\beta_{12})^{3/2}\right.\right.$$

$$+(\gamma_{12})^{3/2} - (\gamma_{12})^{3/2}(\beta_{22})^{3/2} - (\beta_{22})(\gamma_{12})\frac{\varphi_2}{\varphi_1} + (\gamma_{12})(\beta_{22})^{3/2}\frac{\varphi_2}{\varphi_1}\Big\}$$

$$+\frac{5}{2\eta}\left\{\frac{N_1}{\varphi_1} + \frac{N_2\varphi_2}{\varphi_1^2}(\gamma_{12}) - \frac{N_2}{\varphi_1}(\gamma_{12})^{3/2} + \frac{N_2}{\varphi_1}(\gamma_{12})^{3/2}(\beta_{22})^{3/2} - \frac{N_2}{\varphi_1}(\beta_{12})^{3/2}\right.$$

$$\left.\left. - \frac{N_2\varphi_2}{\varphi_1^2}(\gamma_{12})(\beta_{22})^{3/2} + \frac{N_2\varphi_2}{\varphi_1^2}(\gamma_{12})(\beta_{22})\right\}\right]\frac{df_1(x)}{dx}\Big|_{x_1}$$

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$$= \frac{4}{7}\left[\frac{(\beta_{2\bar{2}})^{5/2}(r_{12})^3}{\varphi_2} - \frac{(\beta_{1\bar{2}})^3}{\varphi_2}\right]\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} + \frac{4}{7}\left[\frac{(r_{12})^{5/2}}{\varphi_1} - \frac{(r_{12})^3}{\varphi_2}\right]\sqrt{x}$$

$$(f_2(x))^{5/2}\Big|_{x_2} - \frac{4}{7\varphi_1}\sqrt{x}\,(f_1(x))^{5/2}\Big|_{x_1} + \left[\frac{N_2}{3\eta}\left\{\frac{(\beta_{1\bar{2}})^3}{\varphi_2} + (r_{12})^3(\beta_{2\bar{2}})^2\frac{\varphi_2}{\varphi_2^2}\right.\right.$$

$$\left.\left. - \frac{2(\beta_{1\bar{2}})^{3/2}(r_{12})^{3/2}(\beta_{2\bar{2}})}{\varphi_2}\right\} + \frac{5N_2}{21\eta}\left\{\frac{(\beta_{1\bar{2}})^3}{\varphi_2} - (\beta_{2\bar{2}})^2(r_{12})^3\frac{\varphi_2}{\varphi_2^2}\right\}\right]$$

$$\frac{df_2(x)}{dx}\Big|_{x_2} + \left[\frac{N_2}{3\eta}\left\{\frac{2(\beta_{1\bar{2}})^{3/2}(r_{12})^{3/2}}{\varphi_2} - \frac{2(\beta_{1\bar{2}})^{3/2}(r_{12})}{\varphi_1}\right.\right.$$

$$+ \frac{2(\beta_{2\bar{2}})^{3/2}(r_{12})^{5/2}}{\varphi_1} - \frac{(\beta_{2\bar{2}})^{3/2}(r_{12})^3}{\varphi_2} - (r_{12})^3(\beta_{2\bar{2}})\frac{\varphi_2}{\varphi_2^2}$$

$$\left. - \frac{(r_{12})^{5/2}}{\varphi_1} + (\beta_{2\bar{2}})(r_{12})^2\frac{\varphi_2}{\varphi_1^2} - (\beta_{2\bar{2}})^{3/2}(r_{12})^2\frac{\varphi_2}{\varphi_1^2}\right\}$$

$$+ \frac{N_2}{3\eta}\left\{\frac{(r_{12})^3}{\varphi_2} - \frac{(r_{12})^{5/2}}{\varphi_1} + (r_{12})^2\frac{\varphi_2}{\varphi_1^2}\right\} + \frac{5}{21\eta}\left\{\frac{N_2}{\varphi_2}(r_{12})^3\right.$$

$$+ \frac{N_2\varphi_2}{\varphi_2^2}(\beta_{2\bar{2}})(r_{12})^3 - \frac{N_2}{\varphi_2}(\beta_{2\bar{2}})^{3/2}(r_{12})^3 - \frac{N_2\varphi_2}{\varphi_1^2}(r_{12})^2 - \frac{N_2\varphi_2}{\varphi_1^2}(r_{12})^2$$

$$\left.(\beta_{2\bar{2}}) + \frac{N_2\varphi_2}{\varphi_1^2}(r_{12})^2(\beta_{2\bar{2}})^{3/2}\right\}\Big]\frac{df_2(x)}{dx}\Big|_{x_2} + \left[\frac{N_1}{3\eta\varphi_1}\right.$$

$$- \frac{N_2}{3\eta}(r_{12})\frac{\varphi_2}{\varphi_1^2} + \frac{N_2}{3\eta\varphi_1}\left\{(\beta_{1\bar{2}})^{3/2} + (r_{12})^{3/2} - (r_{12})^{3/2}(\beta_{2\bar{2}})^{3/2}\right.$$

$$\left. - (\beta_{2\bar{2}})(r_{12})\frac{\varphi_2}{\varphi_1} + (r_{12})(\beta_{2\bar{2}})^{3/2}\frac{\varphi_2}{\varphi_1}\right\} + \frac{5}{21\eta}\left\{\frac{N_1}{\varphi_1} + \frac{N_2\varphi_2}{\varphi_1^2}(r_{12})\right.$$

$$- \frac{N_2}{\varphi_1}(r_{12})^{3/2} + \frac{N_2}{\varphi_1}(r_{12})^{3/2}(\beta_{2\bar{2}})^{3/2} - \frac{N_2}{\varphi_1}(\beta_{1\bar{2}})^{3/2} - \frac{N_2\varphi_2}{\varphi_1^2}(r_{12})$$

$$\left.(\beta_{2\bar{2}})^{3/2} + \frac{N_2\varphi_2}{\varphi_1^2}(r_{12})(\beta_{2\bar{2}})\right\}\Big]\frac{df_1(x)}{dx}\Big|_{x_1} \#$$

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$$\int_0^x dx \sqrt{x} \, (f_1(x))^{3/2}$$

In region $0 < x < \overline{x_2}$, $\overline{f_2(x)}$ is dominant

$$(\beta_{12})^{3/2} \int_0^x dx \sqrt{x} \, (\overline{f_2(x)})^{3/2}$$

$$= (\beta_{12})^{3/2} \int_0^x dx \, \frac{x}{\overline{\varphi_2}} \, \frac{d^2 \overline{f_2(x)}}{dx^2}$$

$$= \frac{(\beta_{12})^{3/2}}{\overline{\varphi_2}} \int_0^x dx \, x \, \frac{d^2 \overline{f_2(x)}}{dx^2}$$

$$= \frac{(\beta_{12})^{3/2}}{\overline{\varphi_2}} \left[x \, \frac{d \overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]$$

In region $\overline{x_2} < x < x_2$, $f_2(x)$ is dominant

$$= \int_0^{\overline{x_2}} dx \sqrt{x} \, (f_1(x))^{3/2} + \int_{\overline{x_2}}^x dx \sqrt{x} \, (f_1(x))^{3/2}$$

$$= (\beta_{12})^{3/2} \int_0^{\overline{x_2}} dx \sqrt{x} \, (\overline{f_2(x)})^{3/2} + (\gamma_{12})^{3/2} \int_{\overline{x_2}}^x dx \sqrt{x} \, (f_2(x))^{3/2}$$

$$= \frac{\overline{N_2}}{3\eta} (\beta_{12})^{3/2} + \frac{(\gamma_{12})^{3/2}}{\varphi_2} \int_{\overline{x_2}}^x dx \, x \, \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{\overline{N_2}}{3\eta} (\beta_{12})^{3/2} + \frac{(\gamma_{12})^{3/2}}{\varphi_2} \left[x \, \frac{d f_2(x)}{dx} - f_2(x) \right]_{\overline{x_2}}^x$$

$$= \frac{\overline{N_2}}{3\eta} (\beta_{12})^{3/2} + \frac{(\gamma_{12})^{3/2}}{\varphi_2} \left[x \, \frac{d f_2(x)}{dx} - f_2(x) \right] - \frac{(\gamma_{12})^{3/2}}{\varphi_2} \left[x \, \frac{d f_2(x)}{dx} - f_2(x) \right]_{\overline{x_2}}$$

$$= \frac{\overline{N_2}}{3\eta} (\beta_{12})^{3/2} + \frac{(\gamma_{12})^{3/2}}{\varphi_2} \left[x \, \frac{d f_2(x)}{dx} - f_2(x) \right] - \frac{(\gamma_{12})^{3/2} (\beta_{22})}{\varphi_2} \left[x \, \frac{d \overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_{\overline{x_2}}$$

$$= \frac{\overline{N_2}}{3\eta} (\beta_{12})^{3/2} - \frac{\overline{N_2} (\gamma_{12})^{3/2} (\beta_{22}) \overline{\varphi_2}}{3\eta \varphi_2} + \frac{(\gamma_{12})^{3/2}}{\varphi_2} \left[x \, \frac{d f_2(x)}{dx} - f_2(x) \right]$$

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In region $x_2 < x < x_1$, $f_1(x)$ is dominant

$$= \int_0^{\bar{x}_2} dx \sqrt{x} (f_1(x))^{3/2} + \int_{\bar{x}_2}^{x_2} dx \sqrt{x} (f_1(x))^{3/2} + \int_{x_2}^{x} dx \sqrt{x} (f_1(x))^{3/2}$$

$$= (\beta_{12})^{3/2} \int_0^{\bar{x}_2} dx \sqrt{x} (f_2(x))^{3/2} + (\gamma_{12})^{3/2} \int_{\bar{x}_2}^{x_2} dx \sqrt{x} (f_2(x))^{3/2} + \int_{x_2}^{x} dx \sqrt{x} (f_1(x))^{3/2}$$

$$= \frac{\bar{N}_2}{3\eta} (\beta_{12})^{3/2} + \frac{(\gamma_{12})^{3/2}}{\varphi_2} \int_{\bar{x}_2}^{x_2} dx\, x \frac{d^2 f_2(x)}{dx^2} + \frac{1}{\varphi_1} \int_{x_2}^{x} dx\, x \frac{d^2 f_1(x)}{dx^2}$$

$$= \frac{\bar{N}_2}{3\eta} (\beta_{12})^{3/2} + \frac{(\gamma_{12})^{3/2}}{\varphi_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{\bar{x}_2}^{x_2} + \frac{1}{\varphi_1} \left[x \frac{df_1(x)}{dx} - f_1(x) \right]_{x_2}^{x}$$

$$= \frac{\bar{N}_2}{3\eta} (\beta_{12})^{3/2} + \frac{(\gamma_{12})^{3/2}}{\varphi_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2} - \frac{(\gamma_{12})^{3/2}}{\varphi_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{\bar{x}_2}$$
$$+ \frac{1}{\varphi_1} \left[x \frac{df_1(x)}{dx} - f_1(x) \right] - \frac{1}{\varphi_1} \left[x \frac{df_1(x)}{dx} - f_1(x) \right]_{x_2}$$

$$= \frac{\bar{N}_2}{3\eta} (\beta_{12})^{3/2} - \frac{(\gamma_{12})^{3/2}}{\varphi_2} (\beta_{22}) \frac{\bar{N}_2 \bar{\varphi}_2}{3\eta} + \frac{(\gamma_{12})^{3/2}}{\varphi_2} \left\{ \frac{N_2 \varphi_2}{3\eta} + \beta_{22} \frac{\bar{N}_2 \bar{\varphi}_2}{3\eta} - (\beta_{22})^{3/2} \frac{N_2 \varphi_2}{3\eta} \right\}$$
$$+ \frac{1}{\varphi_1} \left[x \frac{df_1(x)}{dx} - f_1(x) \right] - \frac{\gamma_{12}}{\varphi_1} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2}$$

$$= \frac{\bar{N}_2}{3\eta} (\beta_{12})^{3/2} - \frac{\bar{N}_2}{3\eta} (\gamma_{12})^{3/2} \cancel{(\beta_{22})} \frac{\bar{\varphi}_2}{\varphi_2} + \frac{N_2}{3\eta} (\gamma_{12})^{3/2} + \frac{\bar{N}_2}{3\eta} (\gamma_{12})^{3/2} \cancel{(\beta_{22})} \frac{\bar{\varphi}_2}{\varphi_2}$$
$$- \frac{\bar{N}_2}{3\eta} (\gamma_{12})^{3/2} (\beta_{22})^{3/2} - \frac{N_2 \varphi_2}{3\eta} \cdot \frac{\gamma_{12}}{\varphi_1} - \frac{\bar{N}_2 \bar{\varphi}_2}{3\eta} (\beta_{22}) \cdot \frac{\gamma_{12}}{\varphi_1} + (\beta_{22})^{3/2} \frac{N_2 \varphi_2}{3\eta} \frac{(\gamma_{12})}{\varphi_1}$$
$$+ \frac{1}{\varphi_2} \left[x \frac{df_1(x)}{dx} - f_1(x) \right]$$

$$= \frac{\bar{N}_2}{3\eta} (\beta_{12})^{3/2} + \frac{N_2}{3\eta} (\gamma_{12})^{3/2} - \frac{N_2}{3\eta} (\gamma_{12}) \frac{\varphi_2}{\varphi_1} - \frac{\bar{N}_2}{3\eta} (\gamma_{12})^{3/2} (\beta_{22})^{3/2} - \frac{\bar{N}_2}{3\eta} (\beta_{22}) (\gamma_{12}) \frac{\bar{\varphi}_2}{\varphi_1}$$
$$+ \frac{\bar{N}_2}{3\eta} (\beta_{22})^{3/2} (\gamma_{12}) \frac{\varphi_2}{\varphi_1} + \frac{1}{\varphi_2} \left[x \frac{df_1(x)}{dx} - f_1(x) \right]$$

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$$\int_{x}^{x_1} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}}$$

In region $0 < x < \overline{x_2}$

$$= \int_{x}^{\overline{x_2}} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}} + \int_{\overline{x_2}}^{x_2} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}} + \int_{x_2}^{x_1} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}}$$

$$= (\beta_{1\bar{2}})^{3/2} \int_{x}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} + (r_{12})^{3/2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} + \int_{x_2}^{x_1} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}}$$

$$= \frac{(\beta_{1\bar{2}})^{3/2}}{\overline{\varphi_2}} \int_{x}^{\overline{x_2}} dx \frac{d^2 \overline{f_2(x)}}{dx^2} + \frac{(r_{12})^{3/2}}{\varphi_2} \int_{\overline{x_2}}^{x_2} dx \frac{d^2 f_2(x)}{dx^2} + \frac{1}{\varphi_1} \int_{x_2}^{x_1} dx \frac{d^2 f_1(x)}{dx^2}$$

$$= \frac{(\beta_{1\bar{2}})^{3/2}}{\overline{\varphi_2}} \left[\frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} - \frac{d\overline{f_2(x)}}{dx} \right] + \frac{(r_{12})^{3/2}}{\varphi_2} \left[\frac{df_2(x)}{dx} \Big|_{x_2} - \beta_{2\bar{1}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} \right]$$

$$+ \frac{1}{\varphi_1} \left[\frac{df_1(x)}{dx} \Big|_{x_1} - r_{12} \frac{df_2(x)}{dx} \Big|_{x_2} \right]$$

In region $\overline{x_2} < x < x_2$

$$= \int_{x}^{x_2} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}} + \int_{x_2}^{x_1} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}}$$

$$= (r_{\bar{1}2})^{3/2} \int_{x}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} + \int_{x_2}^{x_1} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}}$$

$$= \frac{(r_{\bar{1}2})^{3/2}}{\varphi_2} \int_{x}^{x_2} dx \frac{d^2 f_2(x)}{dx^2} + \frac{1}{\varphi_1} \int_{x_2}^{x_1} dx \frac{d^2 f_1(x)}{dx^2}$$

$$= \frac{(r_{\bar{1}2})^{3/2}}{\varphi_2} \left[\frac{df_2(x)}{dx} \Big|_{x_2} - \frac{df_2(x)}{dx} \right] + \frac{1}{\varphi_1} \left[\frac{df_1(x)}{dx} \Big|_{x_1} - r_{12} \frac{df_2(x)}{dx} \Big|_{x_2} \right]$$

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In region $x_2 < x < x_1$

$$\int_{x}^{x_1} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}}$$

$$= \frac{1}{Q_1} \int_{x}^{x_1} dx \frac{d^2 f_1(x)}{dx^2}$$

$$= \frac{1}{Q_1} \left[\frac{df_1(x)}{dx} \Big|_{x_1} - \frac{df_1(x)}{dx} \right]$$

$$\int_{x_2}^{x_2} dx \sqrt{x} (f_2(x))^{3/2} = \int_{x_2}^{x_2} dx \frac{x}{Q_2} \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{Q_2} \int_{x_2}^{x_2} dx\, x \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2}^{x_2}$$

$$= \frac{1}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2} - \frac{\beta_{22}}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2}$$

$$\int_{x_2}^{x_1} dx \sqrt{x} (f_1(x))^{3/2} = \int_{x_2}^{x_1} dx \frac{x}{Q_1} \frac{d^2 f_1(x)}{dx^2}$$

$$= \frac{1}{Q_1} \int_{x_2}^{x_1} dx\, x \frac{d^2 f_1(x)}{dx^2}$$

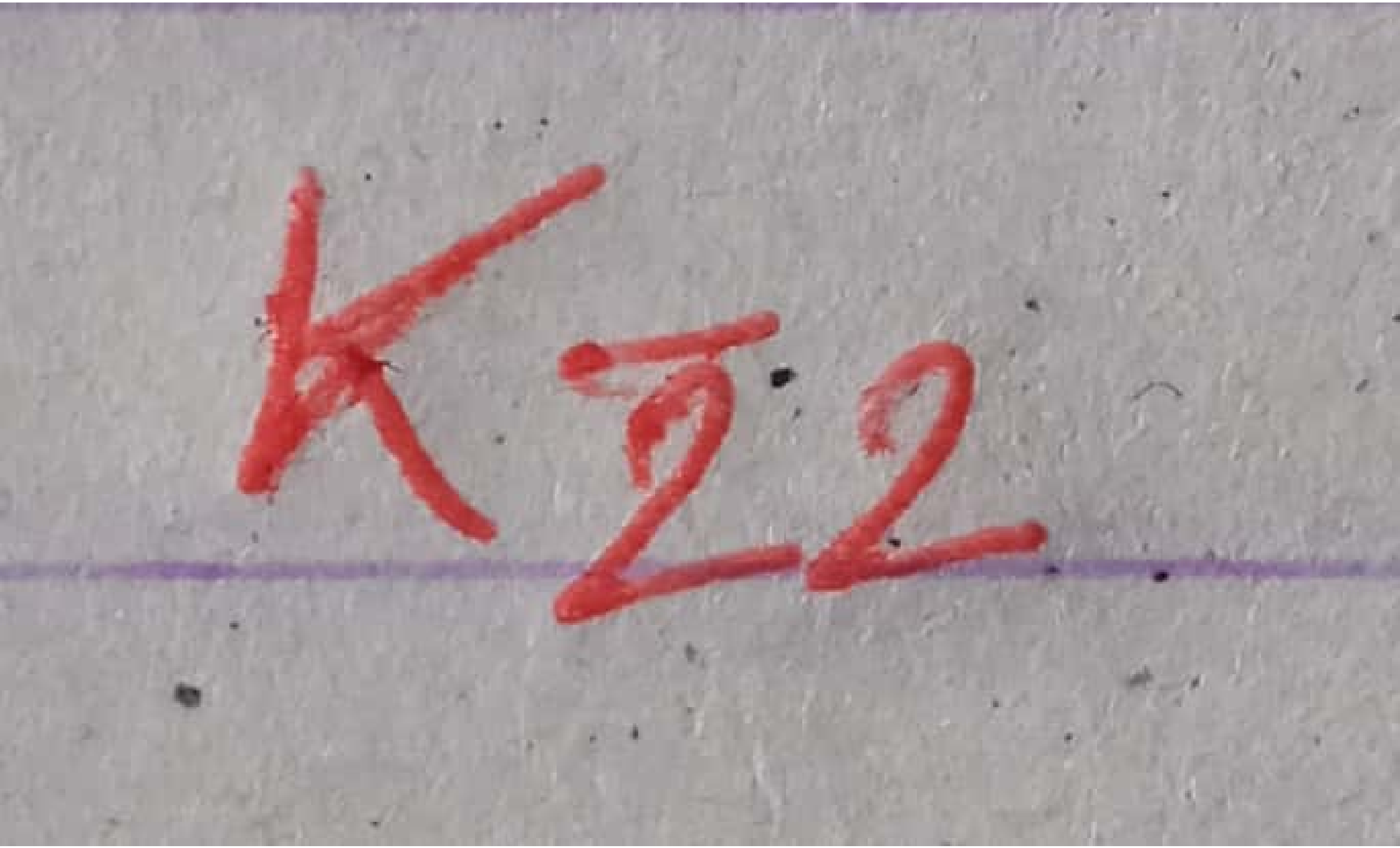
$$= \frac{1}{Q_1} \left[x \frac{df_1(x)}{dx} - f_1(x) \right]_{x_2}^{x_1}$$

$$= \frac{1}{Q_1} \left[x \frac{df_1(x)}{dx} - f_1(x) \right]_{x_1} - \frac{\gamma_{12}}{Q_1} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2}$$

From Normalization condition

$$\int_{0}^{x_2} dx \sqrt{x} (f_2(x))^{3/2} = \frac{N_2}{3\eta}$$

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$$K_{22} = \int_0^{\bar{x}_2} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'}\,(f_2(x'))^{3/2} + \int_0^{\bar{x}_2} dx \sqrt{x}\,(\overline{f_2(x)})^{3/2} \int_x^{x_2} dx' \frac{(f_2(x'))^{3/2}}{\sqrt{x'}}$$

Substituting values from appendix

$$= \int_0^{\bar{x}_2} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_2} \left[2 \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right] + \int_0^{\bar{x}_2} dx \sqrt{x}\,(\overline{f_2(x)})^{3/2}$$

$$\left[\left\{ \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_2} - \frac{\beta_{2\bar{2}}}{\varphi_2} \right\} \frac{d\overline{f_2(x)}}{dx} \Big|_{\bar{x}_2} - \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_2} \frac{d\overline{f_2(x)}}{dx} + \frac{1}{\varphi_2} \frac{df_2(x)}{dx} \Big|_{x_2} \right]$$

$$= \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_2} \left[\int_0^{\bar{x}_2} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} x \frac{d\overline{f_2(x)}}{dx} - \int_0^{\bar{x}_2} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} \overline{f_2(x)} \right]$$

$$+ \int_0^{\bar{x}_2} dx \sqrt{x}\,(\overline{f_2(x)})^{3/2} \left[\left\{ \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_2} - \frac{\beta_{2\bar{2}}}{\varphi_2} \right\} \frac{d\overline{f_2(x)}}{dx} \Big|_{\bar{x}_2} - \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_2} \frac{d\overline{f_2(x)}}{dx} + \frac{1}{\varphi_2} \frac{df_2(x)}{dx} \Big|_{x_2} \right]$$

$$= \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_2} \int_0^{\bar{x}_2} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} \frac{d\overline{f_2(x)}}{dx} - \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_2} \int_0^{\bar{x}_2} dx \frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}}$$

$$+ \int_0^{\bar{x}_2} dx \sqrt{x}\,(\overline{f_2(x)})^{3/2} \left\{ \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_2} - \frac{\beta_{2\bar{2}}}{\varphi_2} \right\} \frac{d\overline{f_2(x)}}{dx} \Big|_{\bar{x}_2}$$

$$- \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_2} \int_0^{\bar{x}_2} dx \sqrt{x}\,(\overline{f_2(x)})^{3/2} \frac{d\overline{f_2(x)}}{dx} + \frac{1}{\varphi_2} \int_0^{\bar{x}_2} dx \sqrt{x}\,(\overline{f_2(x)})^{3/2} \frac{df_2(x)}{dx} \Big|_{x_2}$$

$$= \int_0^{\bar{x}_2} dx \sqrt{x}\,(\overline{f_2(x)})^{3/2} \left\{ \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_2} - \frac{\beta_{2\bar{2}}}{\varphi_2} \right\} \frac{d\overline{f_2(x)}}{dx} \Big|_{\bar{x}_2} + \frac{1}{\varphi_2} \int_0^{\bar{x}_2} dx \sqrt{x}\,(\overline{f_2(x)})^{3/2} \frac{df_2(x)}{dx} \Big|_{x_2}$$

$$- \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_2} \int_0^{\bar{x}_2} dx \frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}}$$

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$$= \frac{\overline{N_2}}{3\eta}\left\{\frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} - \frac{\beta_{2\bar{2}}}{\varphi_2}\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2} + \frac{1}{\varphi_2}\frac{\overline{N_2}}{3\eta}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}}\left[\frac{4}{7}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\bar{x}_2} - \frac{5\overline{N_2}}{21\eta}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}\right]$$

$$= \frac{\overline{N_2}\,(\beta_{2\bar{2}})^{3/2}}{3\eta\,\overline{\varphi_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2} - \frac{\overline{N_2}\,(\beta_{2\bar{2}})}{3\eta\varphi_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2} + \frac{\overline{N_2}}{3\eta\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{4(\beta_{2\bar{2}})^{3/2}}{7\overline{\varphi_2}}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\bar{x}_2} + \frac{5\overline{N_2}\,(\beta_{2\bar{2}})^{3/2}}{21\eta\,\overline{\varphi_2}}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}$$

$$= \frac{\overline{N_2}}{3\eta}\left\{\frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} - \frac{\beta_{2\bar{2}}}{\varphi_2} + \frac{5(\beta_{2\bar{2}})^{3/2}}{7\,\overline{\varphi_2}}\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2}$$

$$- \frac{4(\beta_{2\bar{2}})^{3/2}}{7\,\overline{\varphi_2}}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\bar{x}_2} + \frac{\overline{N_2}}{3\eta\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= -\frac{4}{7\overline{\varphi_2}}(\beta_{2\bar{2}})^{3/2}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\bar{x}_2} + \frac{\overline{N_2}}{3\eta}\left\{\frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} - \frac{\beta_{2\bar{2}}}{\varphi_2}\right.$$

$$\left. + \frac{5(\beta_{2\bar{2}})^{3/2}}{7\overline{\varphi_2}}\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2} + \frac{\overline{N_2}}{3\eta\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2} \quad \#\#$$

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Appendix for K_{22}

$$\frac{d^2\overline{f_2(x)}}{dx^2} = \overline{\Phi_2}\,\frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}}$$

$$\text{or, } \frac{1}{\overline{\Phi_2}}\,\frac{d^2\overline{f_2(x)}}{dx^2} = \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}}$$

$$\int_0^{\overline{x_2}} \frac{1}{\overline{\Phi_2}}\,\frac{d^2\overline{f_2(x)}}{dx^2} = \int_0^{\overline{x_2}} \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}}$$

$$\Rightarrow \int_0^{\overline{x_2}} \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} = \frac{1}{\overline{\Phi_2}}\,\frac{\overline{df_2(x)}}{dx}\Bigg|_{\overline{x_2}}$$

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The normalization condition for antiparticle is given as

$K_{\bar{2}2}-4$

$$\int_0^{x_{max}} dx\sqrt{x}\left(\overline{f^I(x)}\right)^{3/2} = \frac{\overline{N}^I}{3\eta}$$

$$\Rightarrow \int_0^{\overline{x_2}} dx\sqrt{x}\left(\overline{f_2(x)}\right)^{3/2} = \frac{\overline{N_2}}{3\eta}$$

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$$\int_0^x dx \sqrt{x} \left(f_2(x)\right)^{3/2}$$

In region $0 < x < \overline{x_2}$

$$= (\beta_{2\bar{2}})^{3/2} \int_0^x dx \sqrt{x} \left(\overline{f_2(x)}\right)^{3/2}$$

$$= (\beta_{2\bar{2}})^{3/2} \int_0^x dx \cdot \frac{x}{\overline{\varphi_2}} \frac{d^2 \overline{f_2(x)}}{dx^2}$$

$$= \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} \int_0^x dx \; x \frac{d^2 \overline{f_2(x)}}{dx^2}$$

$$= \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_0^x$$

$$= \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]$$

$$\int_x^{x_2} dx \frac{\left(f_2(x)\right)^{3/2}}{\sqrt{x}}$$

$$= \int_x^{\overline{x_2}} dx \frac{\left(f_2(x)\right)^{3/2}}{\sqrt{x}} + \int_{\overline{x_2}}^{x_2} dx \frac{\left(f_2(x)\right)^{3/2}}{\sqrt{x}}$$

$$= (\beta_{2\bar{2}})^{3/2} \int_x^{\overline{x_2}} dx \frac{\left(\overline{f_2(x)}\right)^{3/2}}{\sqrt{x}} + \int_{\overline{x_2}}^{x_2} dx \frac{\left(f_2(x)\right)^{3/2}}{\sqrt{x}}$$

$$= \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} \frac{d\overline{f_2(x)}}{dx} \Bigg|_x^{\overline{x_2}} + \frac{1}{\overline{\varphi_2}} \frac{df_2(x)}{dx} \Bigg|_{\overline{x_2}}^{x_2}$$

$$= \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} \frac{d\overline{f_2(x)}}{dx} \Bigg|_{\overline{x_2}} - \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} \frac{d\overline{f_2(x)}}{dx} \Bigg|_x + \frac{1}{\overline{\varphi_2}} \frac{df_2(x)}{dx} \Bigg|_{x_2} - \frac{1}{\overline{\varphi_2}} \frac{df_2(x)}{dx} \Bigg|_{\overline{x_2}}$$

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$$= \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_2} \frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_2} \frac{df_2(x)}{dx} + \frac{1}{\varphi_2} \frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\beta_{2\bar{2}}}{\varphi_2} \frac{df_2(x)}{dx}\Big|_{x_2}$$

K22T6

$$= \left\{ \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_2} - \frac{\beta_{2\bar{2}}}{\varphi_2} \right\} \frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_2} \frac{df_2(x)}{dx} + \frac{1}{\varphi_2} \frac{df_2(x)}{dx}\Big|_{x_2}$$

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K 21

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$$K_{\bar{2}1} = \int_0^{\bar{x}_2} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{3/2} + \int_0^{\bar{x}_2} dx \sqrt{x} (\overline{f_2(x)})^{3/2} \int_x^{x_1} dx' \frac{(f_1(x'))^{3/2}}{\sqrt{x'}}$$

$K_{\bar{2}1} - 1$

$$K_{\bar{2}1} = K_{\bar{2}1} \cdot 1 + K_{\bar{2}1} \cdot 2$$

Here,

$$K_{\bar{2}1} \cdot 1 = \int_0^{\bar{x}_2} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{3/2}$$

$$= \int_0^{\bar{x}_2} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} \cdot \frac{(\beta_{1\bar{2}})^{3/2}}{\bar{\rho}_2} \left\{ x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right\}$$

$$= \frac{(\beta_{1\bar{2}})^{3/2}}{\bar{\rho}_2} \int_0^{\bar{x}_2} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} \cdot x \frac{d\overline{f_2(x)}}{dx}$$

$$- \frac{(\beta_{1\bar{2}})^{3/2}}{\bar{\rho}_2} \int_0^{\bar{x}_2} dx \frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}}$$

$$= \frac{(\beta_{1\bar{2}})^{3/2}}{\bar{\rho}_2} \int_0^{\bar{x}_2} dx (\overline{f_2(x)})^{3/2} \sqrt{x} \frac{d\overline{f_2(x)}}{dx} - \frac{(\beta_{1\bar{2}})^{3/2}}{\bar{\rho}_2} \int_0^{\bar{x}_2} dx \frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}}$$

(1)

$$= -\frac{(\beta_{1\bar{2}})^{3/2}}{\bar{\rho}_2} \left[\frac{4}{7} \sqrt{x} (\overline{f_2(x)})^{5/2} \Big|_{\bar{x}_2} - \frac{5\bar{N}_2}{2\eta} \frac{d\overline{f_2(x)}}{dx} \Big|_{\bar{x}_2} \right]$$

$$= -\frac{4}{7} \frac{(\beta_{1\bar{2}})^{3/2}}{\bar{\rho}_2} \sqrt{x} (\overline{f_2(x)})^{5/2} \Big|_{\bar{x}_2} + \frac{5\bar{N}_2}{2\eta} \frac{(\beta_{1\bar{2}})^{3/2}}{\bar{\rho}_2} \frac{d\overline{f_2(x)}}{dx} \Big|_{\bar{x}_2}$$

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$K_{21\cdot 2} = \int_0^{x_2} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2} \int_x^{x_1} dx' \frac{(f_1(x'))^{3/2}}{\sqrt{x'}}$ $\quad K_{21}-2$

$= \int_0^{x_2} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2}\Big[\frac{(\beta_{12})^{3/2}}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\beta_{12})^{3/2}}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{(r_{12})^{3/2}}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2}$

$- \frac{(r_{12})^{3/2}(\beta_{22})}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{1}{Q_1}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{(r_{12})^2}{Q_1}\frac{df_2(x)}{dx}\Big|_{x_2}\Big]$

$= \frac{N_2(\beta_{12})^{3/2}}{3\eta Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2(r_{12})^{3/2}}{3\eta Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{N_2(r_{12})^{3/2}(\beta_{22})}{3\eta}\frac{df_2(x)}{dx}\Big|_{x_2}$

$+ \frac{N_2}{3\eta Q_1}\frac{df_2(x)}{dx}\Big|_{x_1} - \frac{N_2(r_{12})^2}{3\eta Q_1}\frac{df_2(x)}{dx}\Big|_{x_2}$

Therefore,

$K_{21} = K_{21\cdot 1} \oplus K_{21\cdot 2}$

$= -\frac{4}{7Q_2}(\beta_{12})^{3/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} + \frac{5N_2}{21\eta Q_2}(\beta_{12})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2}$

$\oplus \frac{N_2(\beta_{22})^{3/2}}{3\eta Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2(r_{12})^{3/2}}{3\eta Q_2}\frac{df_2(x)}{dx}\Big|_{x_2}$

$- \frac{N_2(r_{12})^{3/2}(\beta_{22})}{3\eta}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}$

$- \frac{N_2(r_{12})^2}{3\eta Q_1}\frac{df_2(x)}{dx}\Big|_{x_2}$

$= -\frac{4}{7Q_2}(\beta_{12})^{3/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} + \Big[\frac{5N_2}{21\eta Q_2}(\beta_{12})^{3/2} + \frac{N_2}{3\eta Q_2}(\beta_{12})^{3/2}$

$- \frac{N_2(r_{12})^{3/2}(\beta_{22})}{3\eta}\Big]\frac{df_2(x)}{dx}\Big|_{x_2} + \Big[\frac{N_2(r_{12})^{3/2}}{3\eta Q_2} - \frac{N_2(r_{12})^2}{3\eta Q_1}\Big]\frac{df_2(x)}{dx}\Big|_{x_2}$

$+ \frac{N_2}{3\eta Q_2}\frac{df_1(x)}{dx}\Big|_{x_1}$

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$$= -\frac{4}{7\bar{Q}_2}(\beta_{1\bar{2}})^{3/2}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta}\left\{\frac{5}{7\bar{Q}_2}(\beta_{12})^{3/2} + \frac{(\beta_{1\bar{2}})^{3/2}}{\bar{Q}_2} - (\tilde{r}_{12})^{3/2}(\beta_{2\bar{2}})\right\}^{K_{\bar{2}1}-3}$$

$$\frac{d\,\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{\overline{N_2}\,(\tilde{r}_{12})^{3/2}}{3\eta}\left\{\frac{1}{\bar{Q}_2} - \frac{(\tilde{r}_{12})^{1/2}}{\bar{Q}_1}\right\}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{\overline{N_2}}{3\eta\bar{Q}_1}\frac{df_1(x)}{dx}\Big|_{x_1}^{\#}$$

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Appendix for $K_{21}\cdot 1$

$$\int_0^x dx\sqrt{x}\,(f_1(x))^{3/2}$$

In region $0<x<\overline{x_2}$

$$= (\beta_{12})^{3/2}\int_0^x dx\sqrt{x}\,(\overline{f_2(x)})^{3/2}$$

$$= (\beta_{12})^{3/2}\int_0^x dx\,\frac{x}{\overline{\varphi_2}}\,\frac{d^2\overline{f_2(x)}}{dx^2}$$

$$= \frac{(\beta_{12})^{3/2}}{\overline{\varphi_2}}\int_0^x dx\,x\,\frac{d^2\overline{f_2(x)}}{dx^2}$$

$$= \frac{(\beta_{12})^{3/2}}{\overline{\varphi_2}}\left[x\,\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]$$

Appendix for $K_{21}\cdot 2$

$$\int_x^{x_1} dx'\,\frac{(f_1(x'))^{3/2}}{\sqrt{x'}}$$

$$= \int_x^{\overline{x_2}} dx'\,\frac{(f_1(x'))^{3/2}}{\sqrt{x'}} + \int_{\overline{x_2}}^{x_2} dx'\,\frac{(f_1(x'))^{3/2}}{\sqrt{x'}} + \int_{x_2}^{x_1} dx'\,\frac{(f_1(x'))^{3/2}}{\sqrt{x'}}$$

$$= (\beta_{12})^{3/2}\int_x^{\overline{x_2}} dx'\,\frac{(\overline{f_2(x')})^{3/2}}{\sqrt{x'}} + (\gamma_{12})^{3/2}\int_{\overline{x_2}}^{x_2} dx'\,\frac{(f_2(x'))^{3/2}}{\sqrt{x'}} + \int_{x_2}^{x_1} dx'\,\frac{(f_1(x'))^{3/2}}{\sqrt{x'}}$$

$$= \frac{(\beta_{12})^{3/2}}{\overline{\varphi_2}}\int_x^{\overline{x_2}} dx\,\frac{d^2\overline{f_2(x)}}{dx^2} + \frac{(\gamma_{12})^{3/2}}{\varphi_2}\int_{\overline{x_2}}^{x_2} dx\,\frac{d^2 f_2(x)}{dx^2} + \frac{1}{\varphi_1}\int_{x_2}^{x_1} dx\,\frac{d^2 f_1(x)}{dx^2}$$

$$= \frac{(\beta_{12})^{3/2}}{\overline{\varphi_2}}\,\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{(\beta_{12})^{3/2}}{\overline{\varphi_2}}\,\frac{d\overline{f_2(x)}}{dx} + \frac{(\gamma_{12})^{3/2}}{\varphi_2}\,\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{(\gamma_{12})^{3/2}(\beta_{12})}{\varphi_2}\,\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{1}{\varphi_1}\,\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{\gamma_{12}}{\varphi_1}\,\frac{df_2(x)}{dx}\Big|_{x_2}$$

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K 21

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$$K_{21} = \int_0^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{3/2} + \int_0^{x_2} dx \sqrt{x} (f_2(x))^{3/2} \int_x^{x_1} dx' \frac{(f_1(x'))^{3/2}}{\sqrt{x'}}$$

$$K_{21} = K_{21}\cdot 1 + K_{21}\cdot 2$$

Here,

$$K_{21}\cdot 1 = \int_0^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{3/2}$$

$$= \int_0^{\overline{x_2}} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{3/2} + \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{3/2}$$

← appendix → (K_{21}-9)

$$= (\beta_{2\bar{2}})^{3/2} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} \left\{ \frac{(\beta_{1\bar{2}})^{3/2}}{\overline{Q_2}} \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right] \right\} + \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}}$$

$$\left[\int_0^{\overline{x_2}} dx' \sqrt{x'} (f_1(x'))^{3/2} + \int_{\overline{x_2}}^{x} dx' \sqrt{x'} (f_1(x'))^{3/2} \right]$$

← appendix → (K_{21}-9)

$$= \frac{(\beta_{2\bar{2}})^{3/2} (\beta_{1\bar{2}})^{3/2}}{\overline{Q_2}} \left[\int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} x \frac{d\overline{f_2(x)}}{dx} - \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}} \right]$$

$$+ \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} \left[\frac{\overline{N_2}}{3\eta} (\beta_{1\bar{2}})^{3/2} + \frac{(\tilde{N}_{12})^{3/2}}{Q_2} \left(x \frac{df_2(x)}{dx} - f_2(x) \right) - \frac{(\tilde{N}_{12})^{3/2} (\beta_{2\bar{2}}) \overline{N_2 Q_2}}{3\eta Q_2} \right]$$

$$= \frac{(\beta_{2\bar{2}})^{3/2} (\beta_{1\bar{2}})^{3/2}}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \cancel{(\overline{f_2(x)})^{3/2}} \sqrt{x} \frac{d\overline{f_2(x)}}{dx} - \frac{(\beta_{2\bar{2}})^{3/2} (\beta_{1\bar{2}})^{3/2}}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}}$$

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$$+ \frac{\overline{N_2}}{3\eta} (\beta_{1\bar{2}})^{3/2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} + \frac{(\tilde{N}_{12})^{3/2}}{Q_2} \int_{\overline{x_2}}^{x_2} dx \cancel{(f_2(x))^{3/2}} \sqrt{x} \frac{df_2(x)}{dx}$$

$$- \frac{(\tilde{N}_{12})^{3/2}}{Q_2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}} - \frac{(\tilde{N}_{12})^{3/2} (\beta_{2\bar{2}}) \overline{N_2 Q_2}}{3\eta Q_2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}}$$

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$$= \frac{\overline{N_2}}{3\eta}(\beta_{12})^{3/2}\int_{\bar{x}_2}^{x_2} dx\,\frac{(f_2(x))^{3/2}}{\sqrt{x}} - \frac{(\beta_{22})^{3/2}(\beta_{12})^{3/2}}{Q_2}\int_{0}^{\bar{x}_2} dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}} - \frac{(\tilde{r}_{12})^{3/2}}{Q_2}\int_{\bar{x}_2}^{x_2} dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

$$- \frac{(\tilde{r}_{12})^{3/2}(\beta_{22})\overline{N_2}\,\overline{Q_2}}{3\eta\,Q_2}\int_{\bar{x}_2}^{x_2} dx\,\frac{(f_2(x))^{3/2}}{\sqrt{x}}$$

$$= \frac{\overline{N_2}}{3\eta}(\beta_{12})^{3/2}\int_{\bar{x}_2}^{x_2} dx\,\frac{1}{Q_2}\frac{d^2 f_2(x)}{dx^2} - \frac{(\beta_{22})^{3/2}(\beta_{12})^{3/2}}{Q_2}\left[\frac{4}{7}\sqrt{x}\,(f_2(x))^{5/2}\right]_{\bar{x}_2}$$

$$- \frac{5\overline{N_2}}{21\eta}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}\Big] - \frac{(\tilde{r}_{12})^{3/2}}{Q_2}\left[\frac{4}{7}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} - \frac{4}{7}(\beta_{22})^{5/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\bar{x}_2}\right.$$

$$- \frac{5}{7Q_2}\frac{df_2(x)}{dx}\Big|_{x_2}\left\{\frac{N_2 Q_2}{3\eta} + \beta_{22}\frac{\overline{N_2}\,\overline{Q_2}}{3\eta} - (\beta_{22})^{3/2}\frac{\overline{N_2}\,Q_2}{3\eta}\right\} + \frac{5}{7Q_2}(\beta_{22})^2\frac{\overline{N_2}\,\overline{Q_2}}{3\eta}$$

$$\left.\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}\right] - \frac{(\tilde{r}_{12})^{3/2}(\beta_{22})\overline{N_2}\,\overline{Q_2}}{3\eta Q_2}\int_{\bar{x}_2}^{x_2} dx\,\frac{1}{Q_2}\frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{\overline{N_2}}{3\eta Q_2}(\beta_{12})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{12})^{3/2}(\beta_{22})\frac{df_2(x)}{dx}\Big|_{\bar{x}_2} - \frac{4(\beta_{22})^{3/2}(\beta_{12})^{3/2}}{7\overline{Q_2}}$$

$$\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\bar{x}_2} + \frac{5\overline{N_2}}{21\eta}\frac{(\beta_{22})^{3/2}(\beta_{12})^{3/2}}{Q_2}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2} - \frac{(\tilde{r}_{12})^{3/2}}{Q_2}\left[\frac{4}{7}\sqrt{x}\right.$$

$$(f_2(x))^{5/2}\Big|_{x_2} - \frac{4}{7}(\beta_{22})^{5/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\bar{x}_2} - \frac{5N_2}{21\eta}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{5\overline{N_2}\,\overline{Q_2}}{21\eta Q_2}(\beta_{22})\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5\overline{N_2}}{21\eta}(\beta_{22})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5}{21\eta Q_2}(\beta_{22})^2\overline{N_2}\,\overline{Q_2}$$

$$\left.\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}\right] - \frac{(\tilde{r}_{12})^{3/2}(\beta_{22})\overline{N_2}\,\overline{Q_2}}{3\eta Q_2^2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{(\tilde{r}_{12})^{3/2}(\beta_{22})^2\overline{N_2}\,\overline{Q_2}}{3\eta Q_2^2}$$

$$\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}$$

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$$= \frac{\overline{N_2}}{3\eta\overline{\varphi_2}}(\beta_{12})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\overline{N_2}(\beta_{12})^{3/2}(\beta_{22})}{3\eta\overline{\varphi_2}}\frac{df_2(x)}{dx}\Big|_{\overline{x_2}} - \frac{4(\beta_{22})^{3/2}(\beta_{12})^{3/2}}{7\overline{\varphi_2}}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\overline{x_2}}$$

$$+ \frac{5\overline{N_2}}{21\eta}\frac{(\beta_{22})^{3/2}(\beta_{12})^{3/2}}{\overline{\varphi_2}}\frac{df_2(x)}{dx}\Big|_{\overline{x_2}} - \frac{4(\gamma_{12})^{3/2}}{7\varphi_2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}$$

$$+ \frac{4}{7\varphi_2}(\gamma_{12})^{3/2}(\beta_{22})^{5/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\overline{x_2}} + \frac{5N_2(\gamma_{12})^{3/2}}{21\eta\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+ \frac{5\overline{N_2}\,\overline{\varphi_2}}{21\eta\varphi_2^2}(\beta_{22})(\gamma_{12})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\overline{N_2}}{21\eta\varphi_2}(\beta_{22})^{3/2}(\gamma_{12})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{5}{21\eta\varphi_2^2}(\beta_{22})^2(\gamma_{12})^{3/2}\frac{\overline{N_2}\,\overline{\varphi_2}}{}\frac{df_2(x)}{dx}\Big|_{\overline{x_2}} - \frac{(\gamma_{12})^{3/2}(\beta_{22})\overline{N_2}\,\overline{\varphi_2}}{3\eta\varphi_2^2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+ \frac{(\gamma_{12})^{3/2}(\beta_{22})^2\overline{N_2}\,\overline{\varphi_2}}{3\eta\varphi_2^2}\frac{df_2(x)}{dx}\Big|_{\overline{x_2}}$$

$$= \frac{4}{7\varphi_2}(\gamma_{12})^{3/2}(\beta_{22})^{5/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\overline{x_2}} - \frac{4}{7\overline{\varphi_2}}(\beta_{22})^{3/2}(\beta_{12})^{3/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\overline{x_2}}$$

$$- \frac{4}{7\varphi_2}(\gamma_{12})^{3/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} + \frac{5\overline{N_2}}{21\eta}\frac{(\beta_{22})^{3/2}(\beta_{12})^{3/2}}{\overline{\varphi_2}}\frac{df_2(x)}{dx}\Big|_{\overline{x_2}}$$

$$- \frac{\overline{N_2}}{3\eta\varphi_2}(\beta_{12})^{3/2}(\beta_{22})\frac{df_2(x)}{dx}\Big|_{\overline{x_2}} - \frac{5}{21\eta\varphi_2^2}(\beta_{22})^2(\gamma_{12})^{3/2}\overline{N_2}\,\overline{\varphi_2}\frac{df_2(x)}{dx}\Big|_{\overline{x_2}}$$

$$+ \frac{(\gamma_{12})^{3/2}(\beta_{22})^2\overline{N_2}\,\overline{\varphi_2}}{3\eta\varphi_2^2}\frac{df_2(x)}{dx}\Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta\varphi_2}(\beta_{12})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+ \frac{5N_2(\gamma_{12})^{3/2}}{21\eta\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5\overline{N_2}\,\overline{\varphi_2}}{21\eta\varphi_2^2}(\beta_{22})(\gamma_{12})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{5\overline{N_2}}{21\eta\varphi_2}(\beta_{22})^{3/2}(\gamma_{12})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\gamma_{12})^{3/2}(\beta_{22})\overline{N_2}\,\overline{\varphi_2}}{3\eta\varphi_2^2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

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$$= \frac{4}{7}(\beta_{2\bar{2}})^{3/2}\left\{\frac{(r_{12})^{3/2}(\beta_{2\bar{2}})}{Q_2} - \frac{(\beta_{1\bar{2}})^{3/2}}{Q_2}\right\}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} - \frac{4}{7Q_2}(r_{12})^{3/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}$$

$\Rightarrow$ $K_{21}\cdot 1$

$$+ \frac{N_2}{3\eta}(\beta_{2\bar{2}})\left\{\frac{5(\beta_{2\bar{2}})^{1/2}(\beta_{1\bar{2}})^{3/2}}{7Q_2} - \frac{(\beta_{1\bar{2}})^{3/2}}{Q_2} - \frac{5Q_2(\beta_{2\bar{2}})(r_{12})^{3/2}}{7Q_2^2} + \frac{(r_{12})^{3/2}(\beta_{2\bar{2}})Q_2}{Q_2^2}\right\}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{1}{3\eta Q_2}\left\{N_2(\beta_{1\bar{2}})^{3/2} + \frac{5N_2}{7}(r_{12})^{3/2} + \frac{5N_2Q_2}{7Q_2}(\beta_{2\bar{2}})(r_{12})^{3/2}\right.$$

$$\left. - \frac{5N_2}{7}(\beta_{2\bar{2}})^{3/2}(r_{12})^{3/2} - \frac{N_2Q_2}{Q_2}(r_{12})^{3/2}(\beta_{2\bar{2}})\right\}\frac{df_2(x)}{dx}\Big|_{x_2}$$

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$$K_{21}\cdot 2 = \int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \int_x^{x_1} dx' \frac{(f_1(x'))^{3/2}}{\sqrt{x'}}$$

$$= \int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \int_x^{x_1} dx' \frac{(f_1(x'))^{3/2}}{\sqrt{x'}} + \int_{x_2}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \int_x^{x_1} dx' \frac{(f_1(x'))^{3/2}}{\sqrt{x'}}$$

$$= \int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \left[\int_x^{x_2} dx' \frac{(f_1(x'))^{3/2}}{\sqrt{x'}} + \int_{x_2}^{x_2} dx' \frac{(f_1(x'))^{3/2}}{\sqrt{x'}} + \int_{x_2}^{x_1} dx' \frac{(f_1(x'))^{3/2}}{\sqrt{x'}} \right]$$

$$+ \int_{x_2}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \left[\int_x^{x_2} dx' \frac{(f_1(x'))^{3/2}}{\sqrt{x'}} + \int_{x_2}^{x_1} dx' \frac{(f_1(x'))^{3/2}}{\sqrt{x'}} \right]$$

$$= (\beta_{2\bar{2}})^{3/2} \int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \left[(\beta_{12})^{3/2} \int_x^{x_2} dx' \frac{(f_2(x'))^{3/2}}{\sqrt{x'}} + (\gamma_{12})^{3/2} \int_{x_2}^{x_2} dx' \frac{(f_2(x'))^{3/2}}{\sqrt{x'}} + \int_{x_2}^{x_1} dx' \frac{(f_1(x'))^{3/2}}{\sqrt{x'}} \right]$$

$$+ \int_{x_2}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \left[(\gamma_{12})^{3/2} \int_x^{x_2} dx' \frac{(f_2(x'))^{3/2}}{\sqrt{x'}} + \int_{x_2}^{x_1} dx' \frac{(f_1(x'))^{3/2}}{\sqrt{x'}} \right]$$

$$= (\beta_{2\bar{2}})^{3/2} \int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \left[\frac{(\beta_{12})^{3/2}}{\varphi_2} \int_x^{x_2} dx' \frac{d^2 f_2(x')}{dx'^2} + \frac{(\gamma_{12})^{3/2}}{\varphi_2} \int_{x_2}^{x_2} dx' \frac{d^2 f_2(x')}{dx'^2} + \frac{1}{\varphi_1} \int_{x_2}^{x_1} dx' \frac{d^2 f_1(x')}{dx'^2} \right]$$

$$+ \int_{x_2}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \left[\frac{(\gamma_{12})^{3/2}}{\varphi_2} \int_x^{x_2} dx' \frac{d^2 f_2(x')}{dx'^2} + \frac{1}{\varphi_1} \int_{x_2}^{x_1} dx' \frac{d^2 f_1(x')}{dx'^2} \right]$$

$$= (\beta_{2\bar{2}})^{3/2} \int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \left[\frac{(\beta_{12})^{3/2}}{\varphi_2} \frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{(\beta_{12})^{3/2}}{\varphi_2} \frac{df_2(x)}{dx} + \frac{(\gamma_{12})^{3/2}}{\varphi_2} \frac{df_2(x)}{dx}\bigg|_{x_2} \right.$$

$$\left. - \frac{(\gamma_{12})^{3/2}(\beta_{2\bar{2}})}{\varphi_2} \frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{1}{\varphi_1} \frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{\gamma_{12}}{\varphi_1} \frac{df_2(x)}{dx}\bigg|_{x_2} \right] + \int_{x_2}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}$$

$$\left[\frac{(\gamma_{12})^{3/2}}{\varphi_2} \frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{(\gamma_{12})^{3/2}}{\varphi_2} \frac{df_2(x)}{dx} + \frac{1}{\varphi_1} \frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{\gamma_{12}}{\varphi_1} \frac{df_2(x)}{dx}\bigg|_{x_2} \right]$$

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$$= \frac{(\beta_{2\bar{2}})^{3/2}(\beta_{1\bar{2}})^{3/2}}{\varphi_2}\int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\beta_{2\bar{2}})^{3/2}(\beta_{1\bar{2}})^{3/2}}{\varphi_2}\int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}\frac{df_2(x)}{dx}$$

$$+ \frac{(\beta_{2\bar{2}})^{3/2}(\nu_{12})^{3/2}}{\varphi_2}\int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\nu_{12})^{3/2}(\beta_{2\bar{2}})^{5/2}}{\varphi_2}\int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}$$

$$\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{(\beta_{2\bar{2}})^{3/2}}{\varphi_1}\int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{(\beta_{2\bar{2}})^{3/2}(\nu_{12})}{\varphi_1}\int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}$$

$$\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{(\nu_{12})^{3/2}}{\varphi_2}\int_{x_2}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\nu_{12})^{3/2}}{\varphi_2}\int_{x_2}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}$$

(appendix K_{21}-10)

$$\frac{df_2(x)}{dx} + \frac{1}{\varphi_1}\int_{x_2}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{\nu_{12}}{\varphi_1}\int_{x_2}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

(appendix K_{21}-10) (appendix K_{21}-10)

$$= \frac{N_2(\beta_{2\bar{2}})^{3/2}(\beta_{1\bar{2}})^{3/2}}{3\eta\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2(\beta_{2\bar{2}})^{3/2}(\nu_{12})^{3/2}}{3\eta\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{N_2(\nu_{12})^{3/2}(\beta_{2\bar{2}})^{5/2}}{3\eta\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+ \frac{N_2(\beta_{2\bar{2}})^{3/2}}{3\eta\varphi_1}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{N_2(\beta_{2\bar{2}})^{3/2}(\nu_{12})}{3\eta\varphi_1}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{(\nu_{12})^{3/2}N_2}{3\eta\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{(\beta_{2\bar{2}})^{3/2}(\nu_{12})^{3/2}N_2}{3\eta\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta\varphi_1}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{(\beta_{2\bar{2}})^{3/2}N_2}{3\eta\varphi_1}\frac{df_1(x)}{dx}$$

$$- \frac{\nu_{12}N_2}{3\eta\varphi_1}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{(\beta_{2\bar{2}})^{3/2}(\nu_{12})N_2}{3\eta\varphi_1}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$\Rightarrow$ K_{21}-2

$$= \frac{N_2}{3\eta}(\beta_{2\bar{2}})^{3/2}\left\{\frac{(\beta_{1\bar{2}})^{3/2}}{\varphi_2} - \frac{(\nu_{12})^{3/2}(\beta_{2\bar{2}})}{\varphi_2}\right\}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+ \frac{N_2\nu_{12}}{3\eta}\left\{\frac{(\nu_{12})^{\frac{1}{2}}}{\varphi_2} - \frac{1}{\varphi_1}\right\}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta\varphi_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$

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Therefore,

$$K_{21} = K_{21}\cdot 1 \oplus K_{21}\cdot 2$$

$$= \frac{4}{7}(\beta_{2\bar{2}})^{3/2}\left\{\frac{(\gamma_{12})^{3/2}(\beta_{2\bar{2}})}{\varphi_2} - \frac{(\beta_{1\bar{2}})^{3/2}}{\bar{\varphi}_2}\right\}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\bar{x}_2}$$

$$- \frac{4}{7\varphi_2}(\gamma_{12})^{3/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} + \frac{\overline{N_2}}{3\eta}(\beta_{2\bar{2}})\left\{\frac{5(\beta_{2\bar{2}})^{1/2}(\beta_{1\bar{2}})^{3/2}}{7\bar{\varphi}_2}\right.$$

$$\left. - \frac{(\beta_{1\bar{2}})^{3/2}}{\varphi_2} - \frac{5\bar{\varphi}_2}{7\varphi_2^2}(\beta_{2\bar{2}})(\gamma_{12})^{3/2} + \frac{(\gamma_{12})^{3/2}(\beta_{2\bar{2}})\bar{\varphi}_2}{\varphi_2^2}\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2}$$

$$+ \frac{1}{3\eta\varphi_2}\left[\overline{N_2}(\beta_{1\bar{2}})^{3/2} - \frac{\overline{N_2}\bar{\varphi}_2}{\varphi_2}(\gamma_{12})^{3/2}(\beta_{2\bar{2}}) + \frac{5}{7}(\gamma_{12})^{3/2}\left\{N_2 + \right.\right.$$

$$\left.\left. + \frac{\overline{N_2}\bar{\varphi}_2}{\varphi_2}(\beta_{2\bar{2}}) - \overline{N_2}(\beta_{2\bar{2}})^{\frac{3}{2}}\right\}\right]\frac{df_2(x)}{dx}\Big|_{x_2} \oplus \frac{\overline{N_2}}{3\eta}(\beta_{2\bar{2}})^{3/2}\left\{\frac{(\beta_{1\bar{2}})^{3/2}}{\bar{\varphi}_2}\right.$$

$$\left. - \frac{(\gamma_{12})^{3/2}(\beta_{2\bar{2}})}{\varphi_2}\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2} + \frac{N_2\gamma_{12}}{3\eta}\left\{\frac{(\gamma_{12})^{1/2}}{\varphi_2} - \frac{1}{\varphi_1}\right\}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+ \frac{N_2}{3\eta\varphi_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$

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$$\Rightarrow K_{21} = \frac{4}{7}(\beta_{2\bar{2}})^{3/2}\left\{\frac{(\tilde{r}_{12})^{3/2}(\beta_{2\bar{2}})}{\Phi_2} - \frac{(\beta_{1\bar{2}})^{3/2}}{\overline{\Phi_2}}\right\}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\overline{x_2}}$$

$$-\frac{4}{7\Phi_2}(\tilde{r}_{12})^{3/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} + \frac{\overline{N_2}}{3\eta}(\beta_{2\bar{2}})\left\{5\frac{(\beta_{2\bar{2}})^{7/2}(\beta_{1\bar{2}})^{3/2}}{7\overline{\Phi_2}}\right.$$

$$-\frac{(\beta_{1\bar{2}})^{3/2}}{\Phi_2} - \frac{5\overline{\Phi_2}}{7\Phi_2^2}(\beta_{2\bar{2}})(\tilde{r}_{12})^{3/2} + \frac{(\beta_{2\bar{2}})^{7/2}(\beta_{1\bar{2}})^{3/2}}{\overline{\Phi_2}} - \frac{(\tilde{r}_{12})^{3/2}(\beta_{2\bar{2}})^{3/2}}{\Phi_2}$$

$$+\frac{(\tilde{r}_{12})^{3/2}(\beta_{2\bar{2}})\overline{\Phi_2}}{\Phi_2^2}\left.\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{1}{3\eta\Phi_2}\left[N_2(\tilde{r}_{12})^{3/2} - \frac{N_2\tilde{r}_{12}\Phi_2}{\Phi_1} + \overline{N_2}(\beta_{1\bar{2}})^{3/2}\right.$$

$$-\frac{\overline{N_2}\,\overline{\Phi_2}}{\Phi_2}(\tilde{r}_{12})^{3/2}(\beta_{2\bar{2}}) + \frac{5}{7}(\tilde{r}_{12})^{3/2}\left\{N_2 + \frac{\overline{N_2}\,\overline{\Phi_2}}{\Phi_2}(\beta_{2\bar{2}})\right.$$

$$\left.\left.-\overline{N_2}(\beta_{2\bar{2}})^{3/2}\right\}\right]\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta\Phi_1}\frac{df_1(x)}{dx}\Big|_{x_1} \quad \#$$

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Appendix for K_{21}

$$\int_0^x dx\sqrt{x}\,(f_1(x))^{3/2}$$

In region $0<x<\overline{x_2}$

$$= (\beta_{1\bar{2}})^{3/2}\int_0^x dx\sqrt{x}\,(\overline{f_2(x)})^{3/2}$$

$$= (\beta_{1\bar{2}})^{3/2}\int_0^x dx\,\frac{x}{\overline{\varphi_2}}\,\frac{d^2\overline{f_2(x)}}{dx^2}$$

$$= \frac{(\beta_{1\bar{2}})^{3/2}}{\overline{\varphi_2}}\int_0^x dx\,x\,\frac{d^2\overline{f_2(x)}}{dx^2}$$

$$= \frac{(\beta_{1\bar{2}})^{3/2}}{\overline{\varphi_2}}\left[x\,\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_0^x$$

$$\Rightarrow \int_0^x dx\sqrt{x}\,(f_1(x))^{3/2} = \frac{(\beta_{1\bar{2}})^{3/2}}{\overline{\varphi_2}}\left[x\,\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right] \quad \#$$

The normalization condition gives

$$\int_0^{\overline{x_2}} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2} = \frac{\overline{N_2}}{3\eta} \qquad \left[\because f_1(x) = \beta_{1\bar{2}}\,\overline{f_2(x)}\right]$$

or, $$\frac{1}{(\beta_{1\bar{2}})^{3/2}}\int_0^{\overline{x_2}} dx\sqrt{x}\,(f_1(x))^{3/2} = \frac{\overline{N_2}}{3\eta}$$

$$\Rightarrow \int_0^{\overline{x_2}} dx\sqrt{x}\,(f_1(x))^{3/2} = \frac{\overline{N_2}}{3\eta}(\beta_{1\bar{2}})^{3/2} \quad \#$$

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$$\int_{\bar{x}_2}^{x_2} dx \sqrt{x}\,(f_2(x))^{3/2}$$

$$= \int_{\bar{x}_2}^{x_2} dx \; \frac{x}{Q_2} \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{Q_2} \int_{\bar{x}_2}^{x_2} dx \; x \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{\bar{x}_2}^{x_2}$$

$$= \frac{1}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2} - \frac{\beta_{2\bar{2}}}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{\bar{x}_2}$$

$$= \frac{1}{Q_2} \left[\frac{N_2 Q_2}{3\eta} + \beta_{2\bar{2}} \frac{\bar{N}_2 \bar{Q}_2}{3\eta} - (\beta_{2\bar{2}})^{3/2} \frac{\bar{N}_2 \bar{Q}_2}{3\eta} \right] - \frac{\beta_{2\bar{2}} \bar{N}_2 \bar{Q}_2}{3\eta Q_2}$$

$$= \frac{N_2}{3\eta} + \cancel{\frac{\beta_{2\bar{2}} \bar{N}_2 \bar{Q}_2}{3\eta Q_2}} - \frac{(\beta_{2\bar{2}})^{3/2} \bar{N}_2}{3\eta} - \cancel{\frac{\beta_{2\bar{2}} \bar{N}_2 \bar{Q}_2}{3\eta Q_2}}$$

$$= \frac{N_2}{3\eta} - \frac{(\beta_{2\bar{2}})^{3/2} \bar{N}_2}{3\eta}$$

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K22

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$$K_{22} = \int_0^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{3/2} + \int_0^{x_2} dx \sqrt{x} (f_2(x))^{3/2} \int_x^{x_2} dx' \frac{(f_2(x'))^{3/2}}{\sqrt{x'}}$$

K_{22}-1

$$K_{22} = K_{22}\cdot 1 + K_{22}\cdot 2$$

$$K_{22}\cdot 1 = \int_0^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{3/2}$$

$$= \int_0^{\overline{x_2}} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{3/2} + \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{3/2}$$

$$= \int_0^{\overline{x_2}} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} \left[\frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} \left\{ x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right\} \right] + \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} \left[\left\{ \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} - \frac{\beta_{2\bar{2}}}{\varphi_2} \right\} \right.$$

$$\left. \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_{\overline{x_2}} + \frac{1}{\varphi_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right] \right]$$

$$= \frac{(\beta_{2\bar{2}})^3}{\overline{\varphi_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} \textcircled{1} \; x \frac{d\overline{f_2(x)}}{dx} - \frac{(\beta_{2\bar{2}})^3}{\overline{\varphi_2}} \int_0^{\overline{x_2}} dx \frac{\overline{f_2(x)}^{5/2}}{\sqrt{x}} + \left\{ \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} - \frac{\beta_{2\bar{2}}}{\varphi_2} \right\}$$

$$\left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_{\overline{x_2}} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} + \frac{1}{\varphi_2} \int_{\overline{x_2}}^{x_2} dx \sqrt{x} (f_2(x))^{3/2} \textcircled{2} \frac{df_2(x)}{dx}$$

$$- \frac{1}{\varphi_2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

$$= \left\{ \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} - \frac{\beta_{2\bar{2}}}{\varphi_2} \right\} \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_{\overline{x_2}} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} - \frac{(\beta_{2\bar{2}})^3}{\overline{\varphi_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}}$$

$$- \frac{1}{\varphi_2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

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$$= \frac{1}{\bar{Q}_2}\left\{\frac{(\beta_{2\bar{2}})^{3/2}}{\bar{Q}_2} - \frac{\beta_{2\bar{2}}}{\bar{Q}_2}\right\}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\bar{x}_2}^{x_2} \int_{\bar{x}_2}^{x_2} dx\, \frac{d^2 f_2(x)}{dx^2} - \frac{(\beta_{2\bar{2}})^3}{\bar{Q}_2}\left[\frac{4}{7}\sqrt{x}\right.$$

$$\left.(f_2(x))^{5/2}\Big|_{\bar{x}_2} - \frac{5\bar{N}_2}{2\bar{\eta}}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}\right] - \frac{1}{\bar{Q}_2}\left[\frac{4}{7}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} - \frac{4}{7}(\beta_{2\bar{2}})^{5/2}\sqrt{x}\right.$$

$$(f_2(x))^{5/2}\Big|_{\bar{x}_2} - \frac{5}{7\bar{Q}_2}\frac{df_2(x)}{dx}\Big|_{x_2}\left[\frac{N_2 Q_2}{3\eta} + \beta_{2\bar{2}}\frac{\bar{N}_2\bar{Q}_2}{3\eta} - (\beta_{2\bar{2}})^{3/2}\frac{\bar{N}_2 Q_2}{3\eta}\right]$$

$$\left.+\frac{5}{7\bar{Q}_2}(\beta_{2\bar{2}})^2\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}\frac{\bar{N}_2\bar{Q}_2}{3\eta}\right]$$

$$= \frac{\bar{N}_2\bar{Q}_2}{3\eta\bar{Q}_2}\left\{\frac{(\beta_{2\bar{2}})^{3/2}}{\bar{Q}_2} - \frac{\beta_{2\bar{2}}}{\bar{Q}_2}\right\}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}^{x_2} - \frac{4}{7}\frac{(\beta_{2\bar{2}})^3}{\bar{Q}_2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}$$

$$+\frac{5\bar{N}_2}{2\bar{\eta}}\frac{(\beta_{2\bar{2}})^3}{\bar{Q}_2}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2} - \frac{4}{7\bar{Q}_2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} + \frac{4}{7\bar{Q}_2}(\beta_{2\bar{2}})^{5/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\bar{x}_2}$$

$$+\frac{1}{\bar{Q}_2}\frac{5}{7\bar{Q}_2}\frac{N_2 Q_2}{3\eta}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{1}{\bar{Q}_2}\frac{5}{7\bar{Q}_2}\beta_{2\bar{2}}\frac{\bar{N}_2\bar{Q}_2}{3\eta}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{\bar{Q}_2}\frac{5}{7\bar{Q}_2}$$

$$(\beta_{2\bar{2}})^{3/2}\frac{\bar{N}_2 Q_2}{3\eta}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{\bar{Q}_2}\frac{5}{7\bar{Q}_2}(\beta_{2\bar{2}})^2\frac{\bar{N}_2\bar{Q}_2}{3\eta}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}$$

$$= \frac{\bar{N}_2\bar{Q}_2}{3\eta\bar{Q}_2}\left\{\frac{(\beta_{2\bar{2}})^{3/2}}{\bar{Q}_2} - \frac{\beta_{2\bar{2}}}{\bar{Q}_2}\right\}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\bar{N}_2\bar{Q}_2}{3\eta\bar{Q}_2}\left\{\frac{(\beta_{2\bar{2}})^{5/2}}{\bar{Q}_2} - \frac{(\beta_{2\bar{2}})^2}{\bar{Q}_2}\right\}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}$$

$$-\frac{4}{7}\frac{(\beta_{2\bar{2}})^3}{\bar{Q}_2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\bar{x}_2} + \frac{5\bar{N}_2}{2\bar{\eta}}\frac{(\beta_{2\bar{2}})^3}{\bar{Q}_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{4}{7\bar{Q}_2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}$$

$$+\frac{4}{7\bar{Q}_2}(\beta_{2\bar{2}})^{5/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\bar{x}_2} + \frac{5N_2}{2\bar{\eta}\bar{Q}_2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5\bar{N}_2\bar{Q}_2}{2\bar{\eta}\,\bar{Q}_2^2}\beta_{2\bar{2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$-\frac{5\bar{N}_2}{2\bar{\eta}\bar{Q}_2}(\beta_{2\bar{2}})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\bar{N}_2\bar{Q}_2}{2\bar{\eta}\,\bar{Q}_2^2}(\beta_{2\bar{2}})^2\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}$$

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$$= \frac{4}{7\bar{\varphi}_2}(\beta_{2\bar{2}})^{5/2}\sqrt{x}\,(\overline{f_2(\alpha)})^{5/2}\Big|_{\bar{x}_2} - \frac{4}{7\bar{\varphi}_2}(\beta_{2\bar{2}})^{3}\sqrt{x}\,(\overline{f_2(\alpha)})^{5/2}\Big|_{\bar{\alpha}_2} - \frac{4}{7\bar{\varphi}_2}\sqrt{x}\,(f_2(\alpha))^{5/2}\Big|_{x_2}$$

$$+ \frac{\overline{N_2}\,\overline{\varphi}_2}{3\eta\varphi_2}\left\{\frac{(\beta_{2\bar{2}})^2}{\varphi_2} - \frac{(\beta_{2\bar{2}})^{5/2}}{\bar{\varphi}_2}\right\}\frac{d\overline{f_2(\alpha)}}{dx}\Big|_{\bar{x}_2} + \frac{5\overline{N_2}}{2\eta}\frac{(\beta_{2\bar{2}})^3}{\bar{\varphi}_2}\frac{d\overline{f_2(\alpha)}}{dx}\Big|_{\bar{x}_2}$$

$$- \frac{5\overline{N_2}\,\overline{\varphi}_2}{2\eta\varphi_2^2}(\beta_{2\bar{2}})^2\frac{d\overline{f_2(\alpha)}}{dx}\Big|_{\bar{x}_2} + \frac{\overline{N_2}\,\overline{\varphi}_2}{3\eta\varphi_2}\left\{\frac{(\beta_{2\bar{2}})^{3/2}}{\bar{\varphi}_2} - \frac{\beta_{2\bar{2}}}{\varphi_2}\right\}\frac{df_2(\alpha)}{dx}\Big|_{x_2}$$

$$+ \frac{5N_2}{2\eta\varphi_2}\frac{df_2(\alpha)}{dx}\Big|_{x_2} + \frac{5\overline{N_2}\,\overline{\varphi}_2}{2\eta\varphi_2^2}\beta_{2\bar{2}}\frac{df_2(\alpha)}{dx}\Big|_{x_2} - \frac{5\overline{N_2}}{2\eta\varphi_2}(\beta_{2\bar{2}})^{3/2}\frac{df_2(\alpha)}{dx}\Big|_{x_2}$$

$$= \frac{4}{7}(\beta_{2\bar{2}})^2\left\{\frac{(\beta_{2\bar{2}})^{1/2}}{\varphi_2} - \frac{(\beta_{2\bar{2}})}{\bar{\varphi}_2}\right\}\sqrt{x}\,(\overline{f_2(\alpha)})^{5/2}\Big|_{\bar{x}_2} - \frac{4}{7\varphi_2}\sqrt{x}\,(f_2(\alpha))^{5/2}\Big|_{x_2}$$

$$+ \frac{\overline{N_2}}{3\eta}(\beta_{2\bar{2}})^2\left[\frac{\overline{\varphi}_2}{\varphi_2^2} - \frac{(\beta_{2\bar{2}})^{1/2}}{\varphi_2} + \frac{5}{7\bar{\varphi}_2}(\beta_{2\bar{2}}) - \frac{5\,\bar{\varphi}_2}{7\varphi_2^2}\right]\frac{d\overline{f_2(\alpha)}}{dx}\Big|_{\bar{x}_2}$$

$$+ \left[\frac{\overline{N_2}}{3\eta\varphi_2}(\beta_{2\bar{2}})^{3/2} - \frac{\overline{N_2}\,\overline{\varphi}_2}{3\eta\varphi_2^2}(\beta_{2\bar{2}}) + \frac{5N_2}{2\eta\varphi_2} + \frac{5\overline{N_2}\,\overline{\varphi}_2}{2\eta\varphi_2^2}\beta_{2\bar{2}} - \frac{5\overline{N_2}}{2\eta\varphi_2}(\beta_{2\bar{2}})^{3/2}\right]\frac{df_2(\alpha)}{dx}\Big|_{x_2}$$

$$= \frac{4}{7}(\beta_{2\bar{2}})^2\left\{\frac{(\beta_{2\bar{2}})^{1/2}}{\varphi_2} - \frac{(\beta_{2\bar{2}})}{\bar{\varphi}_2}\right\}\sqrt{x}\,(\overline{f_2(\alpha)})^{5/2}\Big|_{\bar{x}_2} - \frac{4}{7\varphi_2}\sqrt{x}\,(f_2(\alpha))^{5/2}\Big|_{x_2}$$

$$+ \frac{\overline{N_2}}{3\eta}(\beta_{2\bar{2}})^2\left\{\frac{5}{7\bar{\varphi}_2}(\beta_{2\bar{2}}) + \frac{2\,\bar{\varphi}_2}{7\varphi_2^2} - \frac{(\beta_{2\bar{2}})^{1/2}}{\varphi_2}\right\}\frac{d\overline{f_2(\alpha)}}{dx}\Big|_{\bar{\alpha}_2}$$

$$+ \left[\frac{5N_2}{2\eta\varphi_2} + \frac{\overline{N_2}}{3\eta\varphi_2}(\beta_{2\bar{2}})\left\{(\beta_{2\bar{2}})^{1/2} - \frac{2}{7}\frac{\bar{\varphi}_2}{\varphi_2^2} - \frac{5}{7}(\beta_{2\bar{2}})^{1/2}\right\}\right]\frac{df_2(\alpha)}{dx}\Big|_{x_2}$$

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$$K_{22}\cdot 2 = \int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \int_x^{x_2} dx' \frac{(f_2(x'))^{3/2}}{\sqrt{x'}}$$

$$= \int_0^{\overline{x_2}} dx\sqrt{x}\,(f_2(x))^{3/2} \int_x^{x_2} dx' \frac{(f_2(x'))^{3/2}}{\sqrt{x'}} + \int_{\overline{x_2}}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \int_x^{x_2} dx' \frac{(f_2(x'))^{3/2}}{\sqrt{x'}}$$

$$= (\beta_{2\bar{2}})^{3/2} \int_0^{\overline{x_2}} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2} \left[\frac{(\beta_{2\bar{2}})^{3/2}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx} + \frac{1}{Q_2} \frac{df_2(x)}{dx}\Big|_{x_2} \right.$$

$$\left. - \frac{\beta_{2\bar{2}}}{Q_2} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \right] + \int_{\overline{x_2}}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \left[\frac{1}{Q_2} \frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{Q_2} \frac{df_2(x)}{dx} \right]$$

$$= \frac{(\beta_{2\bar{2}})^3}{\overline{Q_2}} \int_0^{\overline{x_2}} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{(\beta_{2\bar{2}})^3}{\overline{Q_2}} \int_0^{\overline{x_2}} dx\sqrt{x}\,(f_2(x))^{3/2} \frac{df_2(x)}{dx} \quad (1)$$

$$+ \frac{(\beta_{2\bar{2}})^{3/2}}{Q_2} \int_0^{\overline{x_2}} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2} \frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\beta_{2\bar{2}})^{5/2}}{Q_2} \int_0^{\overline{x_2}} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$

$$+ \frac{1}{Q_2} \int_{\overline{x_2}}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{Q_2} \int_{\overline{x_2}}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \frac{df_2(x)}{dx} \quad (2)$$

$$= \frac{(\beta_{2\bar{2}})^3}{\overline{Q_2}} \int_0^{\overline{x_2}} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{(\beta_{2\bar{2}})^{3/2}}{Q_2} \int_0^{\overline{x_2}} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2} \frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+ \frac{1}{Q_2} \int_{\overline{x_2}}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} \frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\beta_{2\bar{2}})^{5/2}}{Q_2} \int_0^{\overline{x_2}} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$

$$= \frac{(\beta_{2\bar{2}})^3 \overline{N_2}}{\overline{Q_2}\,3\eta} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{(\beta_{2\bar{2}})^{3/2} N_2}{3\eta Q_2} \frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\beta_{2\bar{2}})^{5/2} \overline{N_2}}{3\eta\,\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$

$$+ \frac{1}{Q_2} \frac{N_2}{3\eta} \frac{df_2(x)}{dx}\Big|_{x_2} + \frac{(\beta_{2\bar{2}}) \overline{N_2}\,\overline{Q_2}}{3\eta\,\overline{Q_2}^2} \frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\beta_{2\bar{2}})^{3/2} N_2}{3\eta Q_2} \frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{\beta_{2\bar{2}}\,\overline{N_2}\,\overline{Q_2}}{3\eta\,\overline{Q_2}^2} \frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= \frac{(\beta_{2\bar{2}})^3 \overline{N_2}}{3\eta\,\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{(\beta_{2\bar{2}})^{5/2} \overline{N_2}}{3\eta Q_2} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{N_2}{3\eta Q_2} \frac{df_2(x)}{dx}\Big|_{x_2}$$

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Therefore,

$$K_{22} = K_{22}\cdot 1 \oplus K_{22}\cdot 2$$

$$= \frac{4}{7}(\beta_{2\bar{2}})^2 \left\{ \frac{(\beta_{2\bar{2}})^{7/2}}{\Phi_2} - \frac{(\beta_{2\bar{2}})}{\overline{\Phi_2}} \right\} \sqrt{x}\,(\overline{f_2(x)})^{5/2} \Big|_{\bar{x}_2} - \frac{4}{7\Phi_2}\sqrt{x}\,(f_2(x))^{5/2} \Big|_{x_2}$$

$$+ \frac{\overline{N_2}}{3\eta}(\beta_{2\bar{2}})^2 \left\{ \frac{5}{7\overline{\Phi_2}}(\beta_{2\bar{2}}) + \frac{2\overline{\Phi_2}}{7\Phi_2^2} - \frac{(\beta_{2\bar{2}})^{7/2}}{\Phi_2} \right\} \frac{d\overline{f_2(x)}}{dx} \Big|_{\bar{x}_2}$$

$$+ \left[\frac{5N_2}{21\eta\Phi_2} + \frac{\overline{N_2}}{3\eta\Phi_2}(\beta_{2\bar{2}}) \left\{ (\beta_{2\bar{2}})^{7/2} - \frac{2}{7}\frac{\overline{\Phi_2}}{\Phi_2^2} - \frac{5}{7}(\beta_{2\bar{2}})^{7/2} \right\} \right] \frac{df_2(x)}{dx} \Big|_{x_2}$$

$$\oplus \frac{(\beta_{2\bar{2}})^3 \overline{N_2}}{3\eta\,\overline{\Phi_2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\bar{x}_2} - \frac{(\beta_{2\bar{2}})^{5/2}\overline{N_2}}{3\eta\Phi_2} \frac{d\overline{f_2(x)}}{dx} \Big|_{\bar{x}_2} + \frac{N_2}{3\eta\Phi_2} \frac{df_2(x)}{dx} \Big|_{x_2}$$

$$= \frac{4}{7}(\beta_{2\bar{2}})^2 \left\{ \frac{(\beta_{2\bar{2}})^{7/2}}{\Phi_2} - \frac{(\beta_{2\bar{2}})}{\overline{\Phi_2}} \right\} \sqrt{x}\,(\overline{f_2(x)})^{5/2} \Big|_{\bar{x}} - \frac{4}{7\Phi_2}\sqrt{x}\,(f_2(x))^{5/2} \Big|_{x_2}$$

$$+ \frac{\overline{N_2}}{3\eta}(\beta_{2\bar{2}})^2 \left\{ \frac{5}{7\overline{\Phi_2}}(\beta_{2\bar{2}}) + \frac{2\overline{\Phi_2}}{7\Phi_2^2} - \frac{(\beta_{2\bar{2}})^{7/2}}{\Phi_2} + \frac{\beta_{2\bar{2}}}{\overline{\Phi_2}} - \frac{(\beta_{2\bar{2}})^{7/2}}{\Phi_2} \right\}$$

$$\frac{d\overline{f_2(x)}}{dx} \Big|_{\bar{x}_2} + \left[\frac{N_2}{3\eta\Phi_2} + \frac{5N_2}{21\eta\Phi_2} + \frac{\overline{N_2}}{3\eta\Phi_2}(\beta_{2\bar{2}}) \left\{ (\beta_{2\bar{2}})^{7/2} - \frac{2}{7}\frac{\overline{\Phi_2}}{\Phi_2^2} - \frac{5}{7}(\beta_{2\bar{2}})^{7/2} \right\} \right]$$

$$\frac{df_2(x)}{dx} \Big|_{x_2}$$

$$= \frac{4}{7}(\beta_{2\bar{2}})^2 \left\{ \frac{(\beta_{2\bar{2}})^{7/2}}{\Phi_2} - \frac{(\beta_{2\bar{2}})}{\overline{\Phi_2}} \right\} \sqrt{x}\,(\overline{f_2(x)})^{5/2} \Big|_{\bar{x}_2} - \frac{4}{7\Phi_2}\sqrt{x}\,(f_2(x))^{5/2} \Big|_{x_2}$$

$$+ \frac{\overline{N_2}}{3\eta}(\beta_{2\bar{2}})^2 \left\{ \frac{12}{7\overline{\Phi_2}}(\beta_{2\bar{2}}) + \frac{2}{7}\frac{\overline{\Phi_2}}{\Phi_2^2} - 2\frac{(\beta_{2\bar{2}})^{7/2}}{\Phi_2} \right\} \frac{d\overline{f_2(x)}}{dx} \Big|_{\bar{x}_2}$$

$$+ \left[\frac{12N_2}{21\eta\Phi_2} + \frac{\overline{N_2}}{3\eta\Phi_2}(\beta_{2\bar{2}}) \left\{ (\beta_{2\bar{2}})^{7/2} - \frac{2}{7}\frac{\overline{\Phi_2}}{\Phi_2^2} - \frac{5}{7}(\beta_{2\bar{2}})^{7/2} \right\} \right] \frac{df_2(x)}{dx} \Big|_{x_2}$$

#

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$$\int_0^x d\alpha \sqrt{\alpha} (f_2(\alpha))^{3/2}$$

In region $0 < \alpha < \overline{\alpha_2}$

$$= (\beta_{2\bar{2}})^{3/2} \int_0^x d\alpha \sqrt{\alpha} (\overline{f_2(\alpha)})^{3/2}$$

$$= (\beta_{2\bar{2}})^{3/2} \int_0^x d\alpha \frac{\alpha}{\overline{\varphi_2}} \frac{d^2 \overline{f_2(\alpha)}}{d\alpha^2}$$

$$= \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} \int_0^x d\alpha \, \alpha \frac{d^2 \overline{f_2(\alpha)}}{d\alpha^2}$$

$$= \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} \left[\alpha \frac{d\overline{f_2(\alpha)}}{d\alpha} - \overline{f_2(\alpha)} \right]$$

In region $\overline{\alpha_2} < \alpha < \alpha_2$

$$\int_0^x d\alpha \sqrt{\alpha} (f_2(\alpha))^{3/2}$$

$$= \int_0^{\overline{\alpha_2}} d\alpha \sqrt{\alpha} (f_2(\alpha))^{3/2} + \int_{\overline{\alpha_2}}^x d\alpha \sqrt{\alpha} (f_2(\alpha))^{3/2}$$

$$= (\beta_{2\bar{2}})^{3/2} \int_0^{\overline{\alpha_2}} d\alpha \sqrt{\alpha} (\overline{f_2(\alpha)})^{3/2} + \int_{\overline{\alpha_2}}^x d\alpha \sqrt{\alpha} (f_2(\alpha))^{3/2}$$

$$= (\beta_{2\bar{2}})^{3/2} \int_0^{\overline{\alpha_2}} d\alpha \frac{\alpha}{\overline{\varphi_2}} \frac{d^2 \overline{f_2(\alpha)}}{d\alpha^2} + \int_{\overline{\alpha_2}}^x d\alpha \frac{\alpha}{\varphi_2} \frac{d^2 f_2(\alpha)}{d\alpha^2}$$

$$= \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} \int_0^{\overline{\alpha_2}} d\alpha \, \alpha \frac{d^2 \overline{f_2(\alpha)}}{d\alpha^2} + \frac{1}{\varphi_2} \int_{\overline{\alpha_2}}^x d\alpha \, \alpha \frac{d^2 f_2(\alpha)}{d\alpha^2}$$

$$= \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} \left[\alpha \frac{d\overline{f_2(\alpha)}}{d\alpha} - \overline{f_2(\alpha)} \right]_{\overline{\alpha_2}} + \frac{1}{\varphi_2} \left[\alpha \frac{df_2(\alpha)}{d\alpha} - f_2(\alpha) \right]_{\overline{\alpha_2}}^{\alpha}$$

$$= \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} \left[\alpha \frac{d\overline{f_2(\alpha)}}{d\alpha} - \overline{f_2(\alpha)} \right]_{\overline{\alpha_2}} + \frac{1}{\varphi_2} \left[\alpha \frac{df_2(\alpha)}{d\alpha} - f_2(\alpha) \right]_{\overline{\alpha_2}}^{\alpha} - \frac{\beta_{2\bar{2}}}{\varphi_2} \left[\alpha \frac{d\overline{f_2(\alpha)}}{d\alpha} - \overline{f_2(\alpha)} \right]_{\overline{\alpha_2}}^{\alpha}$$

$$= \left\{ \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} - \frac{\beta_{2\bar{2}}}{\varphi_2} \right\} \left[\alpha \frac{d\overline{f_2(\alpha)}}{d\alpha} - \overline{f_2(\alpha)} \right]_{\overline{\alpha_2}} + \frac{1}{\varphi_2} \left[\alpha \frac{df_2(\alpha)}{d\alpha} - f_2(\alpha) \right]_{\overline{\alpha_2}}^{\alpha}$$

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$$\int_{x}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}}$$

In region $0<x<\bar{x}_2$

$$= \int_{x}^{\bar{x}_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} + \int_{\bar{x}_2}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}}$$

$$= (\beta_{2\bar{2}})^{3/2} \int_{x}^{\bar{x}_2} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} + \int_{\bar{x}_2}^{x_2} dx \frac{1}{Q_2} \frac{d^2 f_2(x)}{dx^2}$$

$$= (\beta_{2\bar{2}})^{3/2} \int_{x}^{\bar{x}_2} \frac{1}{\bar{Q}_2} \frac{d^2 \overline{f_2(x)}}{dx^2} dx + \frac{1}{Q_2} \int_{\bar{x}_2}^{x_2} dx \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{(\beta_{2\bar{2}})^{3/2}}{\bar{Q}_2} \frac{d\overline{f_2(x)}}{dx}\Big|_{x}^{\bar{x}_2} + \frac{1}{Q_2} \frac{df_2(x)}{dx}\Big|_{\bar{x}_2}^{x_2}$$

$$= \frac{(\beta_{2\bar{2}})^{3/2}}{\bar{Q}_2} \frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2} - \frac{(\beta_{2\bar{2}})^{3/2}}{\bar{Q}_2} \frac{d\overline{f_2(x)}}{dx} + \frac{1}{Q_2} \frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{\beta_{2\bar{2}}}{Q_2} \frac{df_2(x)}{dx}\Big|_{\bar{x}_2}$$

In region $\bar{x}_2<x<x_2$

$$\int_{x}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}}$$

$$= \int_{x}^{x_2} dx \frac{1}{Q_2} \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{Q_2} \frac{df_2(x)}{dx}\Big|_{x}^{x_2}$$

$$= \frac{1}{Q_2} \frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{Q_2} \frac{df_2(x)}{dx}$$

$$\int_{\bar{x}_2}^{x_2} dx \sqrt{x} (f_2(x))^{3/2}$$

$$= \int_{\bar{x}_2}^{x_2} dx \frac{x}{Q_2} \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{Q_2} \int_{\bar{x}_2}^{x_2} dx\, x \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{\bar{x}_2}^{x_2}$$

$$= \frac{1}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2} - \frac{\beta_{2\bar{2}}}{\bar{Q}_2} \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_{\bar{x}_2}$$

$$= \frac{1}{Q_2} \left[\frac{N_2 Q_2}{3\eta} + \beta_{2\bar{2}} \frac{\overline{N_2}\,\overline{Q_2}}{3\eta} - (\beta_{2\bar{2}})^{3/2} \frac{\overline{N_2} Q_2}{3\eta} \right] - \frac{\beta_{2\bar{2}}}{Q_2} \frac{\overline{N_2}\,\overline{Q_2}}{3\eta}$$

$$= \frac{N_2}{3\eta} + \frac{\beta_{2\bar{2}} \overline{N_2}\,\overline{Q_2}}{3\eta Q_2} - (\beta_{2\bar{2}})^{3/2} \frac{\overline{N_2}}{3\eta} - \frac{\beta_{2\bar{2}}}{Q_2} \frac{\overline{N_2}\,\overline{Q_2}}{3\eta}$$

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K 12

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K_{12} = $\int_0^{x_1} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{3/2} + \int_0^{x_1} dx \sqrt{x} (f_1(x))^{3/2} \int_x^{x_2} dx' \frac{(f_2(x'))^{3/2}}{\sqrt{x'}}$ — K_{12-1}

$K_{12} = K_{12}\cdot 1 + K_{12}\cdot 2$

$$K_{12}\cdot 1 = \int_0^{x_1} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{3/2}$$

$$K_{12}\cdot 1 = \int_0^{\overline{x_2}} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}} \underbrace{\int_0^{x} dx' \sqrt{x'} (f_2(x'))^{3/2}}_{\leftarrow K_{12}\cdot 1 \rightarrow} + \int_{x_2}^{x_2} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{3/2}$$

$$+ \int_{x_2}^{x_1} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{3/2}$$

$$= (\beta_{12})^{3/2} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} \left\{ \frac{(\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right] \right\} + (\gamma_{12})^{3/2} \int_{x_2}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}}$$

$$\left[\underbrace{\int_0^{\overline{x_2}} dx' \sqrt{x'} (f_2(x'))^{3/2}}_{\leftarrow K_{12}\cdot 2 \rightarrow} + \underbrace{\int_{\overline{x_2}}^{x} dx' \sqrt{x'} (f_2(x'))^{3/2}}_{\leftarrow K_{12}\cdot 3 \rightarrow} \right] + \int_{x_2}^{x_1} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}}$$

$$\left[\int_0^{\overline{x_2}} dx' \sqrt{x'} (f_2(x'))^{3/2} + \underbrace{\int_{\overline{x_2}}^{x_2} dx' \sqrt{x'} (f_2(x'))^{3/2}}_{\leftarrow K_{12}\cdot 4 \rightarrow} + \int_{x_2}^{x} dx' \sqrt{x'} (f_2(x'))^{3/2} \right]$$

$$= \frac{(\beta_{1\bar{2}})^{3/2} (\beta_{2\bar{2}})^{3/2}}{\overline{\varphi_2}} \left[\int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} \cdot x \frac{d\overline{f_2(x)}}{dx} - \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}} \cdot \overline{f_2(x)} \right]$$

$$+ (\gamma_{12})^{3/2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{3/2}}{\sqrt{x}} \left[\frac{\overline{N_2} (\beta_{2\bar{2}})^{3/2}}{3\eta} + \frac{1}{\varphi_2} \left(x \frac{df_2(x)}{dx} - f_2(x) \right) - \beta_{2\bar{2}} \frac{\overline{N_2 \varphi_2}}{3\eta \varphi_2} \right]$$

$$+ \int_{x_2}^{x_1} dx \frac{(f_1(x))^{3/2}}{\sqrt{x}} \left[\frac{\overline{N_2} (\beta_{2\bar{2}})^{3/2}}{3\eta} + \frac{N_2}{3\eta} - (\beta_{2\bar{2}})^{3/2} \frac{\overline{N_2}}{3\eta} \right]$$

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$$= \frac{(\beta_{12})^{3/2}(\beta_{22})^{3/2}}{\varphi_2}\int_0^{x_2} dx\,\frac{(f_2(x))^{3/2}}{\sqrt{x}} \xrightarrow{①} x\,\frac{df_2(x)}{dx} - \frac{(\beta_{12})^{3/2}(\beta_{22})^{3/2}}{\varphi_2}\int_0^{x_2} dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}} \quad K_{12}-2$$

$$+ \frac{N_2}{3\eta}(\gamma_{12})^{3/2}(\beta_{22})^{3/2}\int_{x_2}^{x_2} dx\,\frac{(f_2(x))^{3/2}}{\sqrt{x}} + \frac{(\gamma_{12})^{3/2}}{\varphi_2}\int_{x_2}^{x_2} dx\,\frac{(f_2(x))^{3/2}}{\sqrt{x}} \xrightarrow{②} x\,\frac{df_2(x)}{dx}$$

$$- \frac{(\gamma_{12})^{3/2}}{\varphi_2}\int_{x_2}^{x_2} dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}} - \frac{N_2\varphi_2}{3\eta\varphi_2}(\gamma_{12})^{3/2}(\beta_{22})\int_{x_2}^{x_2} dx\,\frac{(f_2(x))^{3/2}}{\sqrt{x}}$$

$$+ \frac{N_2}{3\eta}\int_{x_2}^{x_2} dx\,\frac{(f_1(x))^{3/2}}{\sqrt{x}}$$

$$= \frac{N_2}{3\eta}(\gamma_{12})^{3/2}(\beta_{22})^{3/2}\int_{x_2}^{x_2} dx\,\frac{(f_2(x))^{3/2}}{\sqrt{x}} - \frac{(\beta_{12})^{3/2}(\beta_{22})^{3/2}}{\varphi_2}\int_0^{x_2} dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

$$- \frac{(\gamma_{12})^{3/2}}{\varphi_2}\int_{x_2}^{x_2} dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}} - \frac{N_2\varphi_2}{3\eta\varphi_2}(\gamma_{12})^{3/2}(\beta_{22})\int_{x_2}^{x_2} dx\,\frac{(f_2(x))^{3/2}}{\sqrt{x}} + \frac{N_2}{3\eta}\int_{x_2}^{x_1} dx\,\frac{(f_1(x))^{3/2}}{\sqrt{x}}$$

$\leftarrow K_{12}\cdot 6 \rightarrow$ $\qquad$ $\leftarrow K_{12}\cdot 5 \rightarrow$

$$K_{12}\cdot 1 = \frac{N_2}{3\eta}(\gamma_{12})^{3/2}(\beta_{22})^{3/2}\left[\frac{1}{\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\beta_{22}}{\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}\right] - \frac{(\beta_{12})^{3/2}(\beta_{22})^{3/2}}{\varphi_2}$$

$$\left[\frac{4}{7}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} - \frac{5N_2}{21\eta}\frac{df_2(x)}{dx}\Big|_{x_2}\right] - \frac{(\gamma_{12})^{3/2}}{\varphi_2}\left[\frac{4}{7}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}\right.$$

$$\left. - \frac{4}{7}(\beta_{22})^{5/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} - \frac{5}{7\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}\right\{\frac{N_2\varphi_2}{3\eta} + \beta_{22}\frac{N_2\varphi_2}{3\eta} - (\beta_{22})^{3/2}\frac{N_2\varphi_2}{3\eta}\Big\}$$

$$+ \frac{5}{7\varphi_2}(\beta_{22})^2\frac{df_2(x)}{dx}\Big|_{x_2}\frac{N_2\varphi_2}{3\eta}\Big] - \frac{N_2\varphi_2}{3\eta\varphi_2}(\gamma_{12})^{3/2}(\beta_{22})\left[\frac{1}{\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}\right.$$

$$\left. - \frac{\beta_{22}}{\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}\right] + \frac{N_2}{3\eta\varphi_1}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{N_2(\gamma_{12})}{3\eta\varphi_1}\frac{df_2(x)}{dx}\Big|_{x_2}$$

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$$K_{12}\cdot 2 = \int_0^{x_1} dx\sqrt{x}\,(f_1(x))^{3/2} \int_x^{x_2} \frac{dx'\,(f_2(x'))^{3/2}}{\sqrt{x'}}$$

$$= \int_0^{x_2} dx\sqrt{x}\,(f_1(x))^{3/2} \int_x^{x_2} \frac{dx'\,(f_2(x'))^{3/2}}{\sqrt{x'}} + \int_{x_2}^{x_2} dx\sqrt{x}\,(f_1(x))^{3/2} \int_x^{x_2} \frac{dx'\,(f_2(x'))^{3/2}}{\sqrt{x'}}$$

$$+ \int_{x_2}^{x_1} dx\sqrt{x}\,(f_1(x))^{3/2} \int_x^{x_2} \frac{dx'\,(f_2(x'))^{3/2}}{\sqrt{x'}}$$

$$= \int_0^{x_2} dx\sqrt{x}\,(f_1(x))^{3/2}\left[\int_x^{x_2} \frac{dx'\,(f_2(x'))^{3/2}}{\sqrt{x'}} + \int_{x_2}^{x_2} \frac{dx'\,(f_2(x'))^{3/2}}{\sqrt{x'}}\right] + \int_{x_2}^{x_2} dx\sqrt{x}\,(f_1(x))^{3/2} \int_x^{x_2} \frac{dx'\,(f_2(x'))^{3/2}}{\sqrt{x'}}$$

$$= \int_0^{x_2} dx\sqrt{x}\,(f_1(x))^{3/2}\left[\frac{(\beta_{2\bar{2}})^{3/2}}{\rho_2}\int_x^{x_2} dx'\,\frac{d^2 f_2(x')}{dx'^2} + \frac{1}{\rho_2}\int_{x_2}^{x_2} dx'\,\frac{d^2 f_2(x')}{dx'^2}\right]$$

$$+ \int_{x_2}^{x_2} dx\sqrt{x}\,(f_1(x))^{3/2}\left[\frac{1}{\rho_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{\rho_2}\frac{df_2(x)}{dx}\right]$$

$$= \int_0^{x_2} dx\sqrt{x}\,(f_1(x))^{3/2}\left[\frac{(\beta_{2\bar{2}})^{3/2}}{\rho_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\beta_{2\bar{2}})^{3/2}}{\rho_2}\frac{df_2(x)}{dx}\right.$$

$$\left. + \frac{1}{\rho_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{\rho_2}\frac{df_2(x)}{dx}\Big|_{x_2}\right] + \int_{x_2}^{x_2} dx\sqrt{x}\,(f_1(x))^{3/2}$$

$$\left[\frac{1}{\rho_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{\rho_2}\frac{df_2(x)}{dx}\right]$$

$$= (\beta_{1\bar{2}})^{3/2}\int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}\left[\frac{(\beta_{2\bar{2}})^{3/2}}{\rho_2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{1}{\rho_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{\rho_2}\frac{df_2(x)}{dx}\Big|_{x_2}\right]$$

$$- \frac{(\beta_{1\bar{2}})^{3/2}(\beta_{2\bar{2}})^{3/2}}{\rho_2}\int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}\frac{df_2(x)}{dx} + \frac{(\gamma_{12})^{3/2}}{\rho_2}\int_{x_2}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{(\gamma_{12})^{3/2}}{\rho_2}\int_{x_2}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}\frac{df_2(x)}{dx}$$

$$\int_{x_2}^{x_1} dx\sqrt{x}\,(f_1(x))^{3/2}\int_x^{x_2} \frac{dx'\,(f_2(x'))^{3/2}}{\sqrt{x'}} = \int_{x_2}^{x_1} dx\sqrt{x}\,(f_1(x))^{3/2}\left(-\int_{x_2}^{x} \frac{dx'\,(f_2(x'))^{3/2}}{\sqrt{x'}}\right)$$

$= 0$ [$\because f_2(x)$ does not exist in region $> x_2$]

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$$K_{12}\cdot 2 = (\beta_{1\bar{2}})^{3/2} \int_0^{\bar{x}_2} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2}\left[\frac{(\beta_{2\bar{2}})^{3/2}}{\bar{\rho}_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2} + \frac{1}{\rho_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{\rho_2}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}\right] \quad \cdots K_{12}-4$$

$$+\frac{(\gamma_{12})^{3/2}}{\rho_2}\int_{\bar{x}_2}^{x_2} dx\sqrt{x}\,[f_2(x)]^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= (\beta_{1\bar{2}})^{3/2}\frac{N_2}{3\eta}\left[\frac{(\beta_{2\bar{2}})^{3/2}}{\bar{\rho}_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2} + \frac{1}{\rho_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\beta_{2\bar{2}}}{\rho_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2}\right]$$

$$+\frac{(\gamma_{12})^{3/2}}{\rho_2}\left[\frac{1}{\rho_2}\left(x\frac{df_2(x)}{dx} - f_2(x)\right)_{x_2} - \frac{\beta_{2\bar{2}}}{\rho_2}\left(x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right)_{\bar{x}_2}\right]\frac{df_2(x)}{dx}\Big|_{x_2}$$

$\leftarrow K_{12}\cdot 7 \rightarrow$ $\qquad$ $\leftarrow K_{12}\cdot 8 \rightarrow$

$$= \cdots + \frac{(\gamma_{12})^{3/2}}{\rho_2}\left[\frac{1}{\rho_2}\left(\frac{N_2\rho_2}{3\eta} + \beta_{2\bar{2}}\frac{\overline{N_2}\,\overline{\rho_2}}{3\eta} - (\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}\,\rho_2}{3\eta}\right) - \frac{\beta_{2\bar{2}}}{\rho_2}\frac{\overline{N_2}\,\overline{\rho_2}}{3\eta}\right]\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= \cdots + \frac{(\gamma_{12})^{3/2}}{\rho_2}\left[\frac{N_2}{3\eta} + \frac{\beta_{2\bar{2}}\overline{N_2}\,\overline{\rho_2}}{3\eta\rho_2} - (\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}}{3\eta} - \frac{\beta_{2\bar{2}}\overline{N_2}\,\overline{\rho_2}}{3\eta\rho_2}\right]\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= \cdots + \frac{(\gamma_{12})^{3/2}N_2}{3\eta\rho_2}\frac{df_2(x)}{dx}\Big|_{x_2} - (\gamma_{12})^{3/2}(\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}}{3\eta\rho_2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$\Rightarrow K_{12}\cdot 2 = (\beta_{1\bar{2}})^{3/2}\frac{N_2}{3\eta}\left[\frac{(\beta_{2\bar{2}})^{3/2}}{\bar{\rho}_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2} + \frac{1}{\rho_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\beta_{2\bar{2}}}{\rho_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2}\right]$$

$$+\frac{(\gamma_{12})^{3/2}N_2}{3\eta\rho_2}\frac{df_2(x)}{dx}\Big|_{x_2} - (\gamma_{12})^{3/2}(\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}}{3\eta\rho_2}\frac{df_2(x)}{dx}\Big|_{x_2} \quad \#$$

Appendix of above integration

$$\int_{\bar{x}_2}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2}$$

$$= -\frac{1}{\rho_2}\int_{\bar{x}_2}^{x_2} dx\; x\frac{d^2f_2(x)}{dx^2}$$

$$= \frac{1}{\rho_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\bar{x}_2}^{x_2}$$

$$= \frac{1}{\rho_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\bar{x}_2}^{x_2} - \frac{1}{\rho_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\bar{x}_2}$$

$$= \frac{1}{\rho_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2} - \frac{\beta_{2\bar{2}}}{\rho_2}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\bar{x}_2}$$

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Therefore, $K_{12} = K_{12}\cdot 1 \oplus K_{12}\cdot 2$

$$K_{12} = \frac{\overline{N_2}}{3\eta}(\overline{r_{12}})^{3/2}(\beta_{2\bar{2}})^{3/2}\left[\frac{1}{\overline{Q_2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\beta_{2\bar{2}}}{\overline{Q_2}}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}\right] - \frac{(\beta_{1\bar{2}})^{3/2}(\beta_{2\bar{2}})^{3/2}}{\overline{Q_2}}$$

$$\left[\frac{4}{7}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\bar{x}_2} - \frac{5\overline{N_2}}{2\eta}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}\right] - \frac{(\overline{r_{12}})^{3/2}}{\overline{Q_2}}\left[\frac{4}{7}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}\right.$$

$$-\frac{4}{7}(\beta_{2\bar{2}})^{5/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\bar{x}_2} - \frac{5}{7\overline{Q_2}}\frac{df_2(x)}{dx}\Big|_{x_2}\left\{\frac{N_2Q_2}{3\eta} + \beta_{2\bar{2}}\frac{\overline{N_2}\,\overline{Q_2}}{3\eta} - (\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}\,Q_2}{3\eta}\right\}$$

$$\left.+\frac{5}{7\overline{Q_2}}(\beta_{2\bar{2}})^2\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}\frac{\overline{N_2}\,\overline{Q_2}}{3\eta}\right] - \frac{\overline{N_2}\,\overline{Q_2}}{3\eta\overline{Q_2}}(\overline{r_{12}})^{3/2}(\beta_{2\bar{2}})\left[\frac{1}{\overline{Q_2}}\frac{df_2(x)}{dx}\Big|_{x_2}\right.$$

$$\left.-\frac{\beta_{2\bar{2}}}{\overline{Q_2}}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}\right] + \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{N_2}{3\eta Q_1}(\overline{r_{12}})\frac{df_2(x)}{dx}\Big|_{x_2} \oplus (\beta_{1\bar{2}})^{3/2}\frac{\overline{N_2}}{3\eta}\left[\frac{(\beta_{2\bar{2}})^{3/2}}{\overline{Q_2}}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}\right.$$

$$\left.+\frac{1}{\overline{Q_2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\beta_{2\bar{2}}}{\overline{Q_2}}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}\right] + \frac{(\overline{r_{12}})^{3/2}N_2}{3\eta Q_2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$-(\overline{r_{12}})^{3/2}(\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}}{3\eta\overline{Q_2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= -\frac{\overline{N_2}}{3\eta\overline{Q_2}}(\overline{r_{12}})^{3/2}(\beta_{2\bar{2}})^{5/2}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2} - \frac{4}{7\overline{Q_2}}(\beta_{1\bar{2}})^{3/2}(\beta_{2\bar{2}})^{3/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}$$

$$+\frac{5\overline{N_2}(\beta_{1\bar{2}})^{3/2}(\beta_{2\bar{2}})^{3/2}}{2\eta\overline{Q_2}}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2} - \frac{4(\overline{r_{12}})^{3/2}}{7\overline{Q_2}}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}$$

$$+\frac{4}{7\overline{Q_2}}(\beta_{2\bar{2}})^{5/2}(\overline{r_{12}})^{3/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\bar{x}_2} + \frac{5N_2}{2\eta\overline{Q_2}}(\overline{r_{12}})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+\frac{5\beta_{2\bar{2}}\overline{N_2}\,\overline{Q_2}}{2\eta\overline{Q_2}^2}(\overline{r_{12}})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\overline{N_2}}{2\eta\overline{Q_2}}(\beta_{2\bar{2}})^{3/2}(\overline{r_{12}})^{3/2}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}$$

$$-\frac{5\overline{N_2}\,\overline{Q_2}}{2\eta\overline{Q_2}^2}(\beta_{2\bar{2}})^2(\overline{r_{12}})^{3/2}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2} - \frac{\overline{N_2}\,\overline{Q_2}}{3\eta\overline{Q_2}^2}(\overline{r_{12}})^{3/2}(\beta_{2\bar{2}})\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+\frac{\overline{N_2}\,\overline{Q_2}}{3\eta\overline{Q_2}^2}(\overline{r_{12}})^{3/2}(\beta_{2\bar{2}})^2\frac{df_2(x)}{dx}\Big|_{\bar{x}_2} + \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{N_2}{3\eta Q_1}(\overline{r_{12}})\frac{df_2(x)}{dx}\Big|_{x_2}$$

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$$\oplus \frac{\overline{N_2}}{3\eta\overline{\Phi_2}}(\beta_{12}\overline{)}^{3/2}(\beta_{22}\overline{)}^{3/2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta\Phi_2}(\beta_{12}\overline{)}^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$-\frac{\overline{N_2}}{3\eta\Phi_2}(\beta_{12}\overline{)}^{3/2}(\beta_{22}\overline{)}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{N_2}{3\eta\Phi_2}(\overline{r_{12}})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= \frac{4}{7\Phi_2}(\beta_{22}\overline{)}^{5/2}(\overline{r_{12}})^{3/2}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\overline{x_2}} - \frac{4}{7\overline{\Phi_2}}(\beta_{12}\overline{)}^{3/2}(\beta_{22}\overline{)}^{3/2}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\overline{x_2}}$$

$$-\frac{4}{7\Phi_2}(\overline{r_{12}})^{3/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} + \frac{5\overline{N_2}(\beta_{12}\overline{)}^{3/2}(\beta_{22}\overline{)}^{3/2}}{2\eta\overline{\Phi_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$

$$-\frac{\overline{N_2}}{3\eta\Phi_2}(\overline{r_{12}})^{3/2}(\beta_{22}\overline{)}^{5/2}\frac{df_2(x)}{dx}\Big|_{\overline{x_2}} - \frac{5\overline{N_2}\,\overline{\Phi_2}}{2\eta\Phi_2^2}(\beta_{22}\overline{)}^{2}(\overline{r_{12}})^{3/2}\frac{df_2(x)}{dx}\Big|_{\overline{x_2}}$$

$$+\frac{\overline{N_2}\,\overline{\Phi_2}}{3\eta\Phi_2^2}(\overline{r_{12}})^{3/2}(\beta_{22}\overline{)}^{2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{\overline{N_2}}{3\eta\overline{\Phi_2}}(\beta_{12}\overline{)}^{3/2}(\beta_{22}\overline{)}^{3/2}\frac{df_2(x)}{dx}\Big|_{\overline{x_2}}$$

$$-\frac{\overline{N_2}}{3\eta\Phi_2}(\beta_{12}\overline{)}^{3/2}(\beta_{22}\overline{)}\frac{df_2(x)}{dx}\Big|_{\overline{x_2}} + \frac{5N_2}{2\eta\Phi_2}(\overline{r_{12}})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+5\frac{\beta_{22}\overline{N_2}\,\overline{\Phi_2}}{2\eta\Phi_2^2}(\overline{r_{12}})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\overline{N_2}}{2\eta\Phi_2}(\beta_{22}\overline{)}^{3/2}(\overline{r_{12}})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$-\frac{\overline{N_2}\,\overline{\Phi_2}}{3\eta\Phi_2^2}(\overline{r_{12}})^{3/2}(\beta_{22}\overline{)}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{N_2}{3\eta\Phi_2}(\overline{r_{12}})\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+\frac{\overline{N_2}}{3\eta\Phi_2}(\beta_{12}\overline{)}^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta\Phi_2}(\overline{r_{12}})^{3/2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+\frac{N_2}{3\eta\Phi_2}\frac{df_2(x)}{dx}\Big|_{x_1}$$

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$$= \frac{4}{7}(\beta_{2\bar{2}})^{3/2}\left\{\frac{(r_{12})^{3/2}(\beta_{2\bar{2}})}{Q_2} - \frac{(\beta_{1\bar{2}})^{3/2}}{Q_2}\right\}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\bar{x}_2}$$

$$-\frac{4}{7Q_2}(r_{12})^{3/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} + \frac{N_2}{3\eta}(\beta_{2\bar{2}})\left\{\frac{5(\beta_{2\bar{2}})^{1/2}(\beta_{1\bar{2}})^{3/2}}{7\bar{Q}_2}\right.$$

$$-\frac{(r_{12})^{3/2}(\beta_{2\bar{2}})^{3/2}}{Q_2} - \frac{5\bar{Q}_2}{7Q_2^2}(\beta_{2\bar{2}})(r_{12})^{3/2} + \frac{(r_{12})^{3/2}(\beta_{2\bar{2}})\bar{Q}_2}{Q_2^2}$$

$$+\frac{(\beta_{1\bar{2}})^{3/2}(\beta_{2\bar{2}})^{1/2}}{\bar{Q}_2} - \frac{(\beta_{1\bar{2}})^{3/2}}{Q_2}\left.\right\}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2} + \frac{1}{3\eta Q_2}\left[\frac{5N_2}{7}(r_{12})^{3/2}\right.$$

$$+\frac{5\bar{N}_2\bar{Q}_2}{7Q_2}(\beta_{2\bar{2}})(r_{12})^{3/2} - \frac{5\bar{N}_2}{7}(\beta_{2\bar{2}})^{3/2}(r_{12})^{3/2} - \frac{\bar{N}_2\bar{Q}_2}{Q_2}(r_{12})^{3/2}(\beta_{2\bar{2}})$$

$$\left.-\frac{N_2Q_2}{Q_1}(r_{12}) + \bar{N}_2(\beta_{1\bar{2}})^{3/2} + N_2(r_{12})^{3/2}\right]\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$= \frac{4}{7}(\beta_{2\bar{2}})^{3/2}\left\{\frac{(r_{12})^{3/2}(\beta_{2\bar{2}})}{Q_2} - \frac{(\beta_{1\bar{2}})^{3/2}}{\bar{Q}_2}\right\}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\bar{x}_2}$$

$$-\frac{4}{7Q_2}(r_{12})^{3/2}\sqrt{x}(f_2(x))^{5/2}\Big|_{x_2} + \frac{\bar{N}_2}{3\eta}(\beta_{2\bar{2}})\left\{\frac{5(\beta_{2\bar{2}})^{1/2}(\beta_{1\bar{2}})^{3/2}}{7\bar{Q}_2} - \frac{(\beta_{1\bar{2}})^{3/2}}{Q_2}\right.$$

$$-\frac{5\bar{Q}_2}{7Q_2^2}(\beta_{2\bar{2}})(r_{12})^{3/2} + \frac{(\beta_{1\bar{2}})^{3/2}(\beta_{2\bar{2}})^{1/2}}{\bar{Q}_2} - \frac{(r_{12})^{3/2}(\beta_{2\bar{2}})}{Q_2}$$

$$\left.+\frac{(r_{12})^{3/2}(\beta_{2\bar{2}})\bar{Q}_2}{Q_2^2}\right\}\frac{df_2(x)}{dx}\Big|_{\bar{x}_2} + \frac{1}{3\eta Q_2}\left[N_2(r_{12})^{3/2} - \frac{N_2(r_{12})Q_2}{Q_1}\right.$$

$$\left.+\bar{N}_2(\beta_{1\bar{2}})^{3/2} - \frac{\bar{N}_2\bar{Q}_2}{Q_2}(r_{12})^{3/2}(\beta_{2\bar{2}}) + \frac{5}{7}(r_{12})^{3/2}\left\{N_2 + \frac{\bar{N}_2\bar{Q}_2}{Q_2}(\beta_{2\bar{2}})\,\bar{N}_2(\beta_{2\bar{2}})^{3/2}\right\}\right]$$

$$\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1} \quad \#\#$$

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Appendix for K_{12}

$$\int_0^x dx \sqrt{x} (f_2(x))^{3/2}$$

In region $0 < x < \bar{x}_2$ (already done)

$$\Rightarrow \int_0^x dx \sqrt{x} (f_2(x))^{3/2} = \frac{(\beta_{2\bar{2}})^{3/2}}{\bar{\varphi}_2} \left[x \frac{d\bar{f}_2(x)}{dx} - \bar{f}_2(x) \right] \# \quad \text{—— } K_{12} \cdot 1$$

The normalization condition gives

$$\int_0^{\bar{x}_2} dx \sqrt{x} (\bar{f}_2(x))^{3/2} = \frac{\bar{N}_2}{3\eta} \qquad \left[\because f_2(x) = \beta_{2\bar{2}} \bar{f}_2(x) \right]$$

or,

$$\frac{1}{(\beta_{2\bar{2}})^{3/2}} \int_0^{\bar{x}_2} dx \sqrt{x} (f_2(x))^{3/2} = \frac{\bar{N}_2}{3\eta}$$

$$\Rightarrow \int_0^{\bar{x}_2} dx \sqrt{x} (f_2(x))^{3/2} = \frac{\bar{N}_2 (\beta_{2\bar{2}})^{3/2}}{3\eta} \# \quad \text{—— } K_{12} \cdot 2$$

$$\int_{\bar{x}_2}^x dx \sqrt{x} (f_2(x))^{3/2}$$

$$= \int_{\bar{x}_2}^x dx \frac{x}{\varphi_2} \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{\varphi_2} \int_{\bar{x}_2}^x dx \, x \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{\varphi_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{\bar{x}_2}^x$$

$$= \frac{1}{\varphi_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right] - \frac{1}{\varphi_2} \left[x \frac{d\bar{f}_2(x)}{dx} - \bar{f}_2(x) \right]_{\bar{x}_2}$$

$$= \frac{1}{\varphi_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right] - \frac{\beta_{2\bar{2}}}{\varphi_2} \left[x \frac{d\bar{f}_2(x)}{dx} - \bar{f}_2(x) \right]_{\bar{x}_2}$$

$$= \frac{1}{\varphi_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right] - \frac{\beta_{2\bar{2}}}{\varphi_2} \frac{\bar{N}_2 \bar{\varphi}_2}{3\eta}$$

$$\Rightarrow \int_{\bar{x}_2}^x dx \sqrt{x} (f_2(x))^{3/2} = \frac{1}{\varphi_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right] - \frac{\beta_{2\bar{2}} \bar{N}_2 \bar{\varphi}_2}{3\eta \varphi_2} \# \quad \text{—— } K_{12} \cdot 3$$

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$$\int_{\bar{x}_2}^{x_2} dx \sqrt{x}\,(f_2(x))^{3/2}$$

$$= \int_{\bar{x}_2}^{x_2} dx \frac{x}{\varphi_2} \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{\varphi_2} \int_{\bar{x}_2}^{x_2} dx\, x \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{\varphi_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{\bar{x}_2}^{x_2}$$

$$= \frac{1}{\varphi_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2} - \frac{1}{\varphi_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{\bar{x}_2}$$

$$= \frac{1}{\varphi_2} \left[\frac{N_2 \varphi_2}{3\eta} + \beta_{2\bar{2}} \frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta} - (\beta_{2\bar{2}})^{3/2} \frac{N_2 \varphi_2}{3\eta} \right] - \frac{\beta_{2\bar{2}}}{\varphi_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{\bar{x}_2}$$

$$= \frac{N_2}{3\eta} + \beta_{2\bar{2}} \frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta\,\varphi_2} - (\beta_{2\bar{2}})^{3/2} \frac{N_2}{3\eta} - \frac{\beta_{2\bar{2}}}{\varphi_2} \cdot \frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}$$

$$\Rightarrow \int_{\bar{x}_2}^{x_2} dx \sqrt{x}\,(f_2(x))^{3/2} = \frac{N_2}{3\eta} - (\beta_{2\bar{2}})^{3/2} \frac{\overline{N_2}}{3\eta} \quad \# \qquad K_{12}\cdot 4$$

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Appendix for kinetic energy & appendix for K_{12} are same.

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Appendix for Kinetic energy

$$\int dx \frac{(f(x))^{5/2}}{\sqrt{x}} = \frac{4}{7}\sqrt{x}(f(x))^{5/2} - \frac{5}{7\Phi} f'[xf' - f]$$

$$\Rightarrow \int dx \frac{(f(x))^{5/2}}{\sqrt{x}} = \frac{4}{7}\sqrt{x}(f(x))^{5/2} - \frac{5}{7\Phi}\frac{df(x)}{dx}\left[x\frac{df(x)}{dx} - f(x)\right]$$

This means,

$$\int dx \frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}} = \frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{5/2} - \frac{5}{7\overline{\Phi_2}}\frac{d\overline{f_2(x)}}{dx}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]$$

From page (2)

$$\int_0^{\overline{x}_2} dx \frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}} = \frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{5/2}\Big|_{\overline{x}_2} - \frac{5}{7\overline{\Phi_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x}_2}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x}_2}$$

$$= \frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{5/2}\Big|_{\overline{x}_2} - \frac{5}{7\overline{\Phi_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x}_2} \cdot \frac{\overline{N_2}\,\overline{\Phi_2}}{3\eta}$$

$$\Rightarrow \int_0^{\overline{x}_2} dx \frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}} = \frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{5/2}\Big|_{\overline{x}_2} - \frac{5\overline{N_2}}{21\eta}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x}_2} \quad \# \; K_{12}\cdot 5$$

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Normalization condition

$$\int_0^{x_{max}} dx\sqrt{x}\,(f_I(x))^{3/2} = \frac{N_I}{3\eta}$$

$$\int_0^{\overline{x}_{max}} dx\sqrt{x}\,(\overline{f_I(x)})^{3/2} = \frac{\overline{N_I}}{3\eta}$$

$$\Rightarrow \int_0^{\overline{x}_2} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2} = \frac{\overline{N_2}}{3\eta}$$

The Thomas-Fermi equation gives

$$\frac{d^2\overline{f_2(x)}}{dx^2} = \overline{\Phi_2}\,\frac{(\overline{f_2(x)})^{3/2}}{\sqrt{x}}$$

$$x\frac{d^2\overline{f_2(x)}}{dx^2} = \overline{\Phi_2}\sqrt{x}\,(\overline{f_2(x)})^{3/2}$$

$$\int_0^{\overline{x}_2} x\frac{d^2\overline{f_2(x)}}{dx^2} = \overline{\Phi_2}\int_0^{\overline{x}_2} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2}$$

$$\int_0^{\overline{x}_2} x\frac{d^2\overline{f_2(x)}}{dx^2} = \frac{\overline{\Phi_2}\,\overline{N_2}}{3\eta}$$

$$\Rightarrow \left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x}_2} = \frac{\overline{N_2}\,\overline{\Phi_2}}{3\eta} \quad \text{— K·12·8}$$

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$$\int_0^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} = \frac{N_2}{3\eta}$$

$$\text{or, } \int_0^{\overline{x_2}} dx\sqrt{x}(f_2(x))^{3/2} + \int_{\overline{x_2}}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} = \frac{N_2}{3\eta}$$

$$\text{or, } (\beta_{2\bar{2}})^{3/2} \int_0^{\overline{x_2}} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2} + \int_{\overline{x_2}}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} = \frac{N_2}{3\eta}$$

$$\text{or, } (\beta_{2\bar{2}})^{3/2}\,\overline{\frac{N_2}{3\eta}} + \int_{\overline{x_2}}^{x_2} dx\,\frac{x}{\varphi_2}\,\frac{d^2 f_2(x)}{dx^2} = \frac{N_2}{3\eta}$$

$$\text{or, } \frac{1}{\varphi_2}\int_{\overline{x_2}}^{x_2} dx\; x\,\frac{d^2 f_2(x)}{dx^2} = \frac{N_2}{3\eta} - (\beta_{2\bar{2}})^{3/2}\,\overline{\frac{N_2}{3\eta}}$$

$$\text{or, } \frac{1}{\varphi_2}\left[x\,\frac{df_2(x)}{dx} - f_2(x)\right]_{\overline{x_2}}^{x_2} = \frac{N_2}{3\eta} - (\beta_{2\bar{2}})^{3/2}\,\overline{\frac{N_2}{3\eta}}$$

$$\text{or, } \left[x\,\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2} - \left[x\,\frac{df_2(x)}{dx} - f_2(x)\right]_{\overline{x_2}} = \frac{N_2\varphi_2}{3\eta} - (\beta_{2\bar{2}})^{3/2}\,\overline{\frac{N_2\varphi_2}{3\eta}}$$

$$\text{or, } \left[x\,\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2} - \beta_{2\bar{2}}\left[x\,\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}} = \frac{N_2\varphi_2}{3\eta} - (\beta_{2\bar{2}})^{3/2}\,\overline{\frac{N_2\varphi_2}{3\eta}}$$

$$\text{or, } \left[x\,\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2} - \beta_{2\bar{2}}\cdot\overline{\frac{N_2\varphi_2}{3\eta}} = \frac{N_2\varphi_2}{3\eta} - (\beta_{2\bar{2}})^{3/2}\,\overline{\frac{N_2\varphi_2}{3\eta}}$$

$$\text{or, } \left[x\,\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2} = \frac{N_2\varphi_2}{3\eta} + \beta_{2\bar{2}}\,\overline{\frac{N_2\varphi_2}{3\eta}} - (\beta_{2\bar{2}})^{3/2}\,\overline{\frac{N_2\varphi_2}{3\eta}}$$

$$\Rightarrow \left[x\,\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2} = \frac{N_2\varphi_2}{3\eta} + \beta_{2\bar{2}}\,\overline{\frac{N_2\varphi_2}{3\eta}} - (\beta_{2\bar{2}})^{3/2}\,\overline{\frac{N_2\varphi_2}{3\eta}} \;\text{//}$$

— Kt2.7

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$$\int_0^{x_1} dx\sqrt{x}\,(f_1(x))^{3/2} = \frac{N_1}{3\eta}$$

or, $\int_0^{\bar{x}_2} dx\sqrt{x}\,(f_1(x))^{3/2} + \int_{\bar{x}_2}^{x_2} dx\sqrt{x}\,(f_1(x))^{3/2} + \int_{x_2}^{x_1} dx\sqrt{x}\,(f_1(x))^{3/2} = \frac{N_1}{3\eta}$

or, $(\beta_{12})^{3/2}\int_0^{\bar{x}_2} dx\sqrt{x}\,(\overline{f_2(x)})^{3/2} + (\gamma_{12})^{3/2}\int_{\bar{x}_2}^{x_2} dx\sqrt{x}\,(f_2(x))^{3/2} + \int_{x_2}^{x_1} dx\sqrt{x}\,(f_1(x))^{3/2} = \frac{N_1}{3\eta}$

or, $(\beta_{12})^{3/2}\frac{N_2}{3\eta} + (\gamma_{12})^{3/2}\int_{\bar{x}_2}^{x_2} dx\,\frac{x}{\varphi_2}\frac{d^2f_2(x)}{dx^2} + \int_{x_2}^{x_1} dx\sqrt{x}\,(f_1(x))^{3/2} = \frac{N_1}{3\eta}$

or, $\frac{(\gamma_{12})^{3/2}}{\varphi_2}\int_{\bar{x}_2}^{x_2} dx\,x\frac{d^2f_2(x)}{dx^2} + \int_{x_2}^{x_1} dx\sqrt{x}\,(f_1(x))^{3/2} = \frac{N_1}{3\eta} - (\beta_{12})^{3/2}\frac{N_2}{3\eta}$

or, $\frac{(\gamma_{12})^{3/2}}{\varphi_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\bar{x}_2}^{x_2} + \int_{x_2}^{x_1} dx\,\frac{x}{\varphi_1}\frac{d^2f_1(x)}{dx^2} = \frac{N_1}{3\eta} - (\beta_{12})^{3/2}\frac{N_2}{3\eta}$

or, $\frac{(\gamma_{12})^{3/2}}{\varphi_2}\left\{\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2} - \left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\bar{x}_2}\right\} + \frac{1}{\varphi_1}\int_{x_2}^{x_1} dx\,x\frac{d^2f_1(x)}{dx^2}$

$$= \frac{N_1}{3\eta} - (\beta_{12})^{3/2}\frac{N_2}{3\eta}$$

or, $\frac{(\gamma_{12})^{3/2}}{\varphi_2}\left\{\frac{N_2\varphi_2}{3\eta} + \beta_{22}\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta} - (\beta_{22})^{3/2}\frac{\overline{N_2}\varphi_2}{3\eta} - (\beta_{22})\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\bar{x}_2}\right\}$

$$+ \frac{1}{\varphi_1}\int_{x_2}^{x_1} dx\,x\frac{d^2f_1(x)}{dx^2} = \frac{N_1}{3\eta} - (\beta_{12})^{3/2}\frac{N_2}{3\eta}$$

or, $\frac{(\gamma_{12})^{3/2}}{\varphi_2}\left\{\frac{N_2\varphi_2}{3\eta} + \beta_{22}\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta} - (\beta_{22})^{3/2}\frac{\overline{N_2}\varphi_2}{3\eta} - (\beta_{22})\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}\right\}$

$$+ \frac{1}{\varphi_1}\left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_2}^{x_1} = \frac{N_1}{3\eta} - (\beta_{12})^{3/2}\frac{N_2}{3\eta}$$

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$$\text{or, } \frac{(\gamma_{12})^{3/2}}{\varphi_2}\left\{\frac{N_2\varphi_2}{3\eta} - (\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2\varphi_2}}{3\eta}\right\} + \frac{1}{\varphi_1}\left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_2}^{x_1} = \frac{N_1}{3\eta} - (\beta_{1\bar{2}})^{3/2}\frac{N_2}{3\eta}$$

$$\text{or, } \frac{1}{\varphi_1}\left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_2}^{x_1} = \frac{N_1}{3\eta} - (\beta_{1\bar{2}})^{3/2}\frac{N_2}{3\eta} - \frac{(\gamma_{12})^{3/2}}{\varphi_2}\cdot\frac{N_2\varphi_2}{3\eta} + \frac{(\gamma_{12})^{3/2}}{\varphi_2}\cdot(\beta_{2\bar{2}})^{3/2}\frac{N_2\varphi_2}{3\eta}$$

$$\text{or, } \frac{1}{\varphi_1}\left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_2}^{x_1} = \frac{N_1}{3\eta} - (\beta_{1\bar{2}})^{3/2}\frac{N_2}{3\eta} - (\gamma_{12})^{3/2}\frac{N_2}{3\eta} + (\gamma_{12})^{3/2}(\beta_{2\bar{2}})^{3/2}\frac{N_2}{3\eta}$$

$$\text{or, } \left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_1} - \left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_2} = \frac{N_1\varphi_1}{3\eta} - (\beta_{1\bar{2}})^{3/2}\frac{N_2\varphi_1}{3\eta}$$

$$- (\gamma_{12})^{3/2}\frac{N_2\varphi_1}{3\eta} + (\gamma_{12})^{3/2}(\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}\varphi_1}{3\eta}$$

$$\text{or, } \left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_1} - (\gamma_{12})\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2} = \frac{N_1\varphi_1}{3\eta} - (\beta_{1\bar{2}})^{3/2}\frac{\overline{N_2}\varphi_1}{3\eta}$$

$$- (\gamma_{12})^{3/2}\frac{N_2\varphi_1}{3\eta} + (\gamma_{12})^{3/2}(\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}\varphi_1}{3\eta}$$

$$\text{or, } \left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_1} - (\gamma_{12})\left[\frac{N_2\varphi_2}{3\eta} + \beta_{2\bar{2}}\frac{\overline{N_2}\varphi_2}{3\eta} - (\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}\varphi_2}{3\eta}\right] = \frac{N_1\varphi_1}{3\eta}$$

$$- (\beta_{1\bar{2}})^{3/2}\frac{\overline{N_2}\varphi_1}{3\eta} - (\gamma_{12})^{3/2}\frac{N_2\varphi_1}{3\eta} + (\gamma_{12})^{3/2}(\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}\varphi_1}{3\eta}$$

$$\text{or, } \left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_1} = \frac{N_1\varphi_1}{3\eta} - (\beta_{1\bar{2}})^{3/2}\frac{\overline{N_2}\varphi_1}{3\eta} - (\gamma_{12})^{3/2}\frac{N_2\varphi_1}{3\eta} + (\gamma_{12})^{3/2}(\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}\varphi_1}{3\eta}$$

$$+ \gamma_{12}\frac{N_2\varphi_2}{3\eta} + (\gamma_{12})(\beta_{2\bar{2}})\frac{\overline{N_2}\varphi_2}{3\eta} - (\gamma_{12})(\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}\varphi_2}{3\eta}$$

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$$\Rightarrow \left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_1} = \frac{N_1\varphi_1}{3\eta} + \bar{r}_{12}\frac{N_2\varphi_2}{3\eta} - (\bar{r}_{12})^{3/2}\frac{N_2\varphi_1}{3\eta} + \left\{(\bar{r}_{12})^{3/2}(\beta_{22})^{3/2} - (\beta_{12})^{3/2}\right\}$$

$$\frac{\overline{N_2\varphi_1}}{3\eta} - (\bar{r}_{12})(\beta_{22})^{3/2}\frac{\overline{N_2\varphi_2}}{3\eta} + (\bar{r}_{12})(\beta_{22})\frac{\overline{N_2\varphi_2}}{3\eta} \#$$

Continuation of K.E(1)

$$\int_{\bar{x}_2}^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}} = \frac{4}{7}\sqrt{x}(f_2(x))^{5/2}\Big|_{x_2} - \frac{4}{7}(\beta_{22})^{5/2}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\bar{x}_2}$$

$$-\frac{5}{7\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}\left[\frac{N_2\varphi_2}{3\eta} + \beta_{22}\frac{\overline{N_2\varphi_2}}{3\eta} - (\beta_{22})^{3/2}\frac{\overline{N_2\varphi_2}}{3\eta}\right]$$

$$+\frac{5}{7\varphi_2}(\beta_{22})^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\bar{x}_2}\frac{\overline{N_2\varphi_2}}{3\eta} \quad \text{—} \kappa_{12}\cdot 6$$

From K.E(1)

$$\int_{a_2}^{x_1} dx \frac{(f_1(x))^{5/2}}{\sqrt{x}} = \frac{4}{7}\sqrt{x}(f_1(x))^{5/2}\Big|_{x_2}^{x_1} - \frac{5}{7\varphi_1}\frac{df_1(x)}{dx}\left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_2}^{x_1}$$

$$= \frac{4}{7}\sqrt{x}(f_1(x))^{5/2}\Big|_{x_1} - \frac{4}{7}\sqrt{x}(f_1(x))^{5/2}\Big|_{x_2} - \frac{5}{7\varphi_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$\left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_1} + \frac{5}{7\varphi_1}\frac{df_1(x)}{dx}\Big|_{x_2}\left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_2}$$

$$= \frac{4}{7}\sqrt{x}(f_1(x))^{5/2}\Big|_{x_1} - \frac{4}{7}(\bar{r}_{12})^{5/2}\sqrt{x}(f_2(x))^{5/2}\Big|_{x_2} - \frac{5}{7\varphi_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$\left[\frac{N_1\varphi_1}{3\eta} + \bar{r}_{12}\frac{N_2\varphi_2}{3\eta} - (\bar{r}_{12})^{3/2}\frac{N_2\varphi_1}{3\eta} + \left\{(\bar{r}_{12})^{3/2}(\beta_{22})^{3/2} - (\beta_{12})^{3/2}\right\}\right.$$

$$\left.\frac{\overline{N_2\varphi_1}}{3\eta} - (\bar{r}_{12})(\beta_{22})^{3/2}\frac{\overline{N_2\varphi_2}}{3\eta} + (\bar{r}_{12})(\beta_{22})\frac{\overline{N_2\varphi_2}}{3\eta}\right] + \frac{5}{7\varphi_1}(\bar{r}_{12})^2\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$\left[\frac{N_2\varphi_2}{3\eta} + \beta_{22}\frac{\overline{N_2\varphi_2}}{3\eta} - (\beta_{22})^{3/2}\frac{\overline{N_2\varphi_2}}{3\eta}\right] \#$$

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Kinetic Energy

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Calculation of Kinetic energy

$$T=\sum_I \int_0^{r_{max}} d^3r \; \frac{3(6\pi^2\hbar^3)^{5/3}}{20\pi^2\hbar^3 m^I (\xi^I)^{2/3}} (n^I(r))^{5/3} + \sum_I \int_0^{r_{max}} d^3r \; \frac{3(6\pi^2\hbar^3)^{5/3}}{20\pi^2\hbar^3 \bar{m}^I (\bar{\xi}^I)^{2/3}} (\bar{n}^I(r))^{5/3}$$

We have,

$$n^I(r) = \left\{ \frac{2\times(4/3)\alpha_s}{a} \right\}^{3/2} \left(\frac{f^I(r)}{r} \right)^{3/2} \frac{\xi^I}{6\pi^2}$$

$$\bar{n}^I(r) = \left\{ \frac{2\times(4/3)\alpha_s}{a} \right\}^{3/2} \left(\frac{\overline{f^I(r)}}{r} \right)^{3/2} \frac{\bar{\xi}^I}{6\pi^2}$$

So that,

$$T = \sum_I \int_0^{r_{max}} d^3r \; \frac{3(6\pi^2\hbar^3)^{5/3}}{20\pi^2\hbar^3 m^I (\xi^I)^{2/3}} \left[\left\{ \frac{2\times(4/3)\alpha_s}{a} \right\}^{3/2} \left(\frac{f_I(r)}{r} \right)^{3/2} \frac{\xi^I}{6\pi^2} \right]^{5/3}$$

$$+ \sum_I \int_0^{r_{max}} d^3r \; \frac{3(6\pi^2\hbar^3)^{5/3}}{20\pi^2\hbar^3 \bar{m}^I (\bar{\xi}^I)^{2/3}} \left[\left\{ \frac{2\times(4/3)\alpha_s}{a} \right\}^{3/2} \left(\frac{\overline{f^I(r)}}{r} \right)^{3/2} \frac{\bar{\xi}^I}{6\pi^2} \right]^{5/3}$$

$$\Rightarrow T = T_1 + T_2$$

Taking T_1

$$T_1 = \sum_I \int_0^{r_{max}} d^3r \; \frac{3(6\pi^2\hbar^3)^{5/3}}{20\pi^2\hbar^3 m^I (\xi^I)^{2/3}} \left[\left\{ \frac{2\times(4/3)\alpha_s}{a} \right\}^{3/2} \left(\frac{f_I(r)}{r} \right)^{3/2} \frac{\xi^I}{6\pi^2} \right]^{5/3}$$

$$= \sum_I \int_0^{r_{max}} d^3r \; \frac{3(6\pi^2\hbar^3)^{5/3}}{20\pi^2\hbar^3 m^I (\xi^I)^{2/3}} \cdot \left(\frac{\xi^I}{6\pi^2} \right)^{5/3} \left\{ \frac{2\times(4/3)\alpha_s}{a} \right\}^{\frac{3}{2}\times\frac{5}{3}} \left(\frac{f_I(r)}{r} \right)^{\frac{3}{2}\times\frac{5}{3}}$$

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$$= \sum_I \int^{r_{max}} d^3r \; \frac{3\hbar^{7\times\frac{5}{7}} (\xi^I)^{\frac{5}{3}-\frac{5}{3}}}{20\pi^2\hbar^3 m^I} \left\{ \frac{2\times(4/3)\alpha_s}{a} \right\}^{\frac{5}{2}} \left(\frac{f_I(r)}{r} \right)^{\frac{5}{2}}$$

$$= \sum_I \int^{r_{max}} d^3r \; \frac{3\hbar^2 \xi^I}{20\pi^2 m^I} \left\{ \frac{2\times(4/3)\alpha_s}{a} \right\}^{\frac{5}{2}} \left(\frac{f_I(r)}{r} \right)^{\frac{5}{2}}$$

$$= \frac{3\hbar^2}{20\pi^2} \left\{ \frac{2\times(4/3)\alpha_s}{a} \right\}^{\frac{5}{2}} \sum_I \int^{r_{max}} d^3r \frac{\xi^I}{m^I} \left(\frac{f_I(r)}{r} \right)^{5/2}$$

$$= \underbrace{\frac{3\hbar^2}{20\pi^2 m_1} \left\{ \frac{2\times(4/3)\alpha_s}{a} \right\}^{\frac{5}{2}} 4\pi \sum_I}_{\text{Constant}} \int^{r_{max}} dr \; r^2 \alpha_I \xi^I \left(\frac{f_I(r)}{r} \right)^{5/2}$$

$$\left[\because \alpha_I = \frac{m_1}{m_I} \right]$$

$$= \text{Constant} \cdot \sum_I \int^{r_{max}} dr \; r^2 \alpha_I \xi^I \left(\frac{f^I(r)}{r} \right)^{5/2}$$

Put $r = Rx$

$\Rightarrow dr = R\,dx$

$$= \text{Constant} \cdot \sum_I \int^{x_{max}} R\,dx \; R^2 x^2 \alpha_I \xi^I \left(\frac{f^I(x)}{Rx} \right)^{5/2}$$

$$= \text{Constant} \cdot \sum_I \int^{x_{max}} dx \; R^{3-\frac{5}{2}} \alpha_I \xi^I \frac{(f^I(x))^{5/2}}{x^{\frac{5}{2}-2}}$$

$$= \text{Constant} \cdot \sum_I \int^{x_{max}} dx \; \sqrt{R}\, \alpha_I \xi^I \frac{(f^I(x))^{5/2}}{\sqrt{x}}$$

$$= \underbrace{\text{Constant}\sqrt{R}}_{\text{This is also a constant. Let it be represented by term 'c'.}} \cdot \sum_I \int^{x_{max}} dx \; \alpha_I \xi^I \frac{(f^I(x))^{5/2}}{\sqrt{x}}$$

$$= c \sum_I \int^{x_{max}} dx \; \alpha_I \xi^I \frac{(f_I(x))^{5/2}}{\sqrt{x}}$$

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$$\text{or, } \frac{T_I}{c} = \frac{4\pi}{I} \int_0^{x_{max}} dx\, \alpha_I \xi_I^I \frac{(f_I^I(x))^{5/2}}{\sqrt{x}}$$

$$\text{or, } \frac{T_1}{c} = \alpha_1 \xi_1 \int_0^{x_1} dx \frac{(f_1(x))^{5/2}}{\sqrt{x}} + \alpha_2 \xi_2 \int_0^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

$$= \alpha_1 \xi_1 \int_0^{\overline{x}_2} dx \frac{(f_1(x))^{5/2}}{\sqrt{x}} + \alpha_1 \xi_1 \int_{\overline{x}_2}^{x_2} dx \frac{(f_1(x))^{5/2}}{\sqrt{x}} + \alpha_1 \xi_1 \int_{x_2}^{x_1} dx \frac{(f_1(x))^{5/2}}{\sqrt{x}}$$

$$+ \alpha_2 \xi_2 \int_0^{\overline{x}_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}} + \alpha_2 \xi_2 \int_{\overline{x}_2}^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

$$= \alpha_1 \xi_1 (\beta_{1\bar{2}})^{5/2} \int_0^{\overline{x}_2} dx \frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}} + \alpha_1 \xi_1 (\overline{\gamma}_{12})^{5/2} \int_{\overline{x}_2}^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

$$+ \alpha_1 \xi_1 \int_{x_2}^{x_1} dx \frac{(f_1(x))^{5/2}}{\sqrt{x}} + \alpha_2 \xi_2 (\beta_{2\bar{2}})^{5/2} \int_0^{\overline{x}_2} dx \frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}}$$

$$+ \alpha_2 \xi_2 \int_{\overline{x}_2}^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

$$\Rightarrow T_1/c$$

$$= \left[\alpha_1 \xi_1 (\beta_{1\bar{2}})^{5/2} + \alpha_2 \xi_2 (\beta_{2\bar{2}})^{5/2}\right] \int_0^{\overline{x}_2} dx \frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}}$$

$$+ \left[\alpha_1 \xi_1 (\overline{\gamma}_{12})^{5/2} + \alpha_2 \xi_2\right] \int_{\overline{x}_2}^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}} + \alpha_1 \xi_1 \int_{x_2}^{x_1} dx \frac{(f_1(x))^{5/2}}{\sqrt{x}} \quad -(1)$$

From appendix,

$$\int_0^{\overline{x}_2} dx \frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}} = \frac{4}{7} \sqrt{x} (\overline{f_2(x)})^{5/2} \Big|_{\overline{x}_2} - \frac{5\overline{N}_2}{21\eta} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x}_2}$$

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$$\int_{\bar{x}_2}^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}} = \frac{4}{7}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} - \frac{4}{7}(\beta_{2\bar{2}})^{5/2}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\bar{x}_2}$$

$$- \frac{5}{7\varphi_2}\frac{df_2(x)}{dx}\Big|_{x_2}\left[\frac{N_2\varphi_2}{3\eta} + \beta_{2\bar{2}}\frac{\overline{N_2}\overline{\varphi_2}}{3\eta} - (\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}\varphi_2}{3\eta}\right]$$

$$+ \frac{5}{7\varphi_2}(\beta_{2\bar{2}})^2\frac{d\overline{f_2(x)}}{d\bar{x}}\Big|_{\bar{x}_2}\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}$$

$$\int_{x_2}^{x_1} dx \frac{(f_1(x))^{5/2}}{\sqrt{x}} = \frac{4}{7}\sqrt{x}\,(f_1(x))^{5/2}\Big|_{x_1} - \frac{4}{7}(\gamma_{12})^{5/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} - \frac{5}{7\varphi_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$\left[\frac{N_1\varphi_1}{3\eta} + \gamma_{12}\frac{N_2\varphi_2}{3\eta} - (\gamma_{12})^{3/2}\frac{N_2\varphi_1}{3\eta} + \frac{\overline{N_2}\varphi_1}{3\eta}\left\{(\gamma_{12})^{3/2}(\beta_{2\bar{2}})^{3/2} - (\beta_{1\bar{2}})^{3/2}\right\}\right.$$

$$\left. - (\gamma_{12})(\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}\varphi_2}{3\eta} + (\gamma_{12})(\beta_{2\bar{2}})\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}\right] + \frac{5}{7\varphi_1}(\gamma_{12})^2\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$\left[\frac{N_2\varphi_2}{3\eta} + \beta_{2\bar{2}}\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta} - (\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}\varphi_2}{3\eta}\right]$$

Taking T_2

$$T_2 = \sum_I \int_0^{r_{max}} d^3r \frac{3(6\pi^2\hbar^3)^{5/3}}{20\pi^2\hbar^3\bar{m}^I(\bar{\xi}^I)^{2/3}}\left[\left\{\frac{2\times(4/3)\alpha_s}{a}\right\}^{3/2}\left(\frac{\overline{f^I(r)}}{r}\right)^{3/2}\frac{\bar{\xi}^I}{6\pi^2}\right]^{5/3}$$

$$= \sum_I \int_0^{r_{max}} d^3r \frac{3(6\pi^2\hbar^3)^{5/3}}{20\pi^2\hbar^3\bar{m}^I(\bar{\xi}^I)^{2/3}}\cdot\left(\frac{\bar{\xi}^I}{6\pi^2}\right)^{\frac{5}{3}}\left\{\frac{2\times(4/3)\alpha_s}{a}\right\}^{\frac{3}{2}\times\frac{5}{3}}\left(\frac{\overline{f^I(r)}}{r}\right)^{\frac{3}{2}\times\frac{5}{3}}$$

$$= \sum_I \int_0^{r_{max}} d^3r \frac{3\hbar^{3\times\frac{5}{3}}(\bar{\xi}^I)^{\frac{5}{3}-\frac{2}{3}}}{20\pi^2\hbar^3\bar{m}^I}\left\{\frac{2\times(4/3)\alpha_s}{a}\right\}^{\frac{5}{2}}\left(\frac{\overline{f^I(r)}}{r}\right)^{\frac{5}{2}}$$

$$= \sum_I \int_0^{r_{max}} d^3r \frac{3\hbar^2\bar{\xi}^I}{20\pi^2\bar{m}^I}\left\{\frac{2\times(4/3)\alpha_s}{a}\right\}^{\frac{5}{2}}\left(\frac{\overline{f^I(r)}}{r}\right)^{\frac{5}{2}}$$

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$$= \frac{3\hbar^2}{20\pi^2}\left\{\frac{2\times(4/3)\alpha_s}{a}\right\}^{\frac{5}{2}} \sum_I \int_0^{r_{max}} d^3r \frac{\overline{\xi}^I}{m^I}\left(\frac{\overline{f^I(r)}}{r}\right)^{\frac{5}{2}}$$

$$= \underbrace{\frac{3\hbar^2}{20\pi^2 m_1}\left\{\frac{2\times(4/3)\alpha_s}{a}\right\}^{\frac{5}{2}} 4\pi \sum_I}_{\text{Constant}} \int_0^{r_{max}} dr\, r^2 \overline{\alpha_I}\,\overline{\xi}^I \left(\frac{\overline{f^I(r)}}{r}\right)^{5/2} \quad \left[\because \overline{\alpha_I} = \frac{m_1}{m_I}\right]$$

$$= \text{Constant} \cdot \sum_I \int_0^{r_{max}} dr\, r^2 \overline{\alpha_I}\,\overline{\xi}^I \left(\frac{\overline{f^I(r)}}{r}\right)^{5/2}$$

Put $r = R\alpha$

$\Rightarrow dr = R\,d\alpha$

$$= \text{constant} \cdot \sum_I \int_0^{\alpha_{max}} R\,d\alpha\, R^2\alpha^2\, \overline{\alpha_I}\,\overline{\xi}_I \left(\frac{\overline{f^I(\alpha)}}{R\alpha}\right)^{5/2}$$

$$= \text{Constant} \cdot \sum_I \int_0^{\alpha_{max}} d\alpha\, R^{3-\frac{5}{2}}\, \overline{\alpha_I}\,\overline{\xi}_I\, \frac{\left(\overline{f^I(\alpha)}\right)^{5/2}}{\alpha^{\frac{5}{2}-2}}$$

$$= \text{Constant} \cdot \sum_I \int_0^{\alpha_{max}} d\alpha \sqrt{R}\, \overline{\alpha_I}\,\overline{\xi}_I\, \frac{\left(\overline{f^I(\alpha)}\right)^{5/2}}{\sqrt{\alpha}}$$

$$= \underbrace{\text{Constant}\sqrt{R}}_{\text{This is also a constant. Let it be represented by term 'c'.}} \cdot \sum_I \int_0^{\alpha_{max}} d\alpha\, \overline{\alpha_I}\,\overline{\xi}_I\, \frac{\left(\overline{f^I(\alpha)}\right)^{5/2}}{\sqrt{\alpha}}$$

$$= c \sum_I \int_0^{\alpha_{max}} d\alpha\, \overline{\alpha_I}\,\overline{\xi}_I\, \frac{\left(\overline{f^I(\alpha)}\right)^{5/2}}{\sqrt{\alpha}}$$

or, $$\frac{T_2}{c} = \sum_I \int_0^{\alpha_{max}} d\alpha\, \overline{\alpha_I}\,\overline{\xi}_I\, \frac{\left(\overline{f^I(\alpha)}\right)^{5/2}}{\sqrt{\alpha}}$$

$$\Rightarrow \frac{T_2}{c} = \overline{\alpha_2}\,\overline{\xi}_2 \int_0^{\alpha_2} d\alpha\, \frac{\left[\overline{f_2(\alpha)}\right]^{5/2}}{\sqrt{\alpha}} \quad \text{— (2)}$$

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Adding eqⁿ ① & eqⁿ ②

$$\frac{T_1}{c}+\frac{T_2}{c}=\left[\alpha_1\xi_1(\beta_{11})^{5/2}+\alpha_2\xi_2(\beta_{21})^{5/2}\right]\int_0^{\bar{x}_2}dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

$$+\left[\alpha_1\xi_1(\gamma_{12})^{5/2}+\alpha_2\xi_2\right]\int_{\bar{x}_2}^{x_2}dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}}+\alpha_1\xi_1\int_{x_2}^{x_1}dx\,\frac{(f_1(x))^{5/2}}{\sqrt{x}}$$

$$+\bar{\alpha}_2\bar{\xi}_2\int_0^{\bar{x}_2}dx\,\frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}}$$

$$=\left[\alpha_1\xi_1(\beta_{11})^{5/2}+\alpha_2\xi_2(\beta_{21})^{5/2}+\bar{\alpha}_2\bar{\xi}_2\right]\int_0^{\bar{x}_2}dx\,\frac{(\overline{f_2(x)})^{5/2}}{\sqrt{x}}$$

$$+\left[\alpha_1\xi_1(\gamma_{12})^{5/2}+\alpha_2\xi_2\right]\int_{\bar{x}_2}^{x_2}dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}}+\alpha_1\xi_1\int_{x_2}^{x_1}dx\,\frac{(f_1(x))^{5/2}}{\sqrt{x}}$$

$$=\left[\alpha_1\xi_1(\beta_{11})^{5/2}+\alpha_2\xi_2(\beta_{21})^{5/2}+\bar{\alpha}_2\bar{\xi}_2\right]\left[\frac{4}{7}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\bar{x}_2}\right.$$

$$\left.-\frac{5\bar{N}_2}{2\eta}\,\overline{\frac{df_2(x)}{dx}}\Big|_{\bar{x}_2}\right]+\left[\alpha_1\xi_1(\gamma_{12})^{5/2}+\alpha_2\xi_2\right]\left[\frac{4}{7}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}\right.$$

$$-\frac{4}{7}(\beta_{21})^{5/2}\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\bar{x}_2}-\frac{5}{7\varphi_2}\overline{\frac{df_2(x)}{dx}}\Big|_{x_2}\left[\frac{N_2\varphi_2}{3\eta}+\beta_{21}\frac{\bar{N}_2\bar{\varphi}_2}{3\eta}\right.$$

$$\left.-(\beta_{21})^{3/2}\frac{N_2\varphi_2}{3\eta}\right]+\frac{5}{7\varphi_2}(\beta_{21})^2\frac{\bar{N}_2\bar{\varphi}_2}{3\eta}\,\overline{\frac{df_2(x)}{dx}}\Big|_{\bar{x}_2}$$

$$+\alpha_1\xi_1\left[\frac{4}{7}\sqrt{x}\,(f_1(x))^{5/2}\Big|_{x_1}-\frac{4}{7}(\gamma_{12})^{5/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}\right.$$

$$-\frac{5}{7\varphi_1}\frac{df_1(x)}{dx}\Big|_{x_1}\left\{\frac{N_1\varphi_1}{3\eta}+\gamma_{12}\frac{N_2\varphi_2}{3\eta}-(\gamma_{12})^{3/2}\frac{N_2\varphi_1}{3\eta}+\frac{\bar{N}_2\varphi_1}{3\eta}\left((\gamma_{12})^{3/2}(\beta_{21})^{3/2}\right.\right.$$

$$\left.-(\beta_{11})^{3/2}\right)-(\gamma_{12})(\beta_{21})^{3/2}\frac{N_2\varphi_2}{3\eta}+(\gamma_{12})(\beta_{21})\frac{\bar{N}_2\bar{\varphi}_2}{3\eta}\Big\}$$

$$\left.+\frac{5}{7\varphi_1}(\gamma_{12})^2\frac{df_2(x)}{dx}\Big|_{x_2}\left\{\frac{N_2\varphi_2}{3\eta}+\beta_{21}\frac{\bar{N}_2\bar{\varphi}_2}{3\eta}-(\beta_{21})^{3/2}\frac{N_2\varphi_2}{3\eta}\right\}\right]$$

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$$= \frac{4}{7}\left[\alpha_1\xi_1(\beta_{12})^{5/2}+\alpha_2\xi_2(\beta_{22})^{5/2}+\overline{\alpha_2}\,\overline{\xi_2}\right]\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\overline{x_2}}$$

$$-\frac{5\overline{N_2}}{2+\eta}\left[\alpha_1\xi_1(\beta_{12})^{5/2}+\alpha_2\xi_2(\beta_{22})^{5/2}+\overline{\alpha_2}\,\overline{\xi_2}\right]\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}+\frac{4}{7}\left[\alpha_1\xi_1(\gamma_{12})^{5/2}+\alpha_2\xi_2\right]$$

$$\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}-\frac{4}{7}(\beta_{22})^{5/2}\left[\alpha_1\xi_1(\gamma_{12})^{5/2}+\alpha_2\xi_2\right]\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\overline{x_2}}$$

$$-\frac{5}{7\varphi_2}\left[\frac{N_2\varphi_2}{3\eta}+\beta_{22}\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}-(\beta_{22})^{3/2}\frac{\overline{N_2}\varphi_2}{3\eta}\right]\left[\alpha_1\xi_1(\gamma_{12})^{5/2}+\alpha_2\xi_2\right]\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+\frac{5}{7\varphi_2}(\beta_{22})^2\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}\left[\alpha_1\xi_1(\gamma_{12})^{5/2}+\alpha_2\xi_2\right]\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}+\frac{4}{7}\alpha_1\xi_1\sqrt{x}\,(f_1(x))^{5/2}\Big|_{x_1}$$

$$-\frac{4}{7}\alpha_1\xi_1(\gamma_{12})^{5/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}-\frac{5\alpha_1\xi_1}{7\varphi_1}\Big\{\frac{N_1\varphi_1}{3\eta}+\gamma_{12}\frac{N_2\varphi_2}{3\eta}$$

$$-(\gamma_{12})^{3/2}\frac{N_2\varphi_1}{3\eta}+(\gamma_{12})^{3/2}(\beta_{22})^{3/2}\frac{\overline{N_2}\varphi_1}{3\eta}-(\beta_{12})^{3/2}\frac{\overline{N_2}\varphi_1}{3\eta}-(\gamma_{12})(\beta_{22})^{3/2}$$

$$\frac{\overline{N_2}\varphi_2}{3\eta}+(\gamma_{12})(\beta_{22})\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}\Big\}\frac{df_1(x)}{dx}\Big|_{x_1}+\frac{5}{7\varphi_1}(\gamma_{12})^2\Big\{\frac{N_2\varphi_2}{3\eta}+\beta_{22}\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}$$

$$-(\beta_{22})^{3/2}\frac{\overline{N_2}\varphi_2}{3\eta}\Big\}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= \frac{4}{7}\left[\alpha_1\xi_1(\beta_{12})^{5/2}+\alpha_2\xi_2(\beta_{22})^{5/2}+\overline{\alpha_2}\,\overline{\xi_2}\right]\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\overline{x_2}}$$

$$-\frac{4}{7}(\beta_{22})^{5/2}\left[\alpha_1\xi_1(\gamma_{12})^{5/2}+\alpha_2\xi_2\right]\sqrt{x}\,(\overline{f_2(x)})^{5/2}\Big|_{\overline{x_2}}+\frac{4}{7}\left[\alpha_1\xi_1(\gamma_{12})^{5/2}\right.$$

$$\left.+\alpha_2\xi_2\right]\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}-\frac{4}{7}\alpha_1\xi_1(\gamma_{12})^{5/2}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}+\frac{4}{7}\alpha_1\xi_1$$

$$\sqrt{x}\,(f_1(x))^{5/2}\Big|_{x_1}+\frac{5}{7\varphi_2}(\beta_{22})^2\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}\left[\alpha_1\xi_1(\gamma_{12})^{5/2}+\alpha_2\xi_2\right]\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$

$$-\frac{5\overline{N_2}}{2+\eta}\left[\alpha_1\xi_1(\beta_{12})^{5/2}+\alpha_2\xi_2(\beta_{22})^{5/2}+\overline{\alpha_2}\,\overline{\xi_2}\right]\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}-\frac{5\overline{\alpha_1}\xi_1(\gamma_{12})^{5/2}}{7\varphi_2}$$

$$\left.+\alpha_2\xi_2\right]\left[\frac{N_2\varphi_2}{3\eta}+\beta_{22}\frac{\overline{N_2}\,\overline{\varphi_2}}{3\eta}-(\beta_{22})^{3/2}\frac{\overline{N_2}\varphi_2}{3\eta}\right]\frac{df_2(x)}{dx}\Big|_{x_2}+$$

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$$+\frac{5}{7\Phi_1}(\gamma_{12})^2\left\{\frac{N_2\Phi_2}{3\eta}+\beta_{22}\frac{\overline{N_2}\,\overline{\Phi_2}}{3\eta}-(\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}\,\Phi_2}{3\eta}\right\}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$-\frac{5\alpha_1\xi_1}{7\Phi_1}\left\{\frac{N_1\Phi_1}{3\eta}+\gamma_{12}\frac{N_2\Phi_2}{3\eta}-(\gamma_{12})^{3/2}\frac{N_2\Phi_1}{3\eta}+(\gamma_{12})^{3/2}(\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}\,\Phi_1}{3\eta}\right.$$

$$\left.-(\beta_{1\bar{2}})^{3/2}\frac{\overline{N_2}\,\Phi_1}{3\eta}-(\gamma_{12})(\beta_{2\bar{2}})^{3/2}\frac{\overline{N_2}\,\Phi_2}{3\eta}+(\gamma_{12})(\beta_{2\bar{2}})\frac{\overline{N_2}\,\overline{\Phi_2}}{3\eta}\right\}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$\Rightarrow\frac{4}{7}\left[\alpha_1\xi_1(\beta_{1\bar{2}})^{5/2}+\alpha_2\xi_2(\beta_{2\bar{2}})^{5/2}+\overline{\alpha_2}\,\overline{\xi_2}-\alpha_1\xi_1(\gamma_{12})^{5/2}(\beta_{2\bar{2}})^{5/2}\right.$$

$$\left.-\alpha_2\xi_2(\beta_{2\bar{2}})^{5/2}\right]\sqrt{x}\,\overline{(f_2(x))}^{5/2}\Big|_{x_2}+\frac{4}{7}\left[\alpha_1\xi_1(\gamma_{12})^{5/2}+\alpha_2\xi_2\right.$$

$$\left.-\alpha_1\xi_1(\gamma_{12})^{5/2}\right]\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}+\frac{4}{7}\alpha_1\xi_1\sqrt{x}\,(f_1(x))^{5/2}\Big|_{x_1}$$

$$+\frac{5\overline{N_2}}{21\eta}\left[(\beta_{2\bar{2}})^2\frac{\overline{\Phi_2}}{\Phi_2}\left\{\alpha_1\xi_1(\gamma_{12})^{5/2}+\alpha_2\xi_2\right\}-\alpha_1\xi_1(\beta_{1\bar{2}})^{5/2}\right.$$

$$\left.-\alpha_2\xi_2(\beta_{2\bar{2}})^{5/2}-\overline{\alpha_2}\,\overline{\xi_2}\right]\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}+\frac{5}{21\eta}\left[\frac{(\gamma_{12})^2}{\Phi_1}\left\{N_2\Phi_2+\beta_{2\bar{2}}\overline{N_2}\,\overline{\Phi_2}\right.\right.$$

$$\left.\left.-(\beta_{2\bar{2}})^{3/2}\overline{N_2}\Phi_2\right\}-\left\{\alpha_1\xi_1(\gamma_{12})^{5/2}+\alpha_2\xi_2\right\}\left\{N_2+\beta_{2\bar{2}}\frac{\overline{N_2}\,\overline{\Phi_2}}{\Phi_2}\right.\right.$$

$$\left.\left.-(\beta_{2\bar{2}})^{3/2}\overline{N_2}\right\}\right]\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{5\alpha_1\xi_1}{21\eta}\left[N_1+\gamma_{12}\frac{N_2\Phi_2}{\Phi_1}\right.$$

$$+(\gamma_{12})^{3/2}\left\{(\beta_{2\bar{2}})^{3/2}\overline{N_2}-N_2\right\}-(\beta_{1\bar{2}})^{3/2}\overline{N_2}-(\gamma_{12})(\beta_{2\bar{2}})^{3/2}\frac{N_2\Phi_2}{\Phi_1}$$

$$\left.+(\gamma_{12})(\beta_{2\bar{2}})\frac{\overline{N_2}\,\overline{\Phi_2}}{\Phi_1}\right\}\frac{df_1(x)}{dx}\Big|_{x_1}=\frac{(T_1+T_2)}{c}$$

Total kinetic energy

The value of 'c' is already known.

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Therefore, the expression of kinetic energy is

$$\frac{(K.E)}{C} = \frac{4}{7}\left[\alpha_1\xi_1\left\{(\beta_{12})^{5/2} - (\gamma_{12})^{5/2}(\beta_{22})^{5/2}\right\} + \bar{\alpha}_2\bar{\xi}_2\right]\sqrt{x}\,(f_2(x))^{5/2}\Big|_{\bar{x}_2}$$

$$+ \frac{4}{7}\alpha_2\xi_2\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2}^{+} + \frac{4}{7}\alpha_1\xi_1\sqrt{x}\,(f_1(x))^{5/2}\Big|_{x_1}$$

$$+ \frac{5\bar{N}_2}{2\eta}\left[(\beta_{22})^2\frac{\bar{\varphi}_2}{\varphi_2}\left\{\alpha_1\xi_1(\gamma_{12})^{5/2} + \alpha_2\xi_2\right\} - \alpha_1\xi_1(\beta_{12})^{5/2}\right.$$

$$\left. - \alpha_2\xi_2(\beta_{22})^{5/2} - \bar{\alpha}_2\bar{\xi}_2\right]\frac{df_2(x)}{dx}\Big|_{\bar{x}_2}^{+} + \frac{5}{2\eta}\left[\frac{(\gamma_{12})^2}{\varphi_1}\left\{N_2\varphi_2 + \beta_{22}\bar{N}_2\bar{\varphi}_2\right.\right.$$

$$\left. - (\beta_{22})^{3/2}\bar{N}_2\varphi_2\right\} - \left\{\alpha_1\xi_1(\gamma_{12})^{5/2} + \alpha_2\xi_2\right\}\left\{N_2 + \beta_{22}\frac{\bar{N}_2\bar{\varphi}_2}{\varphi_2}\right.$$

$$\left.\left. - (\beta_{22})^{3/2}\bar{N}_2\right\}\right]\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\alpha_1\xi_1}{2\eta}\left[N_1 + \gamma_{12}\frac{N_2\varphi_2}{\varphi_1}\right.$$

$$+ (\gamma_{12})^{3/2}\left\{(\beta_{22})^{3/2}\bar{N}_2 - N_2\right\} - (\beta_{12})^{3/2}\bar{N}_2 - (\gamma_{12})(\beta_{22})^{3/2}\frac{\bar{N}_2\varphi_2}{\varphi_1}$$

$$\left. + (\gamma_{12})(\beta_{22})\frac{\bar{N}_2\bar{\varphi}_2}{\varphi_1}\right\}\frac{df_1(x)}{dx}\Big|_{x_1} \quad \#$$

Where,

$$c = \frac{12\hbar^2\sqrt{R}}{20\pi m_1}\left\{\frac{2\times(4/3)\alpha_s}{9}\right\}^{5/2}$$

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